Industrial Robot

THE400 Safety Manual

(Compatible Controller: TS5000-SS)

Rev. 1

October 1, 2022

Important

- Make sure that this instruction manual is delivered to the final user of Shibaura Machine's industrial robot.
- Be sure to read this manual and that you fully understand its contents before operating the industrial robot.
- After reading this manual, keep it in an easily accessible place for future reference.

Original Instructions

Shibaura Machine

This manual is intended for the following robots.
- THE400

This manual is intended for the following controllers.
- TS5000-SS

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Introduction

Thank you for purchasing Shibaura Machine's robot system.
This manual describes the safety measures for the Shibaura Machine THE-series system robot.
To fully understand the performance of the robot and to use its functions safely for a long time, please read this manual carefully, and operate the robot by following the instructions.

This manual has the following chapters.

- Scope Of Use
This explains the intended scope of use for this product.

- Usage Procedures
This explains the main specifications of this product and how to use it safely.

- Warranty
This explains the terms and conditions of the warranty.

- Standards, laws, and regulations pertaining to the safety of industrial robots
This explains the main standards, laws, and regulations that must be followed when using the robot.

- Safety Information
This provides important information for using the robot safely and correctly.

- Service Network
This provides the contact information for technical support and requesting service.

- Warning Label Locations
This shows the locations where warning labels are affixed to the robot and controller.

- Safety Measures
This explains the safety protection features of the robot and controller itself and important safety information for installing and running robots.

- Applicable Standards and Safety Measures
This shows the compliance of the robot and controller with various standards and safety measures.

Manufacturer

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If you have purchased this robot outside of Japan, please contact the company that you purchased it from.

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Scope of Use

This product is a horizontally articulated industrial robot where the manipulator has two horizontally rotating joints and the mechanical interface on its tip moves vertically and rotates horizontally.
The product has been developed for use in a wide range of automation tasks including transport work such as the movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product for the purpose of automating these tasks and for incorporating into automation equipment.

This product is not intended for use in tasks involving human contact (such as tasks for medical care and nursing care) or to be incorporated in equipment that sustains human life (such as life support devices and incubators), equipment on which people ride (including transportation equipment such as cars and aerospace equipment such as airplanes, and recreational equipment such as roller coasters), and equipment that protects human life or the human body (such as safety devices).
This product is not intended to be incorporated in equipment for controlling large heat quantities (such as nuclear power controllers and incinerator controllers).
This product is prohibited from being used for tasks or incorporated in equipment where a failure or malfunction could likely pose a direct threat to human life or harm to the human body. The customer is responsible for providing any warranties if this product is used.

Usage Procedures

This section describes the usage requirements for the THE-Series. Usage of the product in a state outside the required conditions could result in unexpected risks. Be sure to always use the robot in an environment that meets the requirements.

1. Main Specifications of Product: See "1.4 Specifications Table" in the Robot Manual SPECIFICATIONS and "3.1 Installation Environment" in INSTALLATION.
2. Usage Restrictions: Tables 1 to 7 provide the “Restricted Specifications Sheet” indicating the usage restrictions.

Table 1. Restricted Specifications Sheet (1)

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Clear a transportation route to the unpacking site. [2] Use a forklift or other transfer equipment to hoist the packing box and check its balance. [3] Transport to the unpacking site. (2) Unpacking [1] Clear the space required for unpacking. [2] Unpack. [3] Check that the robot and controller are not damaged. [4] Check the accessories and their quantities. [5] Dispose of, or store and manage the packing materials. (3) Transfer [1] Check the orientation of the robot during transfer. [2] Check the mounting of the robot fixing jigs. [3] Check the tightness of the fixing bolts of the robot fixing jig. [4] Remove the bolts securing the robot base. [5] Hold and carry the robot using at least two people. (4) Installation [1] Secure the robot to the frame and other parts. [2] Remove the robot fixing jigs. [3] Install the controller, and route the wiring. [4] Mount the hand and other parts.

Table 2. Restricted Specifications Sheet (2)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(5) Teaching - Assign a teaching work supervisor who is stationed outside the hazard zone. - Work shall be performed in compliance with the work regulations. [1] Move the movable part of the robot to the desired position by guiding with a teach pendant or by grasping the robot arm or other part by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation - Measures shall be implemented to turn off the servo when someone approaches the robot hazard zone. - Work shall be performed in compliance with the work regulations. [1] Check that no one is inside the safety fence or other enclosure, and then start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [1] Remove the hand and other parts. [2] Mount the robot fixing jigs. [3] Remove the wiring. [4] Pack the robot and controller.

Table 3. Restricted Specifications Sheet (3)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance - Assign a maintenance work supervisor who is stationed outside the hazard zone. - Work shall be performed in compliance with the work regulations. Work performed when the primary power is off: [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check the tension of the timing belt and look for any cracks or other imperfections. [7] Check for any wear on the cable and air tube. [8] Replace the consumable parts for the robot and controller. [8] Replace the memory batteries in the controller. Work performed when the primary power is on: [1] Check the grease state and apply grease (when the servo power is off). [1] Use the teach pendant to check the robot behavior when moving by jog guidance (when the servo power is on). [2] Press each axis by hand to check for any rattling (when the servo power is on). [3] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [4] Check the teaching point for any misalignment (when the servo power is on). [5] Replace the batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Mount the fixing jigs to the robot, and dispose of using a method compliant with your country's laws and regulations.
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or where excessive impact is applied by the robot - Modification of robot, controller, or optional parts, or supplying power while protective covers are removed - Operation in other than the specified installation state (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that a stop to wait for an operation signal during operation was an operation stop - Usage under conditions where the specification values are exceeded or usage under improper settings - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing of a button - Operating without checking the movement in teaching mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm

Table 4. Restricted Specifications Sheet (4)

1. Purpose and application for using product (usage restrictions)	
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Unintended startup due to failure or other cause

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 5. Restricted Specifications Sheet (5)

3. Movable range of product (restrictions on space)		
1) Work envelope	1st axis 2nd axis 3rd axis 4th axis Additional axis — —	See the THE400 TS5000 Series Robot Manual, "1.4 Specifications Table". See the THE400 TS5000 Series Robot Manual, "1.4 Specifications Table". See the THE400 TS5000 Series Robot Manual, "1.4 Specifications Table". See the THE400 TS5000 Series Robot Manual, "1.4 Specifications Table". Compliant with individual specifications. — —
2) Interface	Transportation: Unpacking: Transfer: Installation: Teaching: Operation: Repacking: Maintenance: Disposal: 	The packed robot shall be able to be transferred by a transportation worker. The packed robot shall be able to be carried in. Space of at least 2 m^3 shall be provided. The robot shall be able to be lifted by two or more people and carried or shall be able to be transferred by an overhead crane or other equipment. There shall be no obstacles on the route from the unpacking site to the installation site. The robot shall be able to be moved from the unpacking location to the installation site. Installation shall be performed at a location suitable for the outline and work envelope of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cable shall be installed at a location where it will not be stepped on by operators. Teaching work shall be able to be performed from outside the hazard zone of the robot. Space shall be provided so that, if someone enters the hazard zone, they will not get entrapped by the robot or peripheral equipment. A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Work shall be able to be performed to remove the robot hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. Space required for maintenance shall be obtained. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Space shall be provided so that, in the hazard zone, a person will not get entrapped by the robot or peripheral equipment. Work shall be able to be performed to remove the robot hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the area during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be generated.	

Table 6. Restricted Specifications Sheet (6)

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, risk assessment is carried out for the life cycle of “transfer” (transportation, unpacking, transfer, installation, and repacking), “use” (teaching, operation, and maintenance), and “disposal”.

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for transfer equipment work as defined by the respective country’s laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	—
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	(When transfer equipment is used) Person who has received training for transfer equipment work as defined by the respective country’s laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual
	Requirement	—
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country’s laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	—
7) Repacking worker	Knowledge	Contents of the instruction manual
	Requirement	(When transfer equipment is used) Person who has received training for transfer equipment work as defined by the respective country’s laws and regulations and who has been permitted to perform the work by the employer

Table 7. Restricted Specifications Sheet (7)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot inspection work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual and waste disposal methods
	Requirement	(When transfer equipment is used) Person who has received training for transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If another worker enters the hazard zone of the robot to perform work, robot operation shall be stopped, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

6. Other

Warranty

We conduct strict testing and inspection of this product before delivery to ensure that it satisfies our performance standards.
We warrant this product to be free from defects under the following terms and conditions.

1. Warranty period The warranty period of the product is one of the following, whichever is shorter.
   1) 18 months after shipping from our factory
   2) 12 months after installation at customer site

2. Warranty scope
   1) The warranty is provided for this product. The warranty covers only the specifications and functions included in the product specifications, catalog, and instruction manual. Any secondary or incidental damages caused by failure of this product are not covered by the warranty under any circumstances.
   2) We will repair a faulty product free of charge only if this product has been operated and used in accordance with the instruction manual supplied with the product.
      A fee will be charged for any repairs after the warranty period has expired.

3. Disclaimers The warranty does not include the following cases.
   a) Any failure or damage arising from careless use or misuse, or usage that is not described in the instruction manual.
   b) Any failure caused by aging or normal wear and tear (natural fading of paints, deterioration of consumable parts[*1]).
   c) Irregularities confined to sensory quality such as noise not recognized as affecting functions
   d) When any modification or disassembly without our consent was performed
   e) Any failure or damage arising from inadequate maintenance and inspection or improper repair
   f) Any failure or damage due to an act of God, fire, and other force majeure
   g) Internal data such as programs and points created or modified by customers
   h) When a product purchased in Japan is taken abroad

4. Safety Information
   1) Shibaura Machine shall provide no guarantees for the standard performance of this product when used outside the specifications.
   2) If the customer does not observe the safety information appearing for DANGER, WARNING, and CAUTION contained in this instruction manual, Shibaura Machine will not be liable for any resulting injuries, death, or other accidents, damage, product failure, or any other problems.

3) The safety information appearing for DANGER, WARNING, CAUTION and other explanations in this instruction manual are within the scope of usage assumed by Shibaura Machine, and they do not foresee all possible incidents for all situations that could occur. Please be aware that the scope of this safety information is limited.

*1): Consumable parts refer to the service replacement parts as listed in the instruction manual “THE600-E Robot Manual: Maintenance".

Standards, laws, and regulations pertaining to the safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" specifies risk reduction measures for machinery and equipment.
Step 1: Inherently safe design measures - Limitations on force, speed, energy, etc.
Step 2: Safeguarding - Installation of safety fences, etc.
Complementary protective measures - Installation of emergency stop devices, etc.
Step 3: Information for use - Warning signs, alarms, instruction manuals, etc.

Based on the above, the ISO/IEC has defined "Guide 51", which classifies the various safety standards in a hierarchal structure. Safety standards for industrial robots are standardized as separate safety standards for machinery in Type C.

- ISO 10218, ISO 10218-1

The providing of residual risk information to users after risk assessment has been conducted is standardized.

- IEC 82079-1

Be sure to always follow the standards, laws, and regulations set forth in the country where the industrial robot is used.

Safety Information

This instruction manual contains important information to ensure safe and correct use and to prevent injury to the operator and others and damage to property.
In order to ensure safety from robot installation to disposal, be sure to carefully read this manual before starting the actual work.
Be sure that you fully understand the following information (signal words and symbols) before reading the main text, and always follow the provided safety instructions.

Display of Signal Words and Their Meanings

Signal word and symbol	Meaning
⚠ DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury.
⚠ CAUTION	Indicates the possibility that incorrect operation could cause injury[1] to the operator.
NOTICE	Indicates the possibility that incorrect operation could cause property damage[2] and provides helpful advice about performing work.

1) Injury refer to wounds, burns, electric shocks, and similar cases that do not require hospitalization or prolonged outpatient treatment.
2) Damage to property refers to extensive damage to property or materials.

Work Scope

The expected work scope for each life cycle is shown below.
In actuality, some tasks cannot be clearly classified and others are spread over multiple life cycles. These classifications are intended as a general guideline only, and so please check the safety information for all life cycles, and take protective measures before starting work.

- Transportation: From unloading packing boxes brought in by transportation equipment (truck, etc.) to carrying them to the unpacking location.

- Unpacking: From opening the packing box to taking out the product.

- Transfer: From taking out the product from the packing box to carrying it to the installation site.

- Installation: From installation of the product, wiring, piping, assembly of the hand and optional parts until it is powered on.

- Teaching: Changing parameters, teaching, programming, and checking operation.

- Operation: Operation of equipment incorporating the robot.

- Repacking: Shipping of a product to another installation site after performing operation verification.

- Maintenance: Maintenance and inspection of product.

- Disposal: Disposal of parts and other items that are no longer needed.

Transportation

This section provides important safety information for transporting the robot. Read this information together with "STEA1213: THE400 TS5000 Series Robot Manual: Installation", and be sure to strictly observe the following items.

⚠ DANGER

- When lifting packing boxes, never carry the packing boxes by hand.
  A packing box could fall on a worker.
  - The forklift must be operated by a qualified operator.
  - Align the center of the packing box with the center of the forks when lifting the packing box.
  - Do not allow people to approach the transportation route.

This section provides important safety information for transporting the controller. Read this information together with "STEA1237: TS5000-SS Controller Manual: Installation", and be sure to strictly observe the following items.

⚠ CAUTION

- Do not stack the packing boxes. The load could collapse, resulting in an injury.
- Do not allow the controller to get wet. Moisture could seep inside, resulting in a breakdown.
- Wear protective gloves and shoes when carrying the controller by hand. Dropping could result in crushed hands or feet.
- Use a cart or similar device for transporting the controller. Carrying by hand can lead to dropping, resulting in crushed hands or feet.

NOTICE

- Do not subject the controller to excessive vibrations during transportation. This could damage the controller.
- Do not transport upside down, and do not store on its side. This could damage the controller.

Unpacking

This section provides important safety information for unpacking the robot. Read this information together with "STEA1213: THE400 TS5000 Series Robot Manual: Installation", and be sure to strictly observe the following items.

⚠ DANGER

- After the work is completed, be sure to always check that the robot is not damaged.
  Turning on the power for a malfunctioning robot can lead to an expected accident.
  - Check the appearance based on the external view shown in the manual.
  - Check parts based on the accessory list attached to the controller.
  - If damage is found or parts are missing, please contact the Shibaura Machine Sales Division.
- When carrying the packing boxes to the storage site, never carry the packing boxes by hand.
  A packing box could fall on a worker.
  - Use a hand lifter or similar equipment.
  - Align the center of the packing box with the center of the forks when lifting the packing box.
  - Do not allow people to approach the transportation route.

⚠ WARNING

- When cutting the packing strap with the cutter, never point the blade edge towards a person.
  Otherwise, the person could get cut by the cutter.
  - Use a nippers or scissors.
  - Wear protective gloves.

⚠ CAUTION

- When storing the robot, be sure to insert in the packing box and secure it in place as it was delivered.
  Otherwise, the robot could fall over.
  - The customer must obtain new packing straps (PP bands).
  - If no packing box is available, use the mounting holes in the base to secure in place.

NOTICE

- After the work is completed, be sure to always check that the robot is not damaged.
  A damaged robot could malfunction.
  - Check the appearance based on the external view shown in the manual.
  - Check parts based on the accessory list attached to the controller.
  - If damage is found or parts are missing, please contact the Shibaura Machine Sales Division.
- Be sure to dispose of any unneeded packing materials in accordance with the customer's company regulations.
  When storing the packing materials, be sure to manage them so that they are not used for purposes other than packing the robot.
- When storing the robot, seal it in a plastic bag, and put a desiccant inside the bag.
  This prevents rust from forming and dust from adhering to the ball screw spline unit.
  - Before storing the robot, apply grease to the ball screw spline shaft.
  - For details on how to apply grease, see "STEA1213: THE400 TS5000 Series Robot Manual: Maintenance".
  - Also apply grease to the ball screw spline shaft before use.
- Before operating a robot that was stored, be sure to perform breaking-in operation.
  During storage, the grease viscosity increases. If the robot is suddenly operated at a high speed, the poor lubrication could result in failure.
  - Be sure to perform operation at a low speed (about 20% override) for 5 minutes.
- Before operating a robot that was stored, be sure to replace the encoder batteries. See "STEA1213: THE400 TS5000 Series Robot Manual: Maintenance".
  An alarm occurs before periodic inspection.
  - If power is not supplied to the robot for an extended period of time due to storage or other reasons, the data retention period by the encoder batteries will be shorter.
- When storing the robot, use the supplied fixtures to secure the arm and base, and then firmly secure the mounting sections. If placed directly on the floor, the robot will be unstable and will fall over.
- Never store the robot at a location exposed to direct sunlight, high temperature, or high humidity.
  This could cause the cable ducts, timing belt, and resin parts to deteriorate.
- Do not store the robot at a location where it is exposed directly to rain or water droplets, or at a location exposed to any toxic gas or liquid. This could cause premature deterioration of parts.

This section provides important safety information for unpacking the controller. Read this information together with "STEA1237: TS5000-SS Controller Manual: Installation", and be sure to strictly observe the following items.

⚠ CAUTION

- After unpacking, wear protective gear when disposing of unneeded wooden pallets and cardboard boxes. Otherwise, you could be cut by nails sticking out and by other sharp locations, resulting in an injury.
- Use a box cutter or similar tool when opening the packing box. You could cut your finger or hand if using a cutter with a sharp edge.
- Unpack at a stable location. If unpacked at an unstable location, the controller could fall and crush your hands or feet or strain your back.
- Wear protective gloves when unpacking. Your hand could get trapped between a cardboard box and the controller, resulting in an injury. You could also be injured by the corners of the controller or cardboard box.

NOTICE

- Do not allow the controller to get wet. Moisture could seep inside, resulting in a breakdown.

Transfer

This section provides important safety information for transferring the robot. Read this information together with "STEA1213: THE400 TS5000 Series Robot Manual: Installation", and be sure to strictly observe the following items.

⚠ DANGER

- The robot should never be transferred by only one worker.
  The robot cannot be fully supported and could drop.
  - The robot weight is 31 kg.
  - Lift the robot using more than one person.
  - Use a cart or similar device to move to the transfer location.
  - Clear a transfer route, and make sure that a dropped robot would not hit anyone.
  - Wear protective gloves and safety shoes.
- When hoisting the robot, be sure to always observe the following safety guidelines.
  Otherwise, the robot could drop.
  - Use the attached eyebolts.
  - The crane must be operated by a qualified operator.
  - Use a wire with a load limit that can sufficiently bear the weight of the robot.
- Be sure to always hoist and lower the robot slowly.
  Otherwise, the robot could be tilted and fall.
  - Clear a transfer route, and make sure that no one goes under the suspended load.

⚠ WARNING

- When transferring the robot, be sure to always secure in place using the supplied fixing jigs.
  Otherwise, a worker could get caught in an arm that moved.
  - The weight of the arm section is approximately 17 kg.
  - Wear protective gloves to prevent slippage and to protect the hands.

NOTICE

- The robot should never be transferred by only one worker.
  The robot cannot be fully supported and could drop.
  - Do not use a product that has been dropped.
- Never touch the ball screw spline shaft with your bare hands.
  This could cause rust to form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- Be sure to hoist and lower the robot gently.
  Otherwise, the robot could collide with the peripheral equipment, frame, or other objects, resulting in failure or damage.
- When lifting up the robot by workers, be sure to hold the appropriate sections shown in "STEA1213: THE400 TS5000 Series Robot Manual: Installation".
  If other sections are held, the held sections may be damaged. This could apply excessive force to the mechanical section, resulting in a breakdown.
- After installation is completed, be sure to remove the fixing jig and eyebolts.
  If the robot is operated while the jig is mounted, the robot may be damaged or fail.

This section provides important safety information for transferring the controller. Read this information together with "STEA1237: TS5000-SS Controller Manual: Installation", and be sure to strictly observe the following items.

⚠ CAUTION

- Do not carry by holding the AC power supply unit. Carrying by holding the AC power supply unit could damage the controller or cause it to drop. Also, if AC power is being supplied, it could result in an electric shock.
- Wear protective gloves and shoes when carrying the controller by hand. Dropping could result in crushed fingers or feet.
- Use a cart or similar device for transporting and carrying. Carrying by hand can lead to dropping.
- Carry with the controller cables removed. Carrying with the cables connected could cause cable breakage, resulting in an electric shock.

NOTICE

- Do not subject the controller to excessive vibrations. This could damage the controller.
- Do not allow the controller to get wet. Moisture could seep inside, resulting in a breakdown.

Installation

This section provides important safety information for installing the robot. Read this information together with "STEA1213: THE400 TS5000 Series Robot Manual: Installation", and be sure to strictly observe the following items.

⚠DANGER

- Never turn on the robot in an environment where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  Doing so could result an electric shock due to a leakage current.
- Never turn on the power for the robot in an environment with dust or flammable gas.
  Doing so could lead to a fire.
  Do not place flammable materials near the robot because they could ignite due to a spark if a malfunction occurs.
- Never touch fluid that has leaked from the encoder batteries.
  If the fluid gets on the skin or eyes, it can cause inflammation.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot and frame could fall over together during operation.
  - Prepare the frame by referring to the load applied during horizontal movement in "STEA1213: THE400 TS5000 Series Robot Manual: Installation".
- Never connect a power supply with a voltage exceeding the specified range to the robot hand wiring cable.
  This could cause the cable to overheat, resulting in a fire.
  - Before connecting, use a tester or similar device to check the voltage of the power supply.

⚠ DANGER

- Be sure to connect the cables to the controller after the robot is installed.
  Otherwise, a worker could receive an electric shock due to touching a cable that was broken due to a carried object that was dropped.
  - If a cable is broken, do not turn on the power.
- Be sure to always store the cables in ducts and similar protection.
  Otherwise, a worker could trip and fall over a cable on the floor.
  - Workers should wear a helmet to protect their head.
- Before turning on the power, always make sure that no one is inside the robot's hazard zone.
  An unexpected motion by the robot could damage the mechanical stopper and crash into someone.
  - Make sure that the operation area limited by the mechanical stoppers is also clear of people. Use of the mechanical stoppers does not guarantee stopping of the robot.
- When fabricating the hand wiring cable, be sure to use the compatible wire.
  Otherwise, the cable could overheat or short circuit between the terminals, resulting in a fire.
- Be sure to lock the robot hand connector and secure the cable to the arm.
  If a load is applied to the connector and it is disconnected, the hand could move unpredictably.
  - Adjust the robot hand wiring and piping while checking the status during robot operation to make sure that they are not in contact with the robot or peripheral equipment and that no load is being applied to the connectors.

⚠ WARNING

- Always check the operation status before approaching a stopped robot.
  Otherwise, if the robot was operating, a collision could occur when the robot restarts its motion.
  - Check the operation status displayed by the controller indicator lamps and the teach pendant.
  - The operation status of the robot is output as signals (servo operation ready, external mode in progress, operation ready, automatic operation in progress, fault occurred, and emergency stop in progress). Connect to the display unit or similar device.
- Before turning on the power, be sure to always check that the ground wire is connected.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.
- Be sure to use a robot hand having a structure so that it will not release the workpiece or other object even if the robot power or robot hand driving power supply is cut off.
  Otherwise, the workpiece or other object could fly off or drop.
  - When the robot is turned off, the signal output to the hand is also turned off.
  - The robot hand signals (DOUT201 to DOUT208) are not reset by performing an operation to reset all output signals (such as the system input signal "Output signal reset" or RESET DOUT command).
  - Conduct daily inspection on the air tube connected to the hand, and conduct periodic inspection on the air tubes in the robot, and replace them if they show any damage.
  - Install a fence or other barrier to protect from flying and dropped objects.
  - When using a robot hand that releases the workpiece when the robot power or hand driving power supply is turned off, post notices to not turn off the power or driving power supply.

⚠ CAUTION

- When removing the fixing jig, never grasp the edges strongly.
  If excessive force is applied, a hand or finger could be cut on the fixing jig.
  - Wear protective gloves.
  - The bolts that are fastened to the robot may become hardened due to vibrations during transportation or transfer.
- Always check the working envelope before using the robot.
  If the working envelope is not limited as expected, you could be struck by an operating robot.

NOTICE

- Never turn on the robot in a location where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  This could cause a breakdown of the robot.
- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot could vibrate during operation. This could also lead to a malfunction of the robot.
  - Prepare the frame by referring to the load applied during horizontal movement in "STEA1213: THE400 TS5000 Series Robot Manual: Installation".
- Never reuse a mechanical stopper that has collided with the arm and was used to change the work envelope.
  A damaged mechanical stopper will not function properly.
- Do not route piping together with the body harness.
  Doing so could apply excessive force on the body harness, resulting in breakage of the internal cables.
- Never use the robot under conditions outside the specification values.
  Otherwise, the performance and lifespan of the robot could be adversely affected.
- Never mistake the terminal arrangement on the hand wiring.
  Doing so could cause the connected parts to fail.
  - Before usage, use a tester or similar device to check the terminal arrangement.

NOTICE

- When operating the robot after an extended stop at a low temperature (10°C or less), be sure to perform operation at a low speed (about 20% of override) for 5 minutes.
  The grease viscosity increases in low-temperature environments. If the robot is suddenly operated at a high speed, the poor lubrication could result in failure.
  - Apply grease to the ball screw spline shaft before use.
- Be sure to route the cables after installation is completed.
  Otherwise, a carried object could drop on a cable that was already routed, resulting in breakage.
  - House the wiring in ducts or other protection.
- Be sure to install the robot on a level surface.
  Otherwise, the performance and lifespan of the robot could be adversely affected. This could also lead to a malfunction of the robot.
- Be sure to secure the frame where the robot is installed to a floor or wall.
  Otherwise, the frame will move due to the force applied during robot operation.
- Be sure to always connect a ground wire.
  Otherwise, the robot may fail due to noise.
- When designing, manufacturing, and mounting the mechanical stoppers, be sure to observe the following points.
  - Be sure to set the changed work envelope to within the range shown in the specification table and external view drawing.
    The robot cannot be operated in a work envelope that is outside the specifications.
  - Set the origin position information again by following the procedure in "14.9 [3] Other Origin Position Settings".
    If a mechanical stopper is changed, the function described in "14.9.1.4 Restoring Data by SYSTEM1 or SYSTEM2" is disabled. At installation, enter the origin position data by following the procedure in "14.9.1.2 Setting HOME1 and HOME2".
  - A function is also available to restore the origin position using a setting jig provided by the customer. If using the procedure in "14.9.1.3 Restoring Data by HOME1 or HOME2", enter the origin position data using the procedure in "14.9.1.2 Setting HOME1 and HOME2" during the installation process.
  - A robot failure caused by a mechanical stopper is outside the scope of the warranty.
- Be sure to set the software limits inside the mechanical stoppers.
  If the software limits are not set inside the mechanical stoppers, the robot will collide with the mechanical stoppers during operation.
- If the service tap is used to mount parts, be sure to always check whether the parts will interfere with the robot or peripheral equipment during operation. Otherwise, this could result in a breakdown of the robot or peripheral equipment.
- Be sure to use the PAYLOAD command.
  Doing so could adversely affect the performance and lifespan of the robot.
  - Even if the load center of gravity offset value is 100 mm or less, on robot hands with a long projection length from the 4th axis rotation center, micro-vibrations at the axis center can be amplified to cause large vibrations on the hand and workpiece tip.

NOTICE

- Be sure to always use the PAYLOAD command.
  Otherwise, vibration or overrun could reduce the lifespan of the robot or damage the robot.
  - The moment of load inertia should be used within the tolerance listed in "STEA1213: THE400 TS5000 Series Robot Manual: Installation".
  - Even if there is no load center of gravity offset, when the moment of inertia is large, the robot could vibrate. In these cases, calculate the virtual center of gravity offset L (mm) from the moment of inertia (kg･m^2) and weight M (kg).
    $L=\sqrt{(J\times10^6/M)}$
    Enter this calculation result.
    PAYLOAD={M,L}
  - Micro-vibrations could occur depending on the robot orientation.
  - If the vibrations or overrun does not subside, use together with the SPEED command or DECEL command.
- Even if the weight is within the maximum payload weight, vibrations could occur depending on the robot orientation, such as when a heavy object is held and the 3rd axis is lowered. In these cases, use the SPEED command, ACCEL/DECEL command, LOCUS command, or similar commands to adjust the operating speed and acceleration/deceleration so that vibration does not occur.
  For the SPEED command, ACCEL/DECEL command, and LOCUS command, refer to “3 Explanation of Command Words” in the separate instruction manual “Robot Language Manual”.
- If the robot vibrates when using manual guidance, be sure to set the PAYLOAD in manual mode by referring to “2.1.7 Setting load for manual guide” in the separate instruction manual “STEA1162: TP5000 Operation Manual”.
  Otherwise, vibration could reduce the performance and lifespan of the robot.
  - If the load weight or center of gravity offset is large, the robot will vibrate.
- When fabricating the hand wiring cable, be sure to use the compatible wire.
  Otherwise, overheating or short circuit could occur, resulting in burning of the cable.
- Be sure to use a highly flexible and twisting-resistant robot cable for the hand wiring cable.
  If a general-purpose cable is used, the wire may break during robot operation.
  - Secure the cable to the arm using a saddle or other mechanism to prevent the cable from being swung around.
- Never connect the hand wiring cable to a power supply whose voltage exceeds the specified range.
  This could cause the cable to overheat, resulting in a fire.
  - Before connecting, use a tester or similar device to check the voltage of the power supply.
- Be sure to perform hand wiring and piping while checking the following status during robot operation.
  Otherwise, the robot and hand will break down prematurely.
  - Check whether a load is being applied to the hand connector.
  - Check whether the hand wiring and piping are contacting the robot or peripheral equipment and are worn.
- The cable between the robot and controller is not a movable cable. If the robot is mounted on a traveling axis or similar structure, the cable will be damaged prematurely.
- The solenoid valve air tubes are not supplied with the robot.
  They should be provided by the customer.

This section provides important safety information for installing the controller. Read this information together with "STEA1237: TS5000-SS Controller Manual: Installation", and be sure to strictly observe the following items.

⚠ DANGER

- Never install in locations with water exposure that exceeds the protection rating. Doing so could lead to an electric shock or fire.
- Never install near flammable objects. If sparks occur due to a breakdown, nearby flammable objects could ignite, resulting in a fire.
- Do not place objects on top of the controller. (Stacking is also prohibited.)
  - Lack of ventilation space and heat buildup could cause a fire.
  - Because the design assumes that no objects are placed on top of the controller, the controller could collapse due to being unable to support the object, resulting in a breakdown of the controller or crush someone's hands or feet.
- Never use unauthorized cables. Fabricating and use of unauthorized cables can result in a fire.
- Do not connect to unauthorized connectors or terminals. Unauthorized connectors and terminal wires could result in a breakdown of the controller, electric shock, or a fire.
- Do not insert conductive objects or your fingers inside the power supply terminals. An electric shock could occur.
- Be sure to always install the controller outside of the robot's hazard zone where you can oversee the robot's motion.
  If the controller is operated in a hazard zone or in a location where the robot or peripheral equipment is not visible, an operator or worker in the hazard zone will be struck by the robot or peripheral equipment.
- If the controller must be installed inside the hazard zone for unavoidable reasons, put the teach pendant outside the hazard zone, and set up so that the controller's system signal is used to output the robot status externally to a signal light or LED for displaying the status.
  If there is no means for checking the robot status, a worker who, due to unavoidable reasons, must check the controller installed in the hazard zone will be struck by the robot or peripheral equipment.
- When installing the controller in a closed space, such in a control panel, use air vent holes, forced ventilation, and indirect cooling to prevent heat buildup. If the controller is installed in a closed space, heat will be trapped, resulting in a breakdown or fire.
- Install the controller inside the control panel or in a location where sufficient distance can be provided so that it cannot come into contact with the operator. If the controller is installed within the operator's range of motion, the operator could fall over with arms or legs caught in a cable, or could receive an electric shock due contact with the energized parts.
- Do not install on an unstable location. The controller could fall and crush someone's hands or feet.
- Be sure to always perform the wiring for the robot after installation is completed.
  - If the controller is installed or transported while the wiring is connected, a cable could be broken, resulting in an electric shock.
  - Also, your legs could get caught in a cable, causing you to trip and fall.
- Provide a dedicated duct or similar protection to protect the cable lead-in of the control panel. If the cable is routed without using a dedicated duct, the cable could break, causing a fire or electric shock. The cable could break, causing a fire or electric shock.
- Mount a cover or other protection for the buttons and switches of the control unit connected to the controller. Accidental contact could turn on a signal, causing the robot to move, resulting in a collision.

⚠ DANGER

- Display the execution mode of the respective controller on the operation device. If the operator does not know the execution mode, a collision could occur with the robot due to an unexpected movement.
- Be sure to always connect a ground wire. Usage without connecting the ground wire can result in ground leakage, electric shock, or a fire.
- After wiring, use a tester or other device to check the voltage and for any short-circuits. If the power is turned on while the wiring is incorrect or while the controller is malfunctioning, an electric shock or fire could occur.
- Be sure to connect all connectors securely. If the connector is not engaged properly, static electricity could enter the controller and cause it to malfunction or cause a collision with the robot.

⚠ WARNING

- Turn off the power before connecting and disconnecting cables from the controller. Touching the terminal of a disconnected cable while the power is on could result in an electric shock, and connecting and disconnecting the cable while the power is on could damage the ICs in the board.
- When disconnecting the cable, pull it out by grasping the connector while holding down the controller.
  - Pulling the cable itself could break the cable and result in an electric shock.
  - The controller could fall over, resulting in a breakdown.

NOTICE

- Never supply a current exceeding the rated output. Supplying a current that exceeds the rated output will damage the output elements, circuit board, and other parts, and could result in a fire.
- If the LAN, OUTPUT, and SD connectors will not be used, be sure to attach the connector caps (dummy SD card for the SD connector). If these connectors do not have any protection, electrostatic charges could cause a breakdown.
- Provide ventilation space around the controller of at least 50 mm on the right, left, and rear sides, 100 mm on the top side, and the height of the rubber feet (11 mm) on the bottom side. If no ventilation space is provided, the controller could overheat, resulting in a breakdown.
- When installing the controller in a closed space, such in a control panel, use air vent holes, forced ventilation, and indirect cooling to prevent heat buildup. If the controller is installed in a closed space, heat will be trapped, resulting in a breakdown or fire.
- If the controller will be installed in the control panel, use fixing jigs or other means to secure the controller in place. If the control panel vibrates, its inner components could move, and the controller could drop, resulting in a fire.
- If the controller will be installed in the control panel, use fixing jigs or other means to secure the controller in place. If the control panel vibrates, its inner components could move, and the controller could drop, resulting in a fire.
- Set the robot hand output signal to 0.1 A per output (max. 0.8 A for total 8 outputs). Supplying a current that exceeds the rated output will damage the output elements, circuit board, and other parts, and could result in a fire.
- Apply a voltage in a range of 24V±10% to the input/output external wire. If a voltage exceeding the tolerance is applied, the controller will be damaged, and a fire could result.

Teaching

This section provides important safety information for teaching. Read this information together with "STEA1213: THE400 TS5000 Series Robot Manual: Installation", and be sure to strictly observe the following items.

⚠DANGER

- Never turn on the power for the robot in an environment where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  Doing so could result an electric shock due to a leakage current.
- Never turn on the power for the robot in an environment with dust or flammable gas.
  Doing so could lead to a fire.
  - Do not place flammable materials near the robot because they could ignite due to a spark if a malfunction occurs.
- Never touch fluid that has leaked from the encoder batteries.
  If the fluid gets on the skin or eyes, it can cause inflammation.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot and frame could fall over together during operation.
  - Prepare the frame by referring to the load applied during horizontal movement (Table 6.2).
- Before turning on the power, always make sure that no one is inside the robot's hazard zone.
  An unexpected motion by the robot could damage the mechanical stopper and crash into someone.
  - Make sure that the operation area limited by the mechanical stoppers is also clear of people. Use of the mechanical stoppers does not guarantee stopping of the robot.
- Be sure to lock the robot hand connector and secure the cable to the arm.
  If a load is applied to the connector and it is disconnected, the hand could move unpredictably.
  - Adjust the robot hand wiring and piping while checking the status during robot operation to make sure that they are not in contact with the robot or peripheral equipment and that no load is being applied to the connectors.
- If teaching work is performed while the power is on in the robot's hazard zone, be sure to switch the master mode key inserted in the teach pendant to the "MANUAL" position, remove the key, and carry it with the teach pendant. (for TP5000)
  Otherwise, you could be struck by a robot that has been started by someone else.
  - Post a notice near the controller that work is in progress.
  - Assign a work supervisor who is stationed outside the hazard zone of the robot.
  - The supervisor should ensure the safety of the worker and any other people approaching the robot hazard zone.
  - Make sure that the worker and the supervisor receive safety training on tasks related to the inspection of industrial robots as specified by the laws and regulations of individual countries.
- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could lead to an unpredictable accident due to a robot malfunction.

⚠ WARNING

- Before turning on the power, be sure to always check that the ground wire is connected.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.
- Be sure to use a robot hand having a structure so that it will not release the workpiece or other object even if the robot power or robot hand driving power supply is cut off.
  Otherwise, the workpiece or other object could fly off or drop.
  - When the robot is turned off, the signal output to the hand is also turned off.
  - The robot hand signals (DOUT201 to DOUT208) are not reset by performing an operation to reset all output signals (such as the system input signal "Output signal reset" or RESET DOUT command).
  - Conduct daily inspection on the air tube connected to the hand, and conduct periodic inspection on the air tubes in the robot, and replace them if they show any damage.
  - Install a fence or other barrier to protect from flying and dropped objects.
  - When using a robot hand that releases the workpiece when the robot power or hand driving power supply is turned off, post notices to not turn off the power or driving power supply.
- Always check the operation status before approaching a stopped robot.
  Otherwise, if the robot was operating, a collision could occur when the robot restarts its motion.
  - Check the operation status displayed by the controller indicator lamps and the teach pendant.
  - The operation status of the robot is output as signals (servo operation ready, external mode in progress, operation ready, automatic operation in progress, fault occurred, and emergency stop in progress). Connect to the display unit or similar device.
- Always check the operation status before approaching a stopped robot.
  Otherwise, if the robot was operating, a collision could occur when the robot restarts its motion.
  - Check the operation status displayed by the controller indicator lamps and the teach pendant.
  - The operation status of the robot is output as signals (servo operation ready, external mode in progress, operation ready, automatic operation in progress, fault occurred, and emergency stop in progress). Connect to the display unit or similar device.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50℃.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.
- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.

⚠ CAUTION

- Always check the working envelope before using the robot.
  If the working envelope is not limited as expected, you could be struck by an operating robot.
- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.

NOTICE

- Never touch the ball screw spline shaft with your bare hands.
  This could cause rust to form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could reduce the performance and lifespan of the robot. This could also lead to a malfunction of the robot.
- When pressing the robot by hand to move it, be sure to hold the gray sections shown in Figure 9.1.
  If other sections are held, the held sections may be damaged. This could apply excessive force to the mechanical section, resulting in a breakdown.

This section provides important safety information for teaching. Read this information together with "STEA1237: TS5000-SS Controller Manual, 5.2 Safety Information for Teaching", and be sure to strictly observe the following items.

⚠ DANGER

- Be sure to always perform operations as described in the instruction manual. Otherwise, a collision could occur with the robot, or an electric shock could occur.
- Be sure to perform teaching outside the hazard zone. Otherwise, a collision could occur with the robot. If teaching must be performed within the hazard zone due to unavoidable reasons, be sure follow the guidelines below.

(1) Always perform the work using two people. One person performs the work, and the other person monitors from outside the hazard zone. Each person works to prevent mistaken operation.
(2) When teaching in the hazard zone, the operator should hold the master key of the controller and the teach pendant to prevent operation by a third party.
(3) The operator should work in a position where the emergency stop button can be pressed at any time, and should also work in a location where, after fully checking the robot's work envelope and surrounding shielding, he or she can immediately evacuate in case of an abnormality.
(4) The supervisor monitors the work from a location where he or she can oversee the entire robot and immediately presses the emergency stop button in case of an abnormality. Other people are kept away from the hazard zone. Failure of the operator or others to follow the instructions of the supervisor could result in an accident.

- Be sure to turn off the power if an abnormality occurs when the equipment is turned on or if the controller's "READY" LED does not light up. Otherwise, an electric shock or fire could occur.
- Be sure to always perform daily inspection before operating the equipment.
  Failure to perform daily inspection could lead to an accident.
- When the servo is turned on, be sure to always be outside the hazard zone.
  Otherwise, a collision could occur due to an unexpected movement by the robot.
- In case of malfunction or failure, apply an emergency stop to the robot and turn off the controller.
  Otherwise, a fire or collision with the robot could occur.

⚠ WARNING

- Never enter the hazard zone during test operation. Otherwise, a collision could occur with the robot.
- Never leave any equipment in the work space that could interfere with your work.
  This could cause you to trip and fall. If you concentrate on teaching, you could fail to pay attention to your surroundings and collide with peripheral devices.
- Do not allow people to approach the equipment other than the specified operators. Otherwise, a collision could occur with the robot.
- Never open the cover of the controller during operation. Do not connect or disconnect connectors. An electric shock could occur. A collision could occur due to an unexpected movement by the robot.
- Never change the data in the system configuration files during operation. This could cause the robot to perform an abnormal movement, resulting in a collision.

⚠ WARNING

- Be sure to stop using the robot if any abnormal operation is observed when performing robot guidance. Otherwise, a serious accident could occur.
- Stop operation if there is any unusual noise or odor from the controller or robot, or if it behaves differently than expected. Continued operation while there is an abnormality could lead to a fire or a collision due to an unexpected movement.
- When teaching in a hazard zone, work with the teach pendant or with the master key in your possession. If another person starts operating the robot while you are working, a collision with the robot could occur.
- Turn off the servo when leaving the work area. Otherwise, another worker could mistakenly assume that the robot is stopped, resulting in a collision with the robot.
- In case of malfunction or failure, apply an emergency stop to the robot and turn off the controller. Otherwise, a fire or collision with the robot could occur.

⚠ CAUTION

- All hand output signals are in an OFF state when the controller power is shut off and turned on. Design the end effector so that objects will not drop from the hand due to a power outage or similar situation. Objects dropped from the end effector could crush the hands or feet of the operator.

NOTICE

- Be sure to turn off the servo while the robot is stopped before turning off the main power. Otherwise, if the main power is turned off while the robot is operating, the arm will run idly and collide with peripheral devices.
- Always check the execution status of the program when performing automatic operation. Failure to perform a check could result in interference with the peripheral devices and a breakdown of the equipment.

Operation

This section provides important safety information for teaching. Read this information together with "STEA1237: TS5000-SS Controller Manual: Installation", and be sure to strictly observe the following items.

⚠DANGER

- Never turn on the robot in an environment where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  Doing so could result an electric shock due to a leakage current.
- Never turn on the power for the robot in an environment with dust or flammable gas.
  Doing so could lead to a fire.
  Do not place flammable materials near the robot because they could ignite due to a spark if a malfunction occurs.
- Never touch fluid that has leaked from the encoder batteries.
  If the fluid gets on the skin or eyes, it can cause inflammation.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot and frame could fall over together during operation.
  - Prepare the frame by referring to the load applied during horizontal movement (Table 6.2).
- Before turning on the power, always make sure that no one is inside the robot's hazard zone.
  An unexpected motion by the robot could damage the mechanical stopper and crash into someone.
  - Make sure that the operation area limited by the mechanical stoppers is also clear of people. Use of the mechanical stoppers does not guarantee stopping of the robot.
- Always check the working envelope before using the robot.
  If the working envelope is not limited as expected, you could be struck by an operating robot.
- Be sure to lock the robot hand connector and secure the cable to the arm.
  If a load is applied to the connector and it is disconnected, the hand could move unpredictably.
  - Adjust the robot hand wiring and piping while checking the status during robot operation to make sure that they are not in contact with the robot or peripheral equipment and that no load is being applied to the connectors.
- Always check the operation status before approaching a stopped robot.
  Otherwise, if the robot was operating, a collision could occur when the robot restarts its motion.
  - Check the operation status displayed by the controller indicator lamps and the teach pendant.
  - The operation status of the robot is output as signals (servo operation ready, external mode in progress, operation ready, automatic operation in progress, fault occurred, and emergency stop in progress). Connect to the display unit or similar device.
- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could lead to an unpredictable accident due to a robot malfunction.

⚠ DANGER

- Be sure to switch the master mode key inserted in the teach pendant to the "MANUAL" position, remove the key, and carry it with the teach pendant. (for TP5000)
  Otherwise, you could be struck by a robot that has been started by someone else.
  - If there are no impediments to rescue, apply an emergency stop.

⚠ WARNING

- Be sure to use a robot hand having a structure so that it will not release the workpiece or other object even if the robot power or robot hand driving power supply is cut off.
  Otherwise, the workpiece or other object could fly off or drop.
  - When the robot is turned off, the signal output to the hand is also turned off.
  - The robot hand signals (DOUT201 to DOUT208) are not reset by performing an operation to reset all output signals (such as the system input signal "Output signal reset" or RESET DOUT command).
  - Conduct daily inspection on the air tube connected to the hand, and conduct periodic inspection on the air tubes in the robot, and replace them if they show any damage.
  - Install a fence or other barrier to protect from flying and dropped objects.
  - When using a hand that releases the workpiece when the robot power or hand driving power supply is turned off, post notices to not turn off the power or driving power supply until a safety check is completed.
- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50℃.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.

⚠ CAUTION

- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.

This section provides important safety information for operation. Read this information together with "STEA1237: TS5000-SS Controller Manual, Safety Information for Operation", and be sure to strictly observe the following items.

⚠ DANGER

- Never operate the equipment when it is damaged or has missing parts.
  Doing so could lead to an electric shock or fire.
- Do not insert conductive objects or your fingers inside the power supply terminals. An electric shock could occur.
- Display the operation mode of the respective controller on the operation device. If the operator does not know the operation mode, a collision could occur with the robot due to an unexpected movement.
- Be sure to always perform operations as described in the instruction manual.
  Otherwise, a collision could occur with the robot, or an electric shock could occur.
- Be sure to turn off the power if an abnormality occurs when the equipment is turned on or if the controller's "READY" LED does not light up.
  Otherwise, an electric shock or fire could occur.
- When the servo is turned on, be sure to always be outside the hazard zone.
  Otherwise, a collision could occur due to an unexpected movement by the robot.
- In case of malfunction or failure, apply an emergency stop to the robot and turn off the controller.
  Otherwise, a fire or collision with the robot could occur.

⚠ WARNING

- Never open the cover of the controller during operation. An electric shock could occur.
- Never change the data in the system configuration files during operation. This could cause the robot to perform an abnormal movement, resulting in a collision.
- If there are multiple controllers in the equipment, ensure that they operate the robot using the same operation mode. If the operation modes are different, a collision could occur with the robot due to an unexpected movement.
- Never enter the hazard zone during test operation. A collision could occur with the robot.
- Do not allow people to approach the equipment other than the specified operators. A collision could occur with the robot.
- Be sure to always perform daily inspection before operating the equipment. Failure to perform daily inspection could lead to an accident.
- Stop operation if there is any unusual noise or odor from the controller or robot, or if it behaves differently than expected. Continued operation while there is an abnormality could lead to a fire or a collision due to an unexpected movement.
- Never enter the hazard zone of the robot during automatic operation. If you enter the hazard zone during operation, the robot could collide with you.
- Be sure to check operation of the safety protection devices before starting automatic operation. If the safety protection devices have failed, you will be unable to stop the equipment in the event of an emergency, and a collision with the robot could occur.

NOTICE

- Do not places any obstacles within the robot's work envelope. This could cause a collision with the robot.
- Set the robot hand output signal to 0.1 A per output (max. 0.8 A for total 8 outputs). Supplying a current that exceeds the rated output will damage the output elements, circuit board, and other parts, and could result in a fire.
- Supply a voltage in a range of 24V±10% to the external input/output power supply (24V DC). If a voltage exceeding the tolerance is applied, the controller will be damaged, and a fire could result.
- Be sure to turn off the servo while the robot is stopped before turning off the main power.
  Otherwise, if the main power is turned off while the robot is operating, the arm will run empty and collide with peripheral devices.
- Always check the execution status of the program when performing automatic operation.
  Failure to perform a check could result in interference with the peripheral devices and a breakdown of the equipment.

Repacking

This section provides important safety information for repacking. Read this information together with "STEA1237: TS5000-SS Controller Manual, 5.2 Safety Information for Teaching", and be sure to strictly observe the following items.

⚠ DANGER

- Never turn on the power for the robot in an environment where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  Doing so could result an electric shock due to a leakage current.
- Never turn on the power for the robot in an environment with dust or flammable gas.
  Doing so could lead to a fire.
  - Do not place flammable materials near the robot because they could ignite due to a spark if a malfunction occurs.
- Never touch fluid that has leaked from the encoder batteries.
  If the fluid gets on the skin or eyes, it can cause inflammation.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- The robot should never be transferred by only one worker.
  The robot cannot be fully supported and could drop.
  - The robot weight is 31 kg.
  - Lift the robot using more than one person.
  - Use a cart or similar device to move to the transfer location.
  - Clear a transfer route, and make sure that a dropped robot would not hit anyone.
  - Wear protective gloves and safety shoes.
- Never carry the packing boxes by hand.
  A packing box could fall on a worker.
  - Use a hand lifter or similar equipment.
  - Align the center of the packing box with the center of the forks when lifting the packing box.
  - Do not allow people to approach the transfer route.
- When lifting packing boxes, never carry the packing boxes by hand.
  A packing box could fall on a worker.
  - The forklift must be operated by a qualified operator.
  - Align the center of the packing box with the center of the forks when lifting the packing box.
  - Do not allow people to approach the transportation route.

⚠ DANGER

- Before turning on the power, always make sure that no one is inside the robot's hazard zone.
  An unexpected motion by the robot could damage the mechanical stopper and crash into someone.
  - Make sure that the operation area limited by the mechanical stoppers is also clear of people. Use of the mechanical stoppers does not guarantee stopping of the robot.
- Always check the operation status before approaching a stopped robot.
  Otherwise, if the robot was operating, a collision could occur when the robot restarts its motion.
  - Check the operation status displayed by the controller indicator lamps and the teach pendant.
  - The operation status of the robot is output as signals (servo operation ready, external mode in progress, operation ready, automatic operation in progress, fault occurred, and emergency stop in progress). Connect to the display unit or similar device.
- If work is performed while the power is on in the robot's hazard zone, be sure to switch the master mode key inserted in the teach pendant to the "MANUAL" position, remove the key, and carry it with the teach pendant. (for TP5000)
  Otherwise, you could be struck by a robot that has been started by someone else.
  - Post a notice near the controller that work is in progress.
  Assign a work supervisor who is stationed outside the hazard zone of the robot.
  - The supervisor should ensure the safety of the worker and any other people approaching the robot hazard zone.
- Do not remove the robot until the power is turned off and the cable is removed from the controller.
  Otherwise, a worker could receive an electric shock due to touching a cable that was broken when the robot is dropped.
- When hoisting the robot, be sure to always observe the following safety guidelines.
  Otherwise, the robot could drop.
  - Use the attached eyebolts.
  - The crane must be operated by a qualified operator.
  - Use a wire with a load limit that can sufficiently bear the weight of the robot.
- Be sure to always hoist and lower the robot slowly.
  Otherwise, the robot could be tilted and fall.
  - Clear a transfer route, and make sure that no one goes under the suspended load.

⚠ WARNING

- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.
- Before turning on the power, be sure to always check that the ground wire is connected.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.
  - After turning off the power, remove the ground wire.
- Be sure to check that the robot is using a robot hand having a structure so that it will not release the workpiece or other object before turning off the robot power or robot hand driving power supply.
  Otherwise, the workpiece or other object could drop.
  - When the robot is turned off, the signal output to the hand is also turned off.
  - The robot hand signals (DOUT201 to DOUT208) are not reset by performing an operation to reset all output signals (such as the system input signal "Output signal reset" or RESET DOUT command).
  - When using a hand that releases the workpiece when the robot power or hand driving power supply is turned off, post notices to not turn off the power or driving power supply until a safety check is completed.
- When transferring the robot, be sure to always secure in place using the supplied fixing jigs.
  Otherwise, a worker could get caught in an arm that moved.
  - The weight of the arm section is approximately 17 kg.
  - Wear protective gloves to prevent slippage and to protect the hands.
- Never use an awkward posture when inserting the robot into the packing box.
  Overexertion could injure the worker's back.
  - Provide space where work can be performed easily.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50℃.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.

⚠ CAUTION

- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.
- Always check the working envelope before using the robot.
  If the working envelope is not limited as expected, you could be struck by an operating robot.
- Be sure to secure the robot to be re-transported in a packing box as it was delivered.
  Otherwise, the robot could fall over.
  - The customer must obtain new packing straps (PP bands).
  - If no packing box is available, use the mounting holes in the base to secure in place.

NOTICE

- Never touch the ball screw spline shaft with your bare hands.
  This could cause rust to form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- The robot should never be transferred by only one worker.
  The robot cannot be fully supported and could drop.
  - Do not use a product that has been dropped.
- Be sure to hoist and lower the robot gently.
  Otherwise, the robot could collide with the peripheral equipment, frame, or other objects, resulting in failure or damage.
- Be sure to secure the robot to be shipped in a packing box as it was delivered, and then remove the eyebolts.
  If the robot is moved while the eyebolts are mounted, the robot could be damaged or fail.
  - This is a mechanism indicating that a robot with the eyebolts mounted cannot be operated.
  - Keep the removed eyebolts with the robot.
- When lifting up the robot by workers, be sure to hold the gray portions shown in Figure 5.3.
  If other sections are held, the held sections may be damaged. This could apply excessive force to the mechanical section, resulting in a breakdown.

Maintenance

This section provides important safety information for maintaining the robot. Read this information together with "STEA1213: THE400 TS5000 Series Robot Manual: Maintenance", and be sure to strictly observe the following items.

⚠DANGER

- Never turn on the power for the robot in an environment where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  Doing so could result an electric shock due to a leakage current.
- Never turn on the power for the robot in an environment with dust or flammable gas.
  Doing so could lead to a fire.
  Do not place flammable materials near the robot because they could ignite due to a spark if a malfunction occurs.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a malfunction due to noise could cause an expected accident.
- Never touch fluid that has leaked from the encoder batteries.
  If the fluid gets on the skin or eyes, it can cause inflammation.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- Always check the operation status before approaching the robot.
  Otherwise, a collision could occur with the robot when it starts to move.
  - Check the operation status displayed by the controller indicator lamps and the teach pendant.
  - The operation status of the robot is output as signals (servo operation ready, external mode in progress, operation ready, automatic operation in progress, fault occurred, and emergency stop in progress). Connect to the display unit or similar device.
- Be sure to always turn off the power before performing maintenance on the robot.
  If work must be performed on the robot within the robot's hazard zone due to unavoidable circumstances, be sure to switch the master mode to "MANUAL". Carry the teach pendant and, if using the TP5000, remove the master mode key, and manage it properly.
  Otherwise, you could be struck by a robot that has been started by someone else.
  - Post a notice near the controller that work is in progress.
  - Assign a work supervisor who is stationed outside the hazard zone of the robot.
  - The supervisor should ensure the safety of the worker and any other people approaching the robot hazard zone.
  - If the teach pendant is not used, carry an emergency stop device with you when performing work.
  - If the power will be turned on, make sure that the worker and the supervisor receive safety training on tasks related to the inspection of industrial robots as specified by the laws and regulations of the respective country.
- Before turning on the power, always make sure that no one is inside the robot's hazard zone.
  A collision could occur due to an unexpected movement by the robot.
  - Make sure that the operation area limited by the mechanical stoppers is also clear of people. Use of the mechanical stoppers does not guarantee stopping of the robot.

⚠ DANGER

- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could lead to an unpredictable accident due to a robot malfunction.
- Never modify parts except those that are expressly permitted.
  Otherwise, the robot could fail, and cause an expected accident.
- Be sure to always check that the polarity of the encoder batteries is correct.
  If the polarity is wrong, electrolyte could leak from the encoder batteries.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- Be sure to replace parts with the Shibaura Machine-specified service replacement parts.
  Use of unauthorized parts could lead to an unpredictable accident.
  - All items other than the encoder batteries are special-specification products. To order them, please contact Shibaura Machine.
  - Be sure to contact the Shibaura Machine servicing company to request any replacement work other than replacement of the encoder batteries. A fee will be charged for any replacement work.
  - Shibaura Machine does not cover any accidents or damage resulting from replacement work performed by the customer or its results.

⚠ WARNING

- Be sure to check that the robot is using a robot hand having a structure so that it will not release the workpiece or other object even if the robot power or robot hand driving power supply is cut off.
  Otherwise, the workpiece or other object could drop.
  - When the robot is turned off, the signal output to the hand is also turned off.
  - The robot output signals (DOUT201 to DOUT208) are not reset by performing an operation to reset all output signals (such as the system input signal "Output signal reset" or RESET DOUT command).
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other barrier to protect from dropped and flying objects.
  - When using a hand that releases the workpiece when the robot power or hand driving power supply is turned off, post notices to not turn off the power or driving power supply until a safety check is completed.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50℃.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.
- Before turning on the power, be sure to always check that the ground wire is connected.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.
- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.

⚠ CAUTION

- Always check the working envelope before operating the robot.
  If the working envelope is not limited as expected, you could be struck by a moving robot.
- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.

NOTICE

- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could reduce the performance and lifespan of the robot. This could also lead to a malfunction of the robot.
- Never modify parts except those that are expressly permitted.
  Otherwise, the performance and lifespan of the robot could be adversely affected. This could also lead to a malfunction of the robot.
- Be sure to always perform daily inspection for the air tube.
  Replace the air tube immediately if it is damaged.
- Never touch the ball screw spline unit with your bare hands.
  Touching it with bare hands causes the ball screw spline to rust.
  - Be sure to always wear gloves.
- When greasing and applying corrosion inhibitor, be sure to cover the peripheral equipment.
  Grease and corrosion inhibitor could drip on the peripheral equipment.
  - Wipe off any excess grease and corrosion inhibitor with a waste cloth.
- Be sure to check the grease application state daily.
  Insufficient grease or rust formation could cause scratches on the sliding surface, resulting in reduced performance and lifespan of the robot. This could also lead to a malfunction of the robot.
- Be sure to use the specified types of grease.
  Otherwise, the performance and lifespan of the robot could be adversely affected. This could also lead to a malfunction of the robot.
- If the origin is set again using the ZEROP function or REORG function, be sure to perform teaching of the teaching points again.
  The zero position shifts slightly due to error when setting. If the origin position is used without setting it again, the robot could collide with the peripheral equipment.
- Be sure to always replace the encoder batteries during the annual inspection.
  If the position detector data is lost, the robot can no longer be operated.
  - If the encoder battery voltage drops even further, a “Battery alarm” will occur. At this stage, data is not lost yet. If the encoder batteries are replaced, the alarm is automatically cleared.
  - If the battery voltage drops even further, a “Battery alarm” will occur. At this stage, the data is lost. A position detector error alarm will also occur.
  - If the data is lost, the origin position must be set again. For details on setting the origin position, refer to "14. Robot Origin Position and Position Detector Error".

NOTICE

- When the robot is not used while the power turned off, be sure to replace the encoder batteries every two months.
  Because the encoder batteries use a large amount of power while the power is turned off, the position detector data will be lost before the periodic inspection is conducted.
- To set the origin position again using the ZEROP function or REORG function, perform the following operation procedure.
  Because the 3rd axis operates in tandem with the 4th axis, the origin position is not restored if the steps are performed in the wrong order.
  Operation procedure
  1) Move the robot to the specified position.
  2) Move the cursor to E4 on the screen, enter the data, and restore the origin position of the 4th axis.
  3) Move the cursor to E3 on the screen, enter the data, and restore the origin position of the 3rd axis.
  - E1 (1st axis) and E2 (2nd axis) do not have any specific order for setting their origin position again.
- Be sure to enter data to HOME1 and HOME2 for the REORG function when installing the robot.
  Even if the data is entered after the origin position has shifted, it cannot be used as data for restoring the origin position.
- Be sure to replace parts with the Shibaura Machine-specified service replacement parts.
  Use of unauthorized parts could lead to reduced robot performance and lifespan. This could also lead to a malfunction of the robot.

This section provides important safety information for maintaining the controller. Read this information together with "STEA1237: TS5000-SS Controller Manual: Maintenance", and be sure to strictly observe the following items.

⚠ DANGER

- Never perform any parts replacement or modification except those described in the instruction manual. Reduced performance or malfunctions could result in electric shock, fire, or collision due to unexpected robot movements.
- When opening the cover of the controller, do not insert liquids such as water or foreign objects inside. This could cause a short-circuit inside the controller, resulting in an electric shock or fire.
- Do not incinerate, disassemble, short-circuit, or charge the batteries. This could rupture, resulting in a fire. Dispose of the batteries in accordance with your company's regulations.
- Shibaura Machine special parts must be used for some service replacement parts. Unauthorized connectors and terminal wires could result in a breakdown of the controller, electric shock, or a fire.
- Be sure to always request part replacement from the Shibaura Machine servicing company for parts replacement that is not described in this manual. (A fee will be charged for this work.) Shibaura Machine will not be liable for any accidents or damage that arises due to replacement work performed by the customer or its results.
- Install the switching power supply as described in the instruction manual. Installing the unit upside-down or with the right and left sides reversed, or using a different connector could cause a short-circuit or disconnection, which could result in an electric shock or fire.
- Be sure to replace parts with the Shibaura Machine-specified service replacement parts. Use of unauthorized parts could lead to an unpredictable accident.
- All items other than the position detector batteries are special-specification products. To purchase and order them, please contact Shibaura Machine.
- Except when replacing button batteries and fuses, be sure to always request replacement work from the Shibaura Machine servicing company. A fee will be charged for this work. Shibaura Machine will not be liable for any accidents or damage that arises due to replacement work performed by the customer or its results.

⚠ WARNING

- Wait about two minutes after disconnecting the power supply before touching the terminals of the controller or the internal circuit board. Residual voltage may remain in the high-capacity capacitors in the controller, which could result in an electric shock.
- Use Shibaura Machine-authorized spare parts when replacing parts. Use of unauthorized parts could cause a fire or a collision due to an unexpected robot movement.
- Be sure to wear protective gear and remove the power plug before starting work. If the power is turned on, the power supply board, servo board, switching power supply, and other components will be electrically charged, and could cause an electric shock.
- Turn off the power before opening the cover of the controller. If the cover is opened while power is being supplied, an electric shock could result.
- Perform the following actions whenever entering the movable range of the robot.
  - Wear protective gear
  - Place the teach pendant in a location accessible to the worker
  - Change to MANUAL mode (remove the mode selector key so that the mode cannot be changed)

⇒ If the above actions are not taken, an emergency stop could be cleared by a third party during daily inspection work within the moving range, and that person could be struck by a robot running under continuous operation.

- Use the voltage measurement and LEDs to check that a PN voltage is not remaining in the capacitor. An electric shock will occur if a PN voltage remains.
- Do not use a metal scale while the power is being supplied. An electric shock will occur if contacting the energized parts.

NOTICE

- Be sure to replace parts with the Shibaura Machine-specified service replacement parts. Use of unauthorized parts could lead to reduced robot performance and lifespan. This could also lead to a malfunction of the robot.
- The files in the TS5KBKUP folder of the SD card will be overwritten. To prevent overwriting of valuable data, move the target file from the TS5KBKUP folder to another storage location.
- All files in the controller will be deleted by USER BACKUP LOAD. Before executing, use the file transfer or other function to save the files in the controller to an external device.

Disposal

This section provides important safety information for disposal of the robot. Read this information together with "STEA1213: THE400 TS5000 Series Robot Manual: Maintenance", and be sure to strictly observe the following items.

⚠ DANGER

- Never incinerate, short-circuit, charge, disassemble, or heat the batteries.
  They could rupture and leak, and if the battery fluid gets on your skin or eyes, it will cause inflammation.
  - If battery fluid gets on your skin, immediately flush with water to wash off.
  - If battery fluid gets in your eyes, immediately flush with water to wash off, and then seek medical attention.
  - Dispose of the batteries in accordance with the customer's regulations.

NOTICE

- Be sure to dispose of any unneeded packing materials in accordance with your company's rules.
  When storing the packing materials, be sure to manage them so that they are not used for purposes other than packing the robot.

This section provides important safety information for disposal of the controller. Read this information together with "STEA1237: TS5000-SS Controller Manual: Disposal", and be sure to strictly observe the following items.

⚠DANGER

- Never incinerate, short-circuit, charge, disassemble, or heat the batteries. They could rupture and leak, and if the battery fluid gets on your skin or eyes, it will cause inflammation.
  - If battery fluid gets on your skin, immediately flush with water to wash off.
  - If battery fluid gets in your eyes, immediately flush with water to wash off, and then seek medical attention.
  - Dispose of the batteries in accordance with the customer's regulations.

NOTICE

- Be sure to dispose of any unneeded packing materials in accordance with your company's rules. When storing the packing materials, be sure to manage them so that they are not used for purposes other than packing the robot.

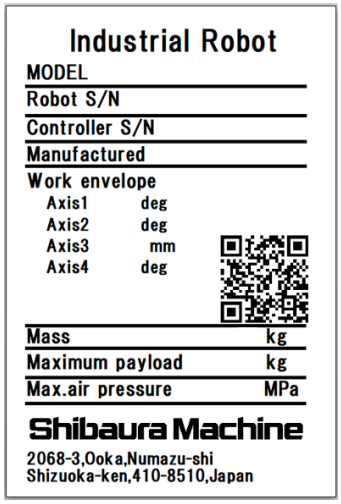
Service Network

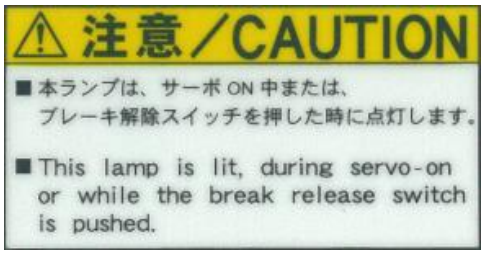
Please refer to our distributor's handouts

1. Warning Labels

1.1 Warning Label Content

This section explains the information contained in the warning labels and other labels affixed to the robot and controller.

Label layout	Entry contents
	MODEL: Robot model Robot S/N: Robot serial number Controller S/N: Controller serial number Manufactured: Month of manufacture. Year of manufacture Work envelope: Maximum movable range of each joint axis MASS: Robot weight Maximum payload: Maximum payload * QR code: MODEL, SER.NO., and manufacturing instruction number can be read.
• MODEL, SER.NO., and Mfd.DATE are information required for receiving after-sale service. When calling us at the after-sale service agent, notify the service representative of these information. • During installation, transportation, and disposal, be sure to confirm the mass before carrying the robot. Increase the number of persons and use the transport equipment as required. The transport equipment should be operated by an operator who has received training pertaining to the transport equipment.	

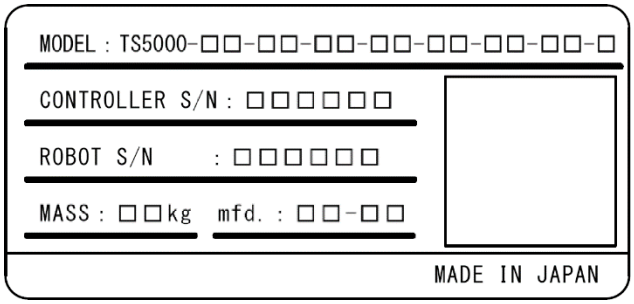
Label layout	Entry contents
	• The lamp on the robot arm 2 is lit during servo ON or when the brake release switch is pressed.
• The lamp on the robot arm 2 is lit during servo ON or when the brake release switch is pressed. If the brake release switch is not pressed, the robot may wait for operation instruction during program operation. Because the robot may start moving suddenly, do not get close to the robot. When getting close to the robot, confirm that the program has stopped, and then switch the operation mode to TEACH. If the brake release switch is pressed, the axis 3 may drop. While using the support that prevents drop, perform the operation.	

Label layout	Description
	- Keep your hands away from the operating parts of the robot. Your fingers could get entangled, resulting in an injury.
- If you put your hand near the operating parts of the robot, it could get caught in the robot arm or shaft, resulting in an injury. During transfer for installation, transportation, and disposal, secure the arm and shaft in place using the supplied fixture so that they do not move. - If you must enter the moving range of the robot to perform teaching or maintenance work for unavoidable reasons, the work must be performed by a trained worker. - Install a protective fence to prevent people from entering (entering the moving range of the robot or having a body part get near the operating parts) during operation.	

Label layout	Description
	- Do not enter into the robot's movable range. Otherwise, an injury could result.
- If you enter the movable range of the robot, you could be killed or injured due to contact with the moving robot. If you must enter the moving range of the robot for unavoidable reasons to perform teaching or maintenance work, this must be performed by a trained operator. Implement measures so that no one other than workers performing teaching or maintenance can enter the movable range.	

Label layout	Description
	-High temperature This is affixed to locations that could become hot when operating the robot. Touching these locations could result in a burn. Wait until these high-temperature locations have cooled before performing any maintenance, inspection, or adjustment work.
Wait until these high-temperature locations are fully cooled before performing any work where the robot is touched. If necessary, protect your hands using thermal-resistant gloves or similar protection when performing the work.	

Label layout	Description
	- Be sure to always turn off the controller before performing any maintenance or inspection work. Otherwise, death or serious injury could result due to an electric shock or unexpected movement of the robot.
- If maintenance or inspection work will be performed without turning off the primary power of the controller, the worker could accidentally touch an energized part or charged part, resulting in an electric shock. Also, the robot could start to move due to an unexpected command issued from an external device, contacting the worker and resulting in death or serious injury. Before performing any maintenance or inspection work, be sure to always turn off the primary power, and disconnect the power connector. Also, be sure to post signage that work is in progress so that the primary power will not be turned on during the work. If you must turn on the power to perform work for unavoidable reasons, the work must be performed by a trained worker.	

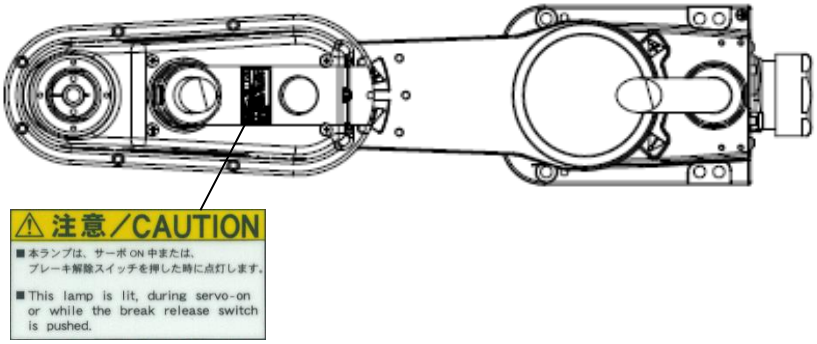
Label layout	Description
	- MODEL: Controller model - CONTROLLE S/N: Controller serial number - ROBOT S/N: Robot serial number - MASS: Controller weight - mfd.: Month of manufacture. Year of manufacture *QR code: This scans the model, serial number, and production order number.
- If operation is performed using a combination with the wrong serial number, the individual robot parameters will not match, and this could result in shifted coordinates and other problems. Check the serial numbers of the robot and controller, and install by connecting the correct combination. - The model, serial number, and date of manufacture information are required when receiving after-sales service. When contacting the Shibaura Machine Servicing Company, please provide this information to the service technician.	

Label layout	Description
	- Use the robot body and controller of this product in the correct serial number combination shown on the label. - If an abnormality occurs, turn off the controller power, and contact the Shibaura Machine Servicing Company. - This robot and controller must not be modified without prior written consent from Shibaura Machine.
- If operation is performed using a combination with the wrong serial number, the individual robot parameters will not match, and this could result in shifted coordinates and other problems. Check the serial numbers of the robot and controller, and install by connecting the correct combination. - If the robot or controller is modified without approval, it could break down or run out of control. Do not perform any unapproved modifications. For any technical inquiries, please contact the Shibaura Machine Servicing Company.	

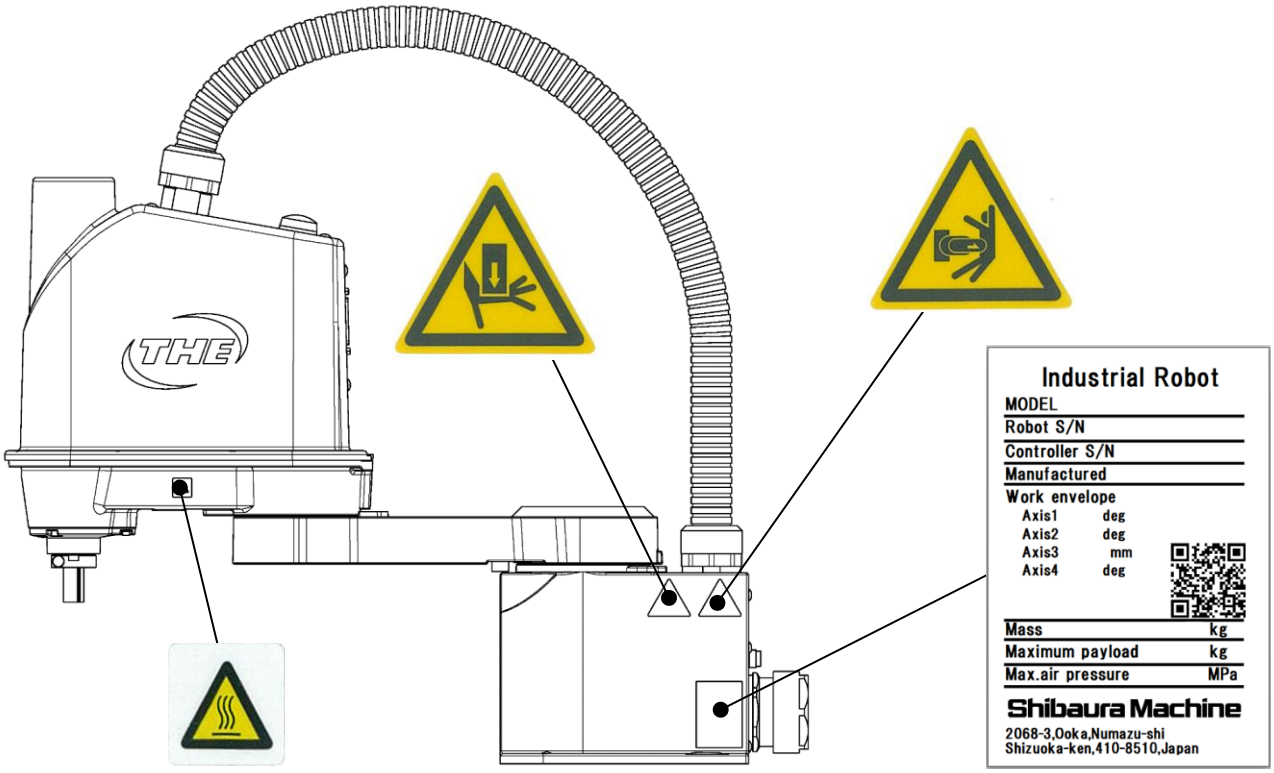
Label layout	Description
	- If an abnormality occurs, turn off the controller power, and contact Shibaura Machine. - This robot and controller must not be modified without prior written consent from Shibaura Machine. - Be sure to perform the wiring by connecting the power cable and ground cable for the connector (ACIN) as shown in the drawing. Improper wiring could result in an electric shock or equipment damage. - Death or serious injury could result due to an electric shock. - Before starting any maintenance or inspection work, be sure to always turn off the power, and disconnect the controller power connector (ACIN). - Do not touch the heat sink while the power is turned on or for a short time after turning off. A burn could occur.

- If the robot or controller is modified without approval, it could break down or run out of control. Do not perform any unapproved modifications. For any technical inquiries, please contact the Servicing Company.
- Usage while the power cable or ground cable is incorrectly wired could result in an electric shock or equipment damage.
  Before connecting, use a tester or other device to check that they are wired correctly. Also, check that the primary supply voltage is within the usage range.
- If maintenance or inspection work will be performed without turning off the primary power of the controller, the worker could accidentally touch an energized part or charged part, resulting in a death or serious injury due to electric shock.
  Before performing any maintenance or inspection work, be sure to always turn off the primary power, and disconnect the power connector. Also, be sure to post signage that work is in progress so that the primary power will not be turned on during the work.
  If you must turn on the power to perform work for unavoidable reasons, the work must be performed by a trained worker.
- Always check the weight of the robot before transferring it for installation, transportation, or disposal. If necessary, increase the number of people, or use transportation equipment or other means. The transportation equipment must be operated by someone who has received training in operation of the equipment.

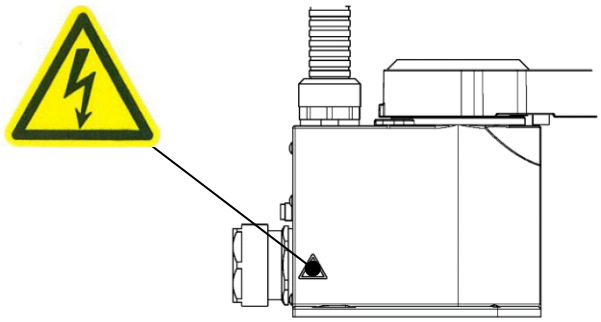
1.2 Robot Warning Label Affixing Locations (THE400)



Top view of robot

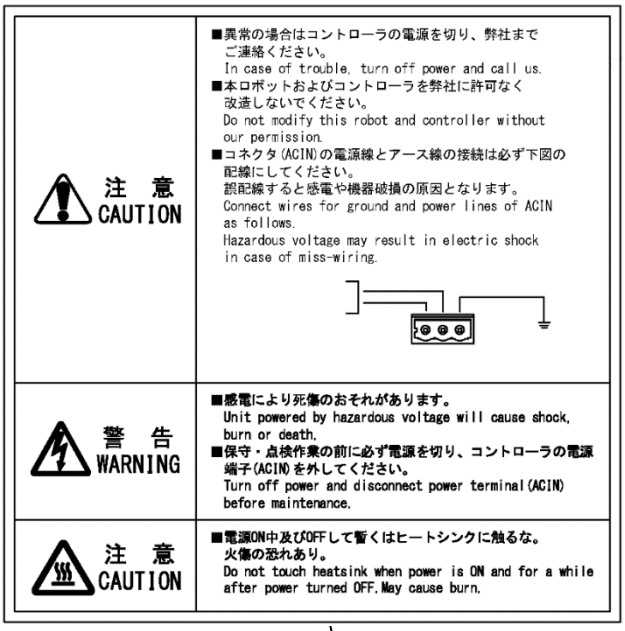


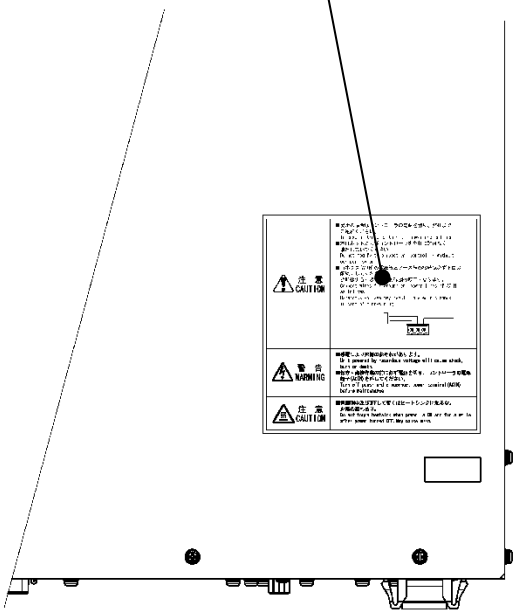
Right side view of robot



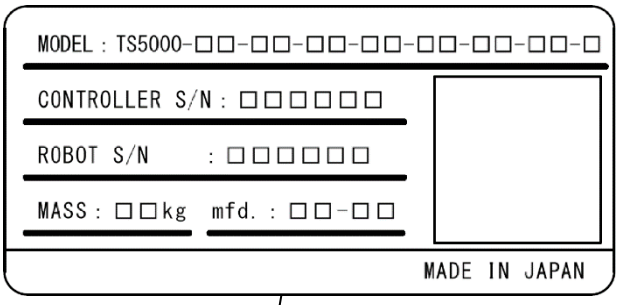
Left side view of robot

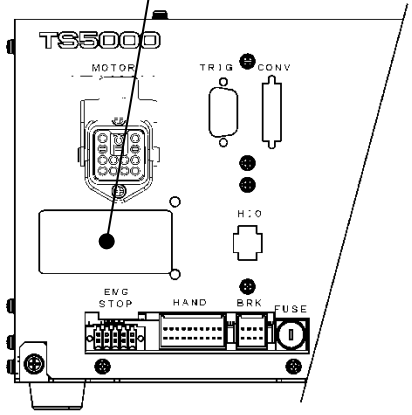
1.3 Controller Warning Label Affixing Locations (TS5000-SS)





Controller Top Right Side View





Controller Front View

2. Safety Measures

This chapter describes the need for safety protection, the safety protection functions of the robot and controller, and the general safety measures that should be taken.

2.1 Safety Information

The Shibaura Machine SCARA robot and controller are equipped with various safety protection features. However, the following potential hazards may occur in the actual operation of the robot.

a) Potential hazards under normal automatic operation
- Mistaken operation or misjudgment by operator, programming mistakes
- Unexpected movement of the robot due to failure of control electronic components or other cause, release or dropping of workpieces, etc.

b) Potential hazards in teaching, inspection, and other work
- Hazards due to entering the robot's movable range
- Mistaken operation due to panic caused by sudden abnormalities, skipping procedures due to familiarity, mistaken operation due to lack of knowledge about robot movement
- Coming into close proximity to a robot due to its unpredictable and complex movements
- Abnormal operation caused by incorrect wiring, poor contact, deterioration, noise, or other cause

c) Potential hazards in related machinery
- Robot suddenly starts moving due to a command from related machinery or other source
- Related machinery suddenly starts moving due to operation of the robot
- Hazards of becoming crushed or entangled while teaching, inspecting, adjusting, or performing other operations on the robot while related machinery is operating

Consequently, in order to use the robot safely, safety protection measures that match the usage conditions must be implemented, and failure to do so could entail a risk of an unexpected accident. If there are any legal regulations pertaining to safety, please follow the regulations of the respective country, and refer to all manuals pertaining to this robot and controller.

2.2 Safety Protection

The robot and controller are equipped with various protective functions and additional safeguards for ensuring safety.

2.2.1 Emergency Stop Function

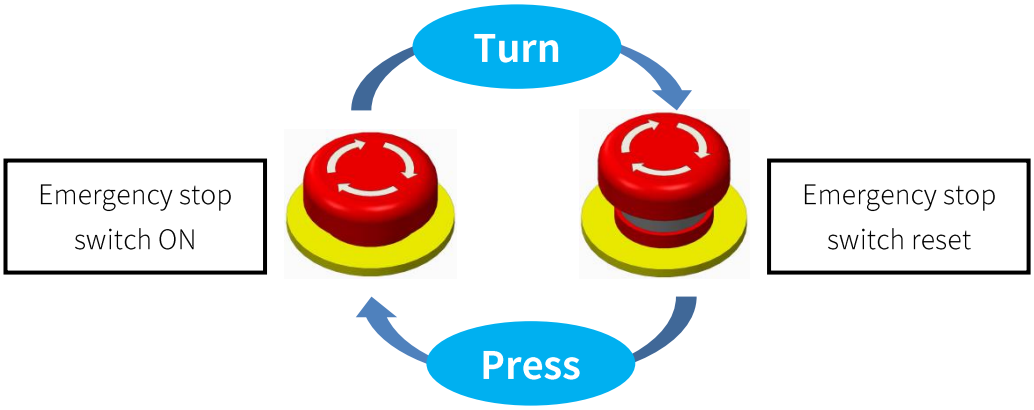
This robot is equipped with an emergency stop function to stop the robot in case of an emergency.

2.2.1.1 Emergency Stop Switch

The teach pendant is equipped with an emergency stop pushbutton switch, which can be used to immediately stop the robot and turn off the power source for the robot. (Stop category 1) The emergency stop function is always enabled regardless of the robot's operating status.

The emergency stop button uses a switch with a locking mechanism, and so to release it, turn it and pull up the switch top as shown in the figure below.

It also has an external emergency stop contact, and so an emergency stop switch can be added as needed. Select an emergency stop switch compliant with ISO 13850.



Important information on emergency stop switches

Select the emergency stop button by examining the following important points.

- Emergency stop devices shall be installed at locations where emergency stops are required and shall be located in easily accessible areas.
- Once an emergency stop device has been activated, the circuit will not be reset automatically until it is reset manually, and so select a structure that allows the contacts to be reset only after the device has been reset.
- Select an emergency stop device with a suitable shape as shown below so that it can eliminate hazardous situations based on the structure and characteristics of the machine.
  1) Mushroom head pushbutton
  2) Rope pull type, bar type
  3) Hip or knee activated type
  4) Pedal switch without protective cover
- The actuator of the pushbutton-type emergency stop device shall be red, and the surrounding background color shall be yellow.
- Rope pull emergency stop devices should maintain the proper tension of the rope at all times, and the rope should be identifiable by its red and yellow colors.

2.2.2 Robot Behavior When Emergency Stop Is Applied

If an emergency stop signal is received during robot operation, the robot comes to a stop, and the power source is disconnected. However, in an emergency stop, the robot does not immediately turn off the servo, but instead, the servo is turned off after decelerating.
Graphs are provided showing the approximate stopping distance and stopping time for PAYLOAD settings of "5 kg load with 0 mm offset", "3.3 kg load with 0 mm offset", and "1.7 kg load with 0 mm offset" when an emergency stop is performed.
The stopping distances and stopping times shown in the graph are the measured results for each joint axis. Use these values to install and determine the distances for the robot and light curtain and other components.

MEMO

- *For details on the principles and setting methods for the PAYLOAD command, refer to "STEA1213: THE400 TS5000 Series Robot Manual, Chapter 7.4".*

2.2.2.1 Robot Behavior When Emergency Stop Is Applied (for PAYLOAD (5,0))

The stopping distances and stopping times for each joint axis when the PAYLOAD is set to "5 kg load with 0 mm offset" are shown below.

When operating 1st axis (for PAYLOAD (5,0))

The robot behavior when an emergency stop is applied during 1st axis single-axis operation is shown below. Table 1 shows the operating conditions, and Figures 1 to 6 show the relationship between the stopping distance and stopping time when an emergency stop is applied for various operating conditions.

Table 1 Operating Conditions (1st Axis)

Operation axis	PAYLOAD setting value	3rd axis position	Approach start position	Stop signal issuing position	2nd axis angle	Override setting value	Load	Drawing
1st axis	(5,0)	0 mm (lower limit)	130°	0°	0° (100% of maximum displacement)	100%	5 kg	Fig. 1 Fig. 2
							3.3 kg	
							1.7 kg	
						66%	5 kg	
							3.3 kg	
							1.7 kg	
						33%	5 kg	
							3.3 kg	
							1.7 kg	
					49.3° (66% of maximum displacement)	100%	5 kg	Fig. 3 Fig. 4
							3.3 kg	
							1.7 kg	
						66%	5 kg	
							3.3 kg	
							1.7 kg	
						33%	5 kg	
							3.3 kg	
							1.7 kg	
					97.15° (33% of maximum displacement)	100%	5 kg	Fig. 5 Fig. 6
							3.3 kg	
							1.7 kg	
						66%	5 kg	
							3.3 kg	
							1.7 kg	
						33%	5 kg	
							3.3 kg	
							1.7 kg	

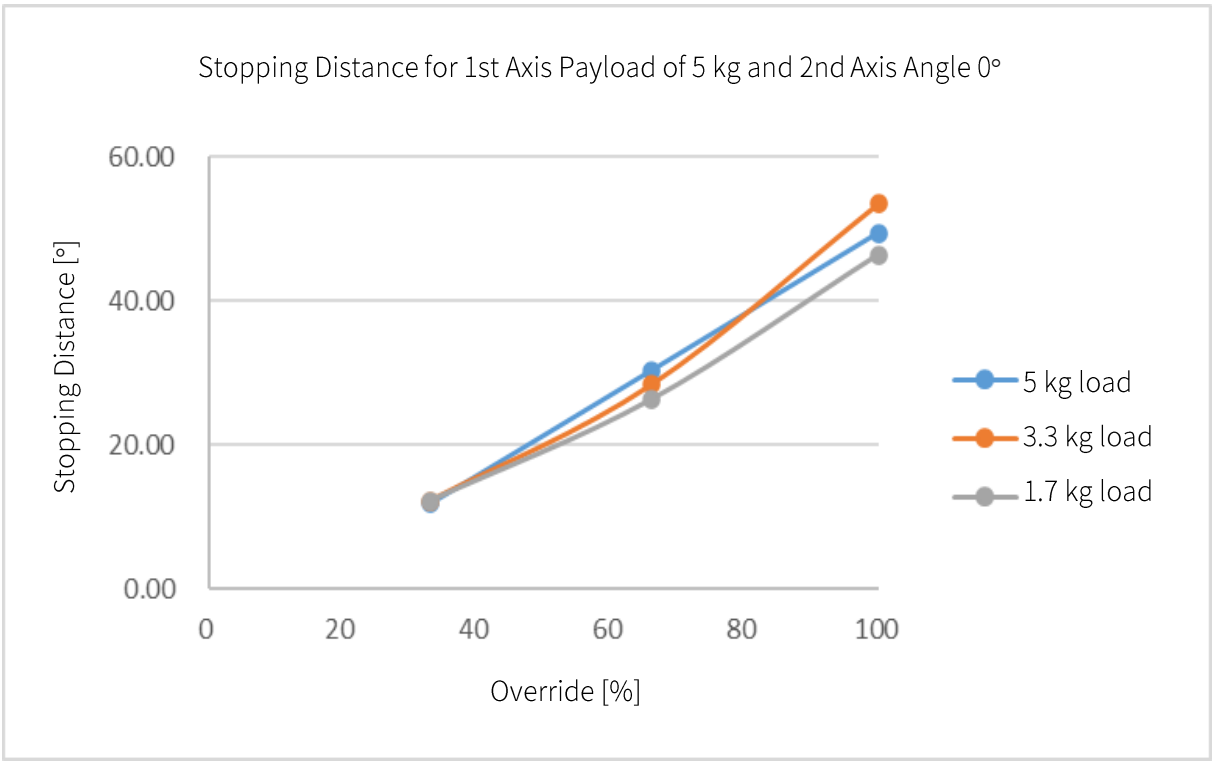


Fig. 1. 1st Axis Stopping Angle for Payload (5,0) and 2nd Axis Angle 0°

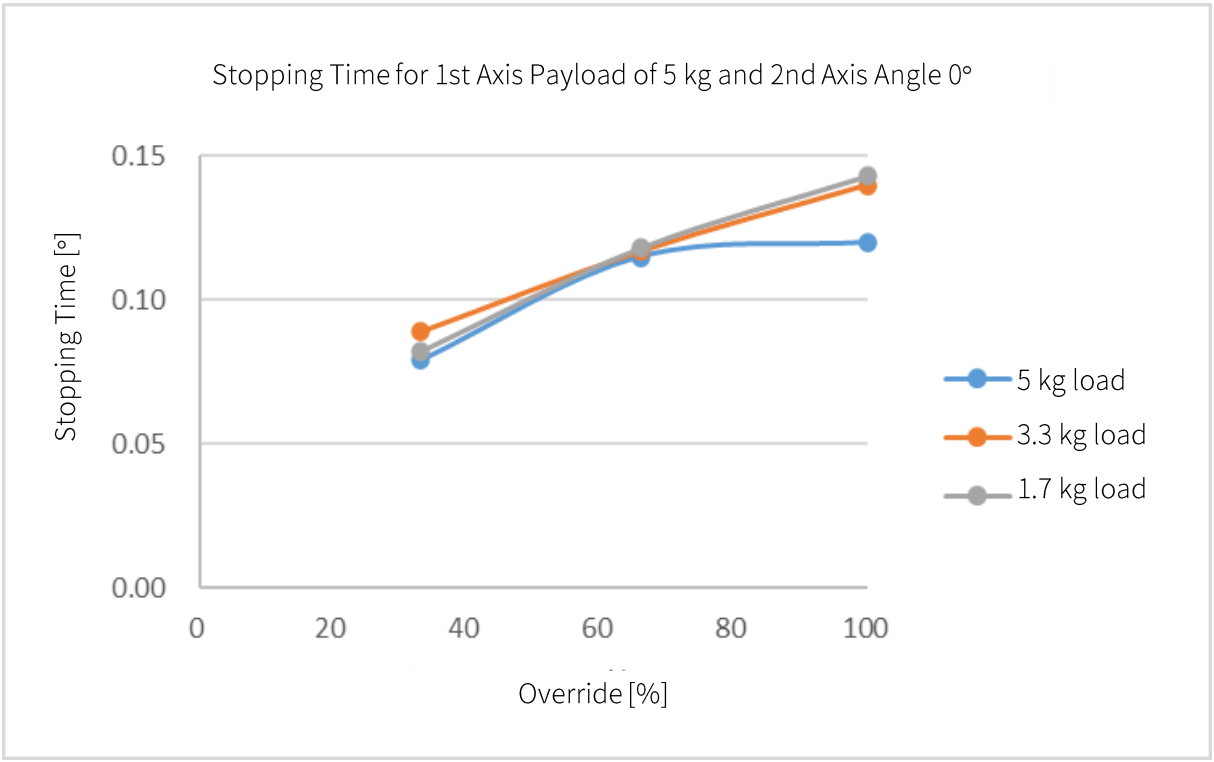


Fig. 2. 1st Axis Stopping Time for Payload (5,0) and 2nd Axis Angle 0°

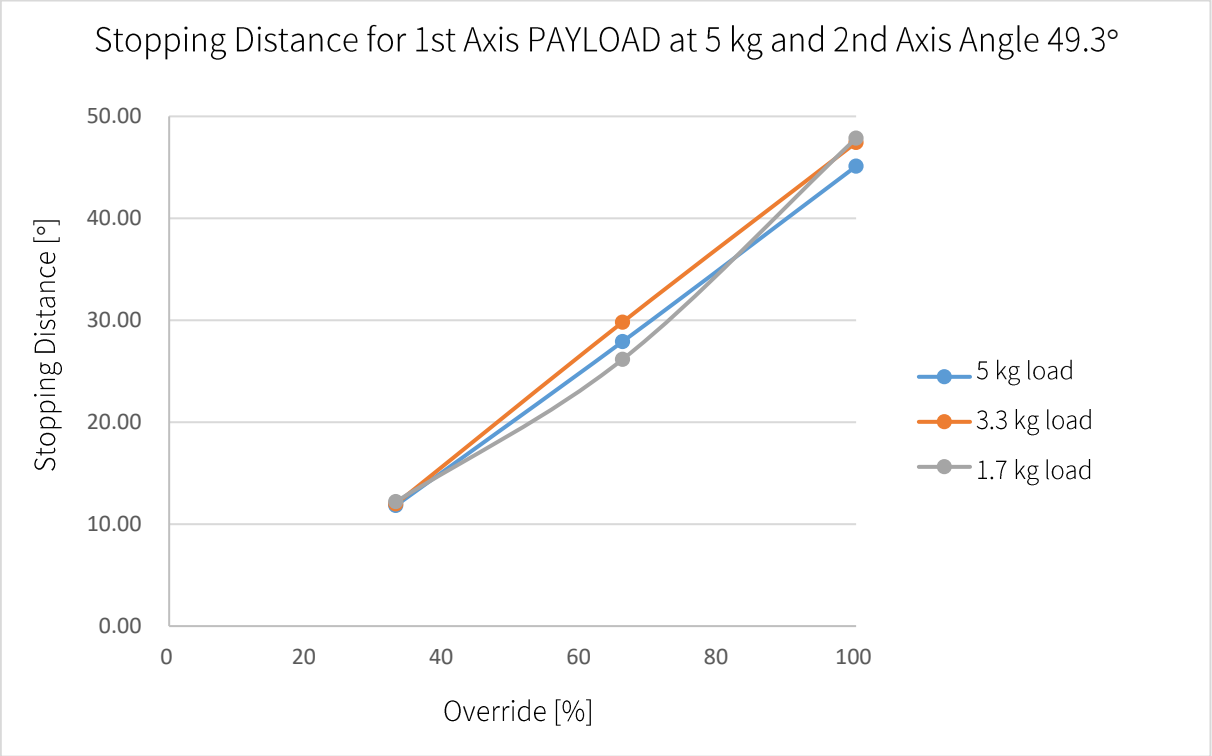


Fig. 3. 1st Axis Stopping Angle for Payload (5,0) and 2nd Axis Angle 49.3°

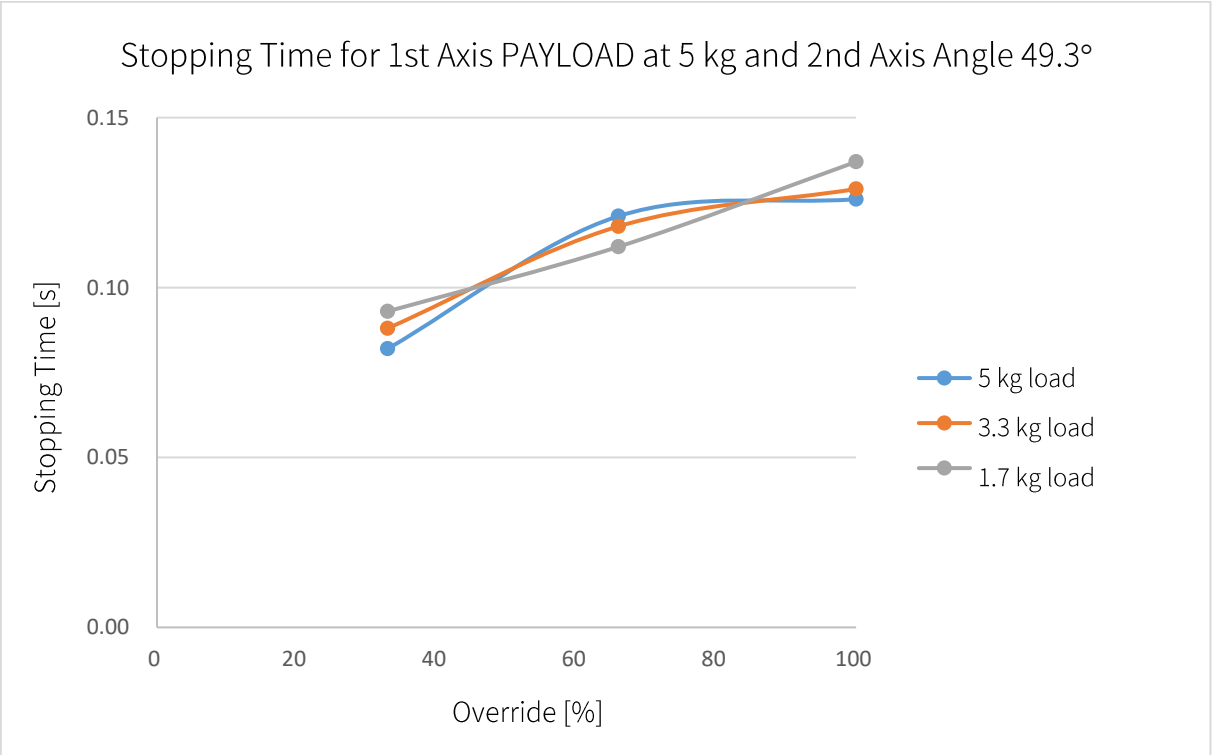


Fig. 4. 1st Axis Stopping Time for Payload (5,0) and 2nd Axis Angle 49.3°

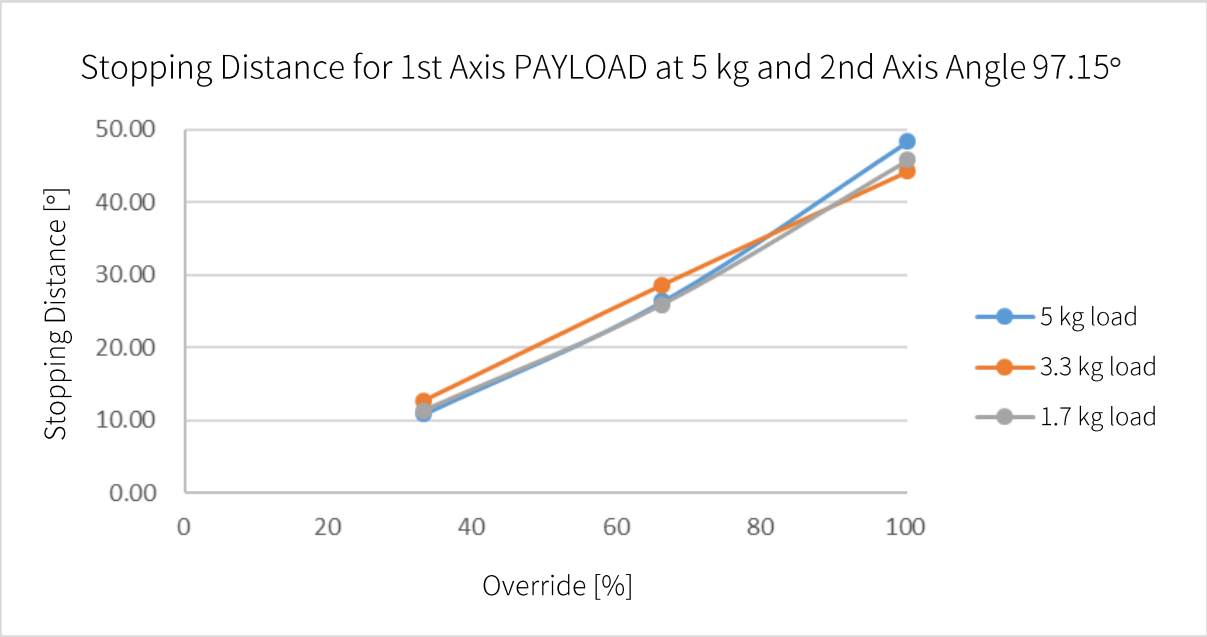


Fig. 5. 1st Axis Stopping Angle for Payload (5,0) and 2nd Axis Angle 97.15°

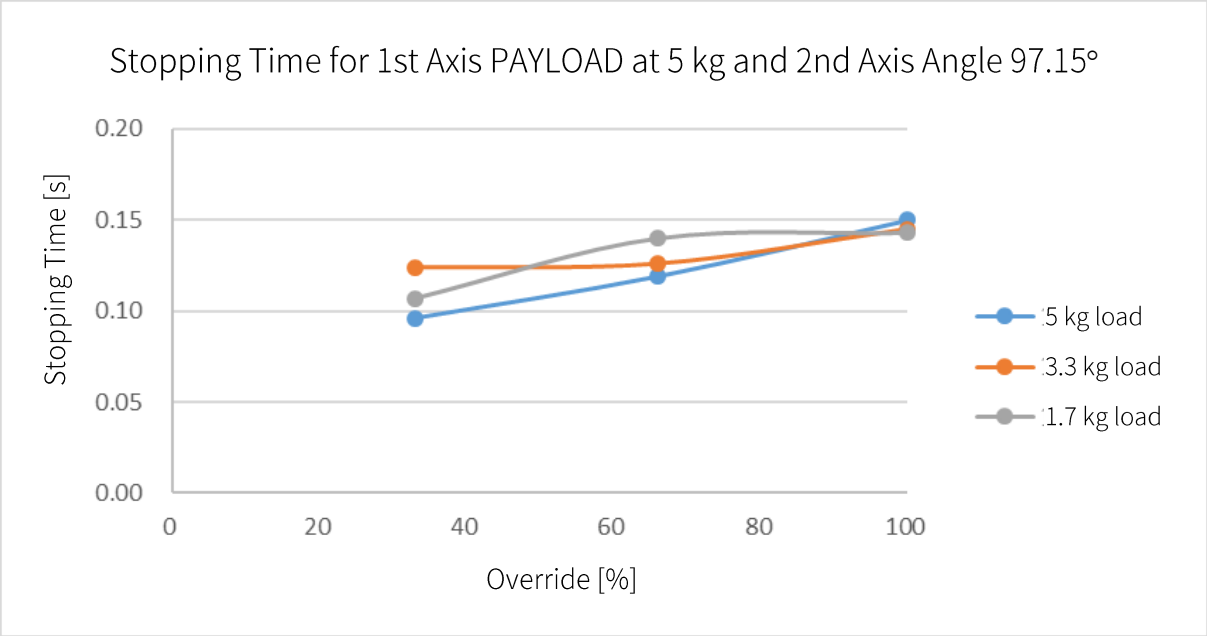


Fig. 6. 1st Axis Stopping Time for Payload (5,0) and 2nd Axis Angle 97.15°

When operating 2nd, 3rd, or 4th axis (for PAYLOAD (5,0))

The robot behavior when an emergency stop is applied during 2nd axis, 3rd axis, or 4th axis single-axis operation is shown below. Table 2 shows the operating conditions, and Figures 7 to 12 show the relationship between the stopping distance and stopping time when an emergency stop is applied for various operating conditions.

Table 2 Operating Conditions (2nd, 3rd, or 4th Axis)

Operation axis	PAYLOAD setting value	3rd axis position	Approach start position	Stop signal issuing position	Override setting value	Load	Drawing
2nd axis	(5,0)	0 mm	145°	0°	100%	5 kg	Fig. 7 Fig. 8
						3.3 kg	
						1.7 kg	
					66%	5 kg	
						3.3 kg	
						1.7 kg	
					33%	5 kg	
						3.3 kg	
						1.7 kg	
3rd axis			160 mm (Upper limit)	80 mm	100%	5 kg	Fig. 9 Fig. 10
						3.3 kg	
						1.7 kg	
					66%	5 kg	
						3.3 kg	
						1.7 kg	
					33%	5 kg	
						3.3 kg	
						1.7 kg	
4th axis		0 mm	180°	0°	100%	5 kg	Fig. 11 Fig. 12
						3.3 kg	
						1.7 kg	
					66%	5 kg	
						3.3 kg	
						1.7 kg	
					33%	5 kg	
						3.3 kg	
						1.7 kg	

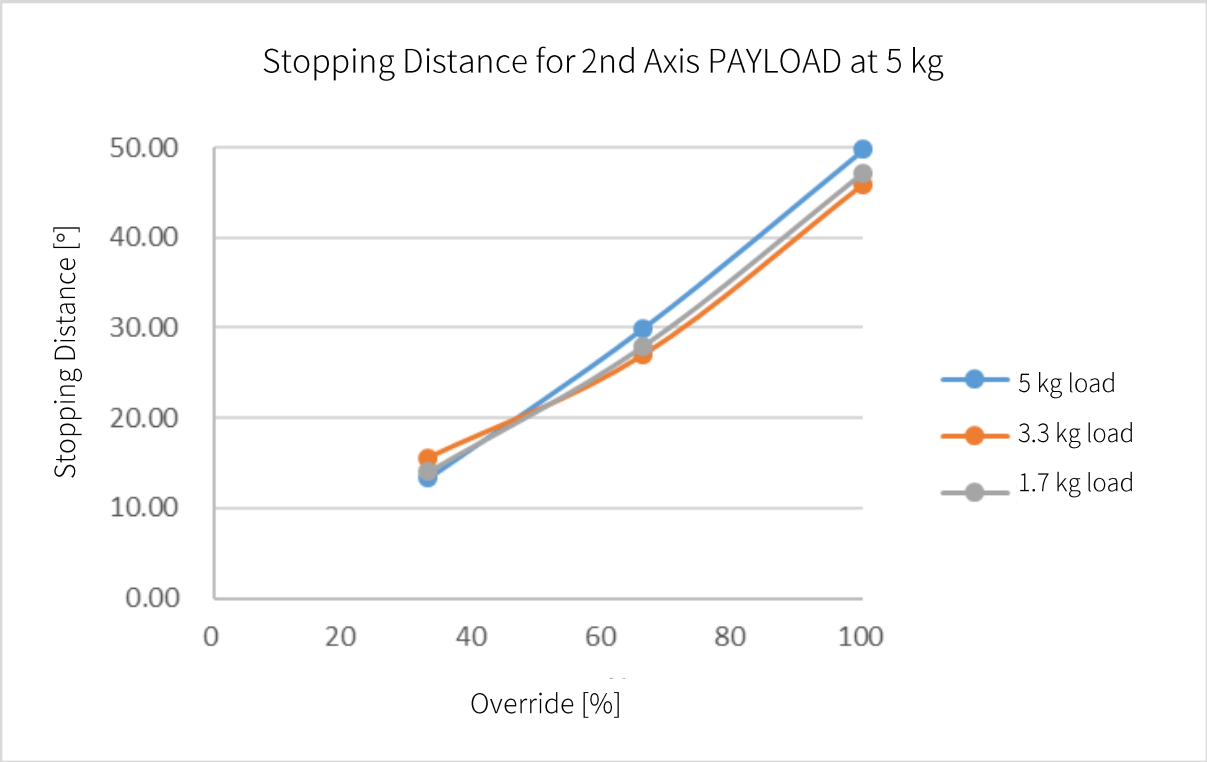


Fig. 7. 2nd Axis Stopping Angle for Payload (5,0)

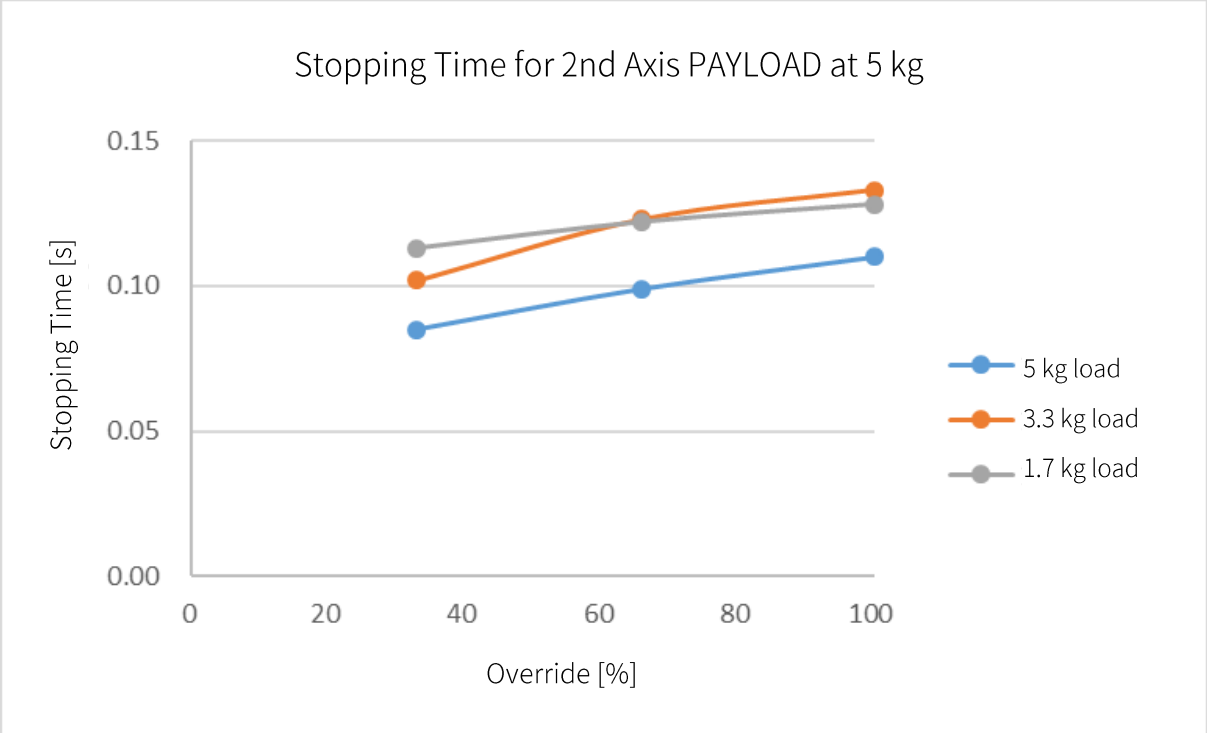


Fig. 8. 2nd Axis Stopping Time for Payload (5,0)

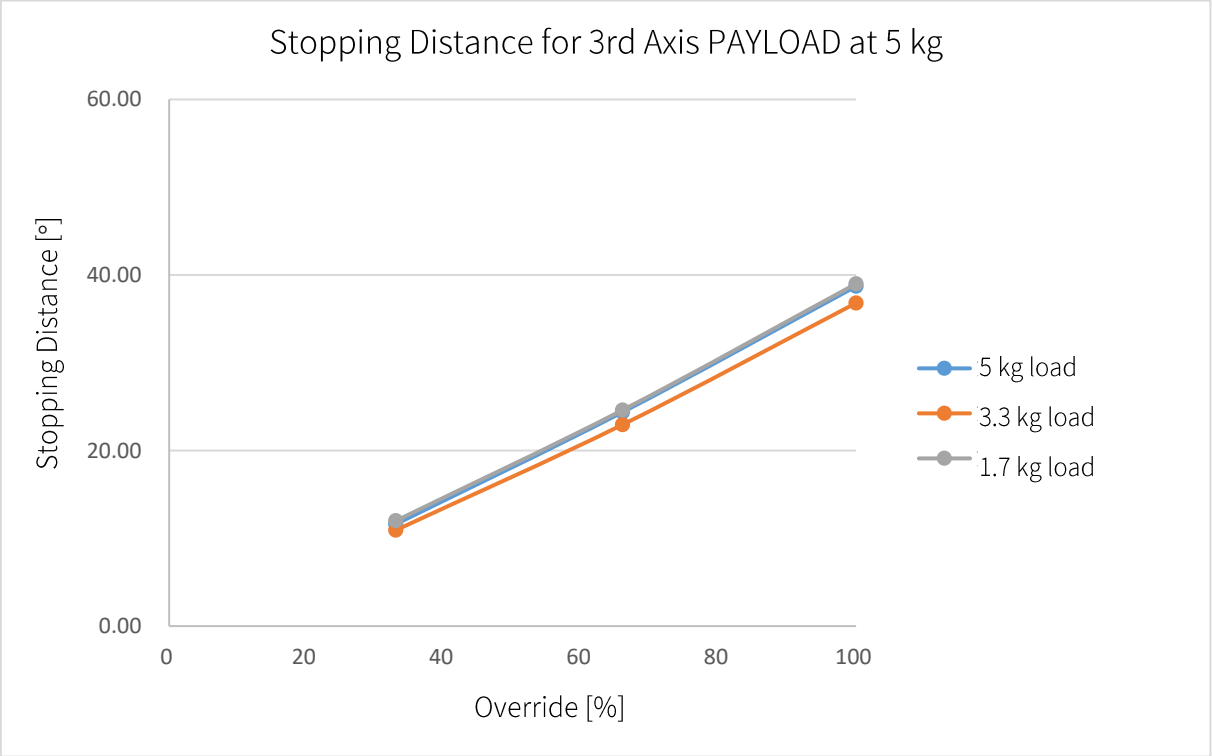


Fig. 9. 3rd Axis Stopping Angle for Payload (5,0)

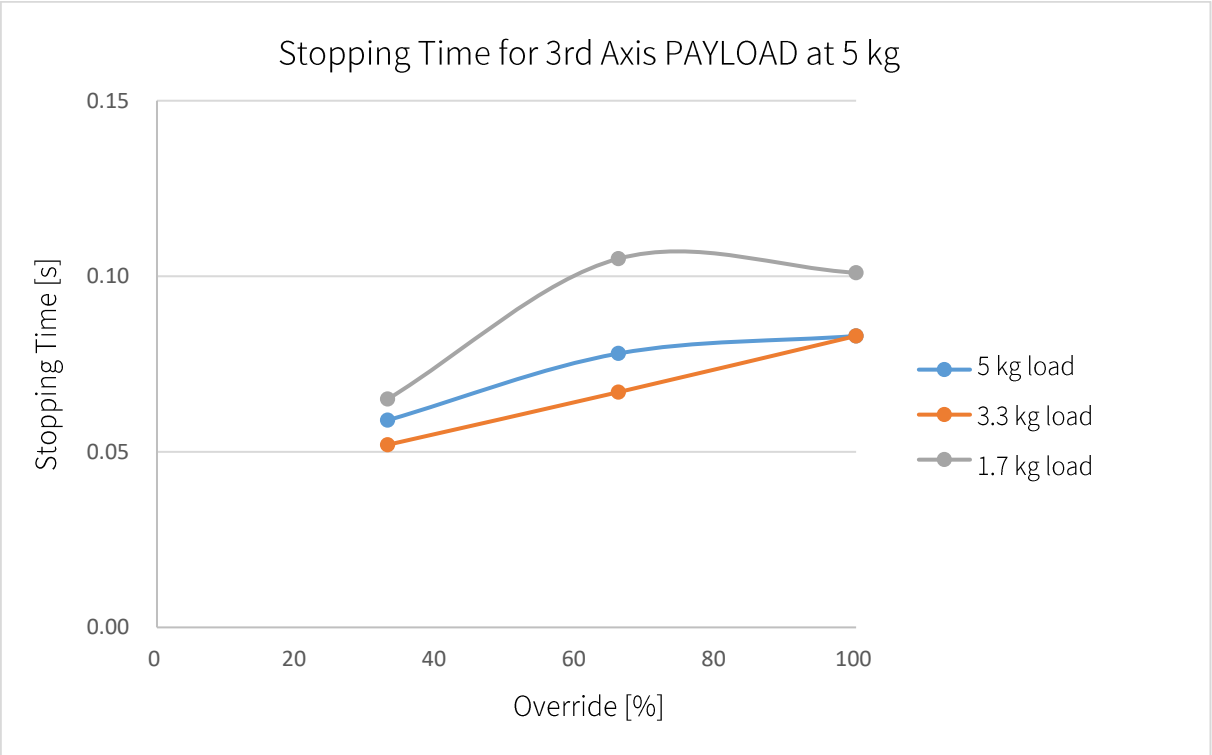


Fig. 10. 3rd Axis Stopping Time for Payload (5,0)

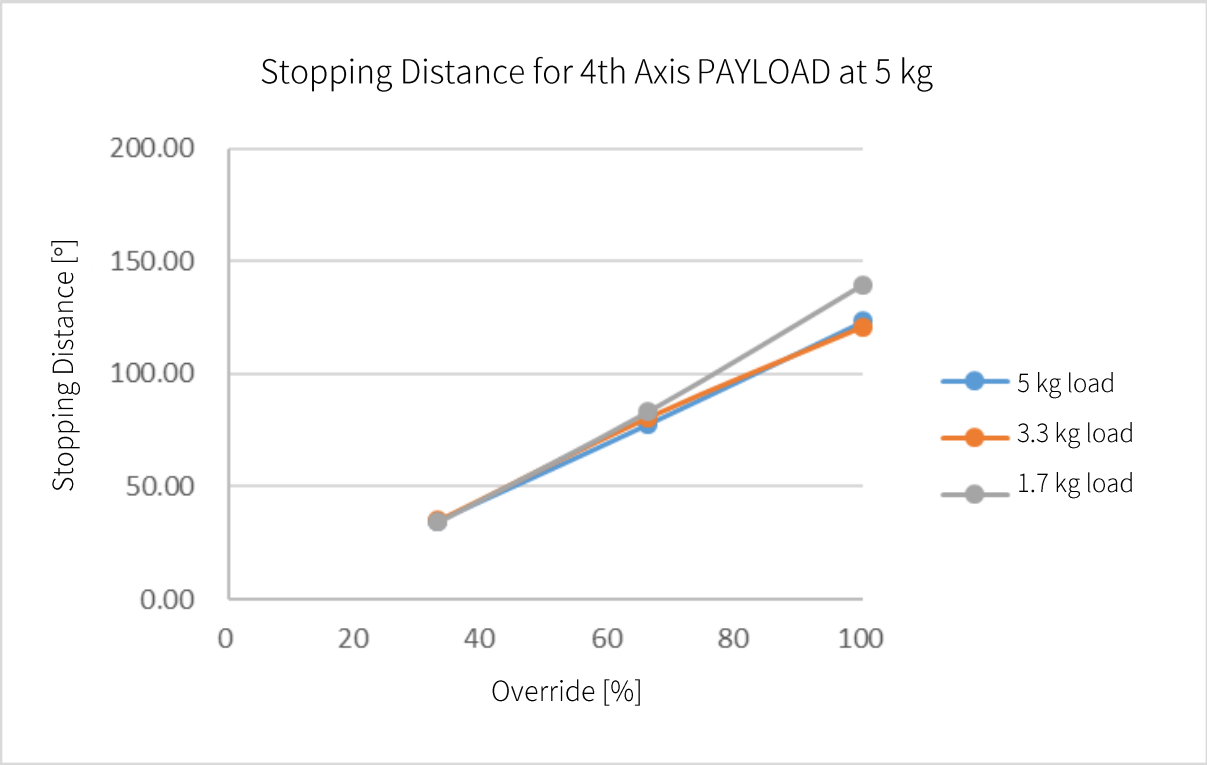


Fig. 11. 4th Axis Stopping Angle for Payload (5,0)

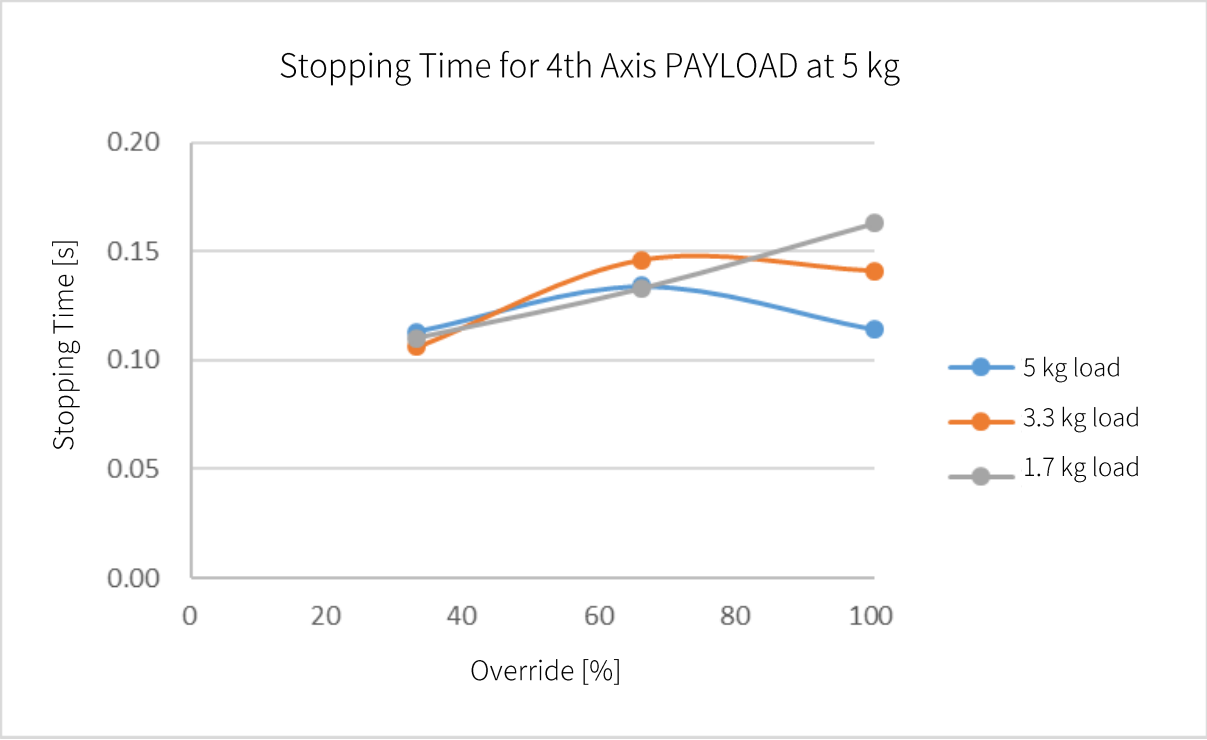


Fig. 12. 4th Axis Stopping Time for Payload (5,0)

2.2.2.2 Robot Behavior When Emergency Stop Is Applied (for PAYLOAD (3.3,0))

The stopping distances and stopping times for each joint axis when the PAYLOAD is set to "3.3 kg load with 0 mm offset" are shown below.

When operating 1st axis (for PAYLOAD (3.3,0))

The robot behavior when an emergency stop is applied during 3rd axis single-axis operation is shown below. Table 3 shows the operating conditions, and Figures 13 to 18 show the relationship between the stopping distance and stopping time when an emergency stop is applied for various operating conditions.

Table 3 Operating Conditions (1st Axis)

Operation axis	PAYLOAD setting value	3rd axis position	Approach start position	Stop signal issuing position	2nd axis angle	Override setting value	Load	Drawing
1st axis	(3.3,0)	0 mm (lower limit)	130°	0°	0° (100% of maximum displacement)	100%	3.3 kg	Fig. 13 Fig. 14
							2.2 kg	
							1.1 kg	
						66%	3.3 kg	
							2.2 kg	
							1.1 kg	
						33%	3.3 kg	
							2.2 kg	
							1.1 kg	
					49.3° (66% of maximum displacement)	100%	3.3 kg	Fig. 15 Fig. 16
							2.2 kg	
							1.1 kg	
						66%	3.3 kg	
							2.2 kg	
							1.1 kg	
						33%	3.3 kg	
							2.2 kg	
							1.1 kg	
					97.15° (33% of maximum displacement)	100%	3.3 kg	Fig. 17 Fig. 18
							2.2 kg	
							1.1 kg	
						66%	3.3 kg	
							2.2 kg	
							1.1 kg	
						33%	3.3 kg	
							2.2 kg	
							1.1 kg	

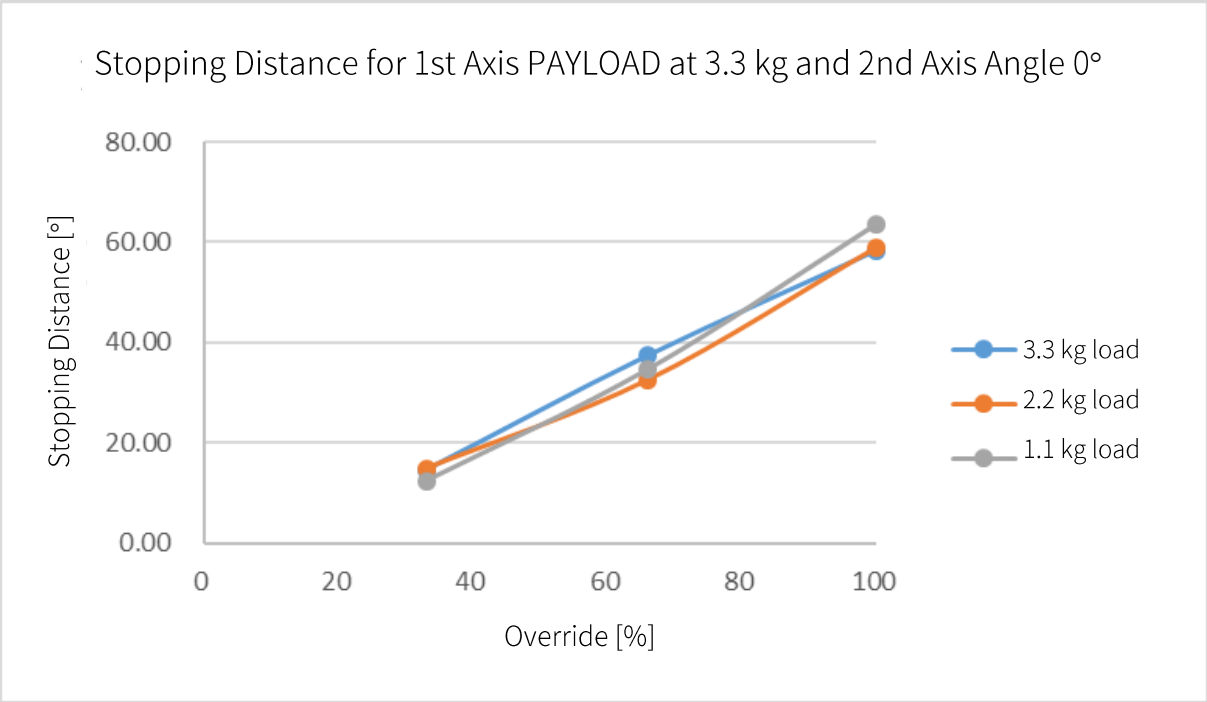


Fig. 13. 1st Axis Stopping Angle for Payload (3.3,0) and 2nd Axis Angle 0°

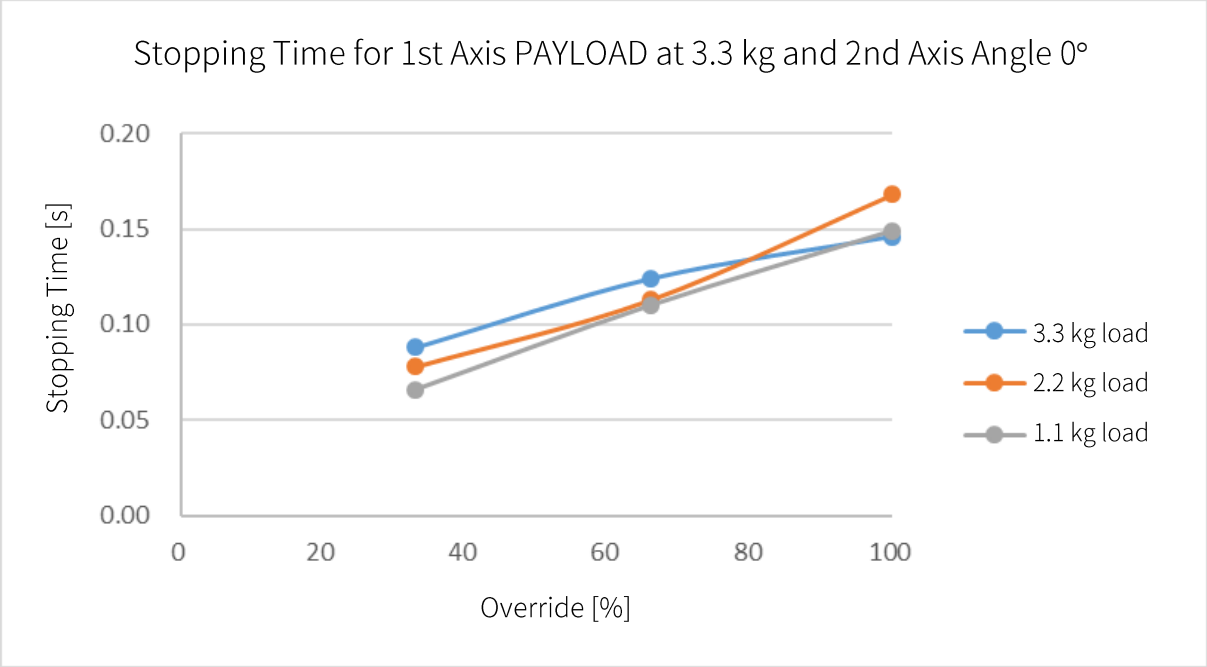


Fig. 14. 1st Axis Stopping Time for Payload (3.3,0) and 2nd Axis Angle 0°

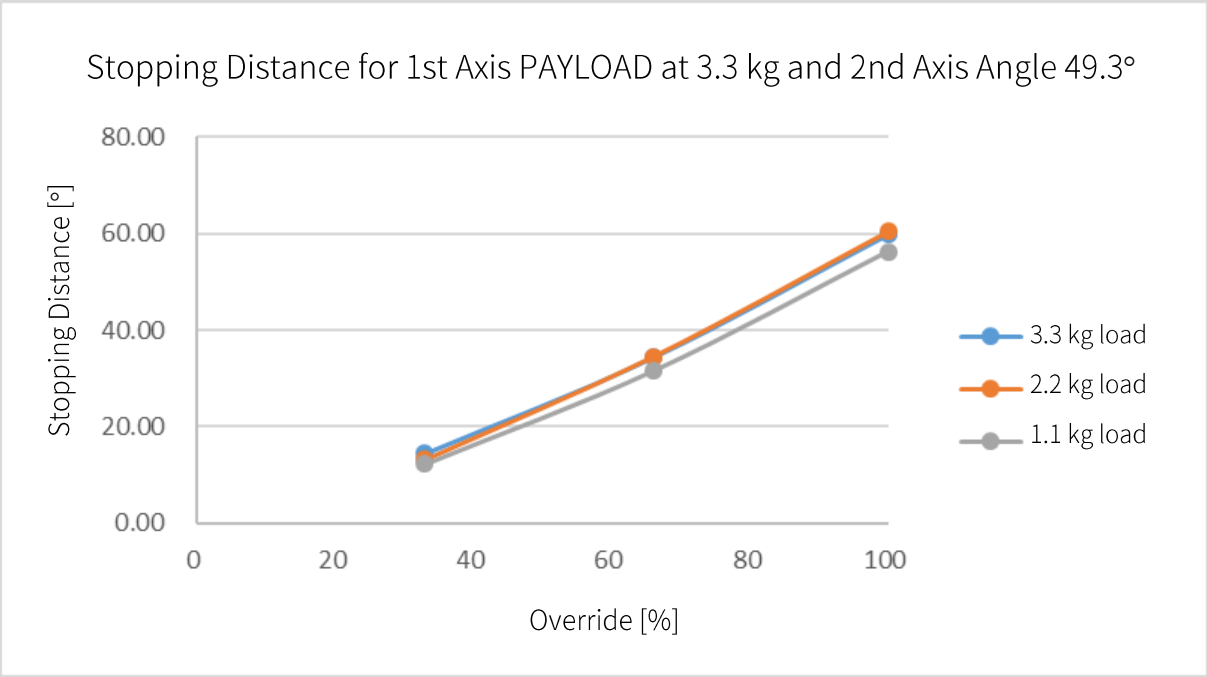


Fig. 15. 1st Axis Stopping Angle for Payload (3.3,0) and 2nd Axis Angle 49.3°

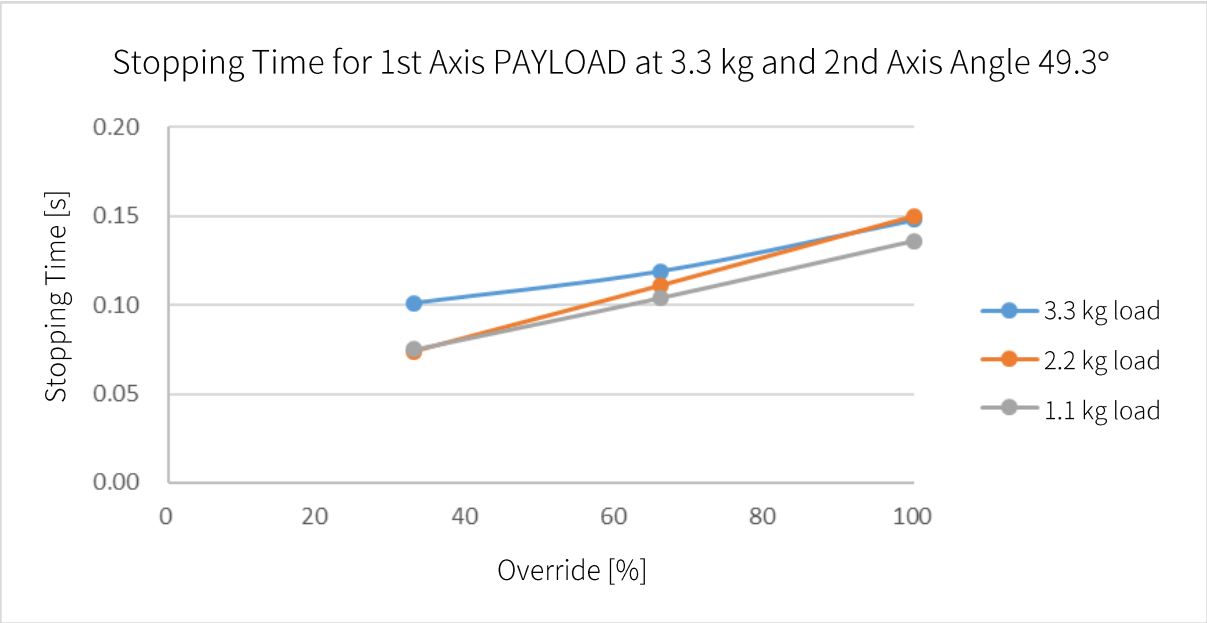


Fig. 16. 1st Axis Stopping Time for Payload (3.3,0) and 2nd Axis Angle 49.3°

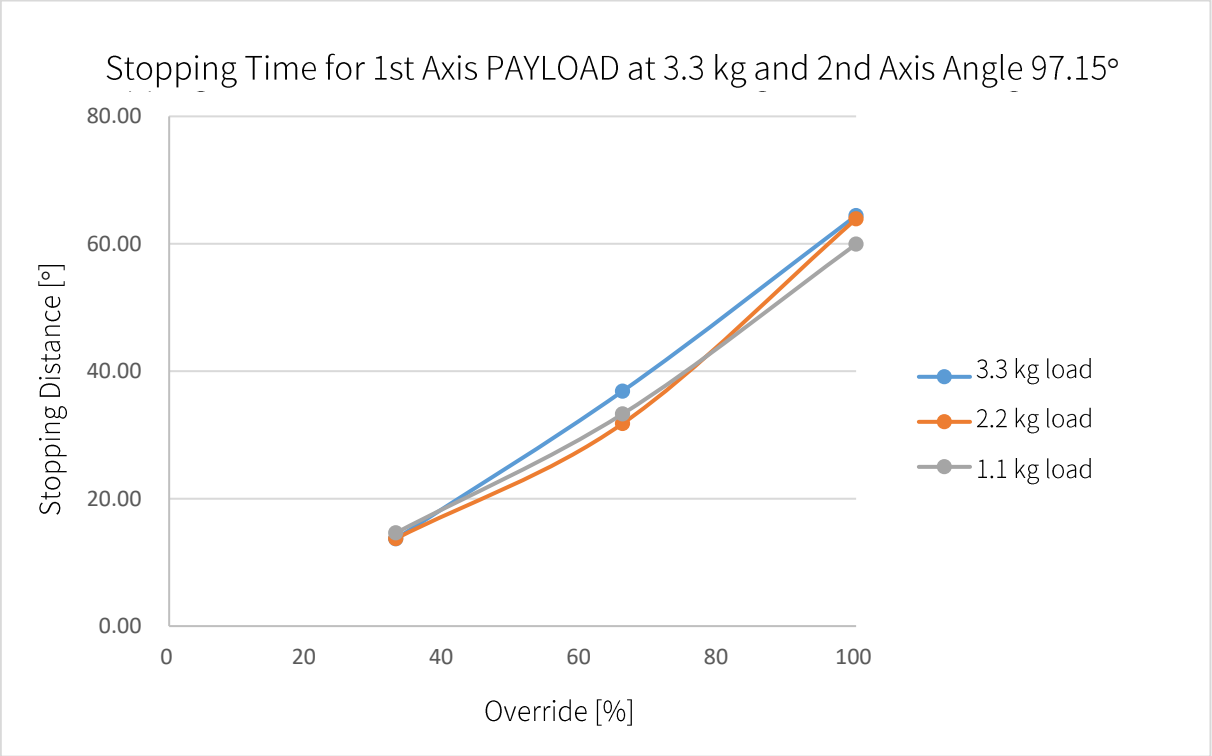


Fig. 17. 1st Axis Stopping Angle for Payload (3.3,0) and 2nd Axis Angle 97.15°

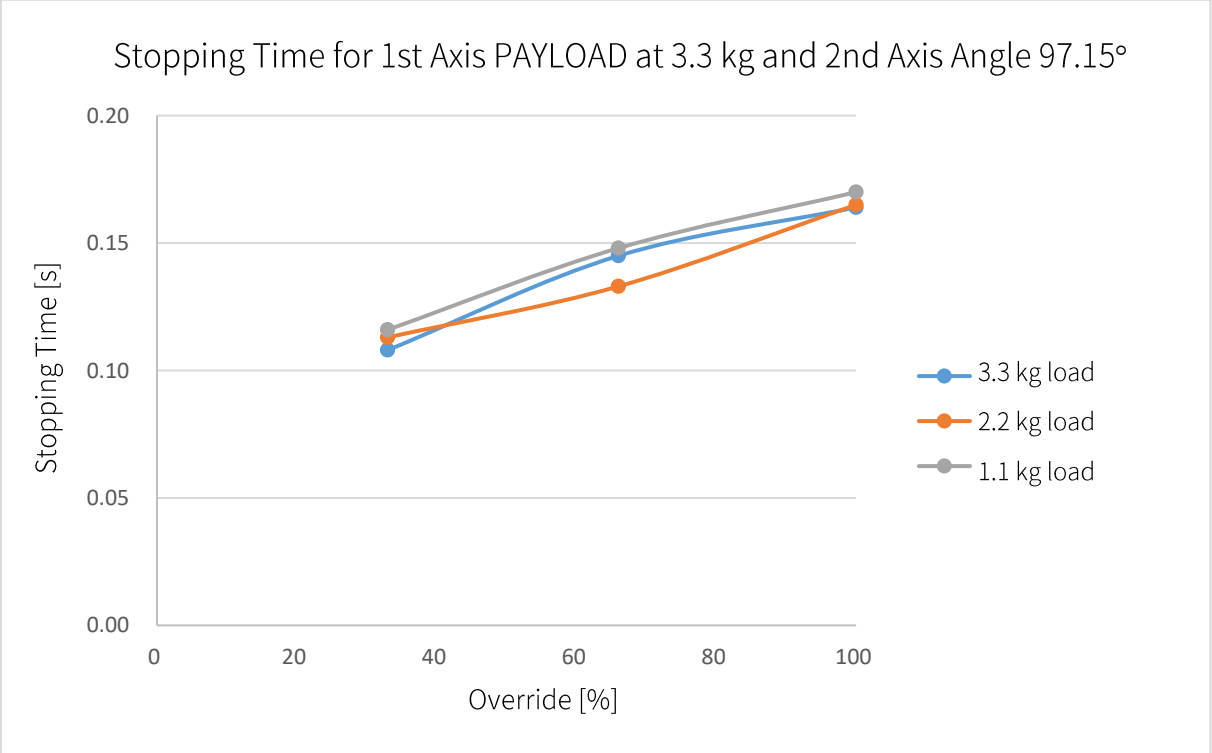


Fig. 18. 1st Axis Stopping Time for Payload (3.3,0) and 2nd Axis Angle 97.15°

When operating 2nd, 3rd, or 4th axis (for PAYLOAD (3.3,0))

The robot behavior when an emergency stop is applied during 2nd axis, 3rd axis, or 4th axis single-axis operation is shown below. Table 4 shows the operating conditions, and Figures 19 to 24 show the relationship between the stopping distance and stopping time when an emergency stop is applied for various operating conditions.

Table 4 Operating Conditions (2nd, 3rd, or 4th Axis)

Operation axis	PAYLOAD setting value	3rd axis position	Approach start position	Stop signal issuing position	Override setting value	Load	Drawing
2nd axis	(3.3,0)	0 mm	145°	0°	100%	3.3 kg	Fig. 19 Fig. 20
						2.2 kg	
						1.1 kg	
					66%	3.3 kg	
						2.2 kg	
						1.1 kg	
					33%	3.3 kg	
						2.2 kg	
						1.1 kg	
3rd axis			160 mm (Upper limit)	80 mm	100%	3.3 kg	Fig. 21 Fig. 22
						2.2 kg	
						1.1 kg	
					66%	3.3 kg	
						2.2 kg	
						1.1 kg	
					33%	3.3 kg	
						2.2 kg	
						1.1 kg	
4th axis		0 mm	180°	0°	100%	3.3 kg	Fig. 23 Fig. 24
						2.2 kg	
						1.1 kg	
					66%	3.3 kg	
						2.2 kg	
						1.1 kg	
					33%	3.3 kg	
						2.2 kg	
						1.1 kg	

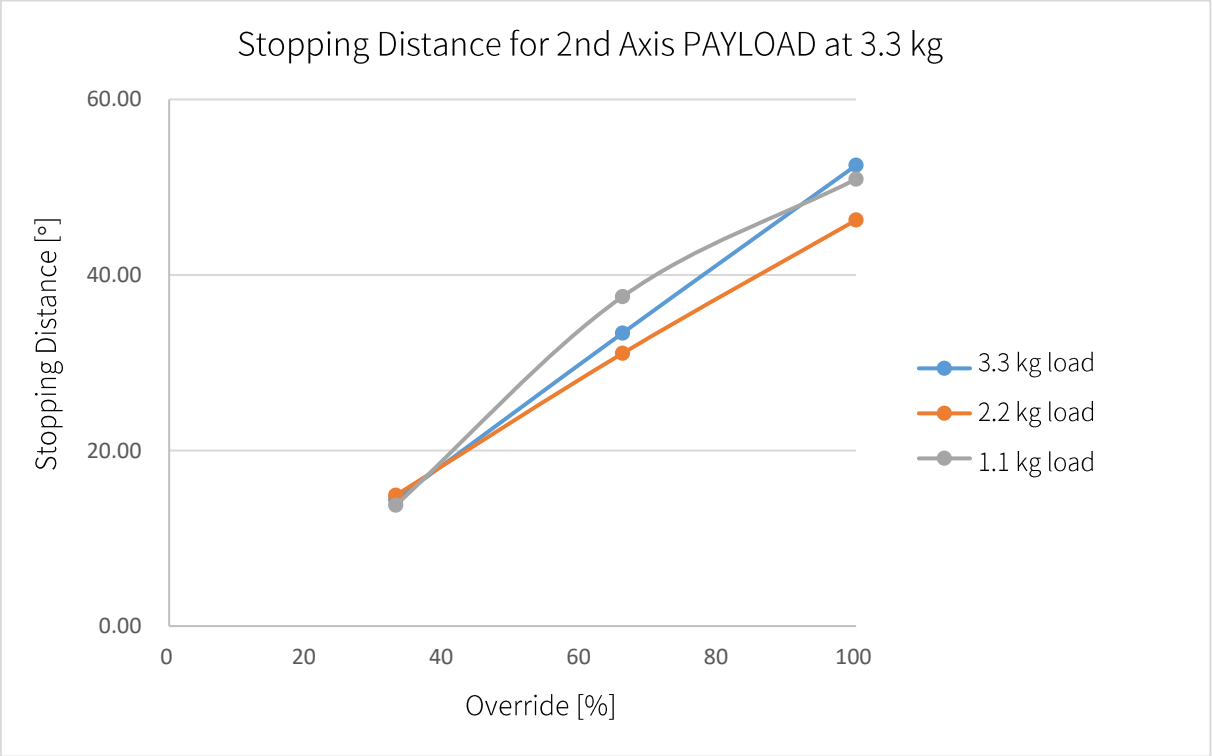


Fig. 19. 2nd Axis Stopping Angle for Payload (3.3,0)

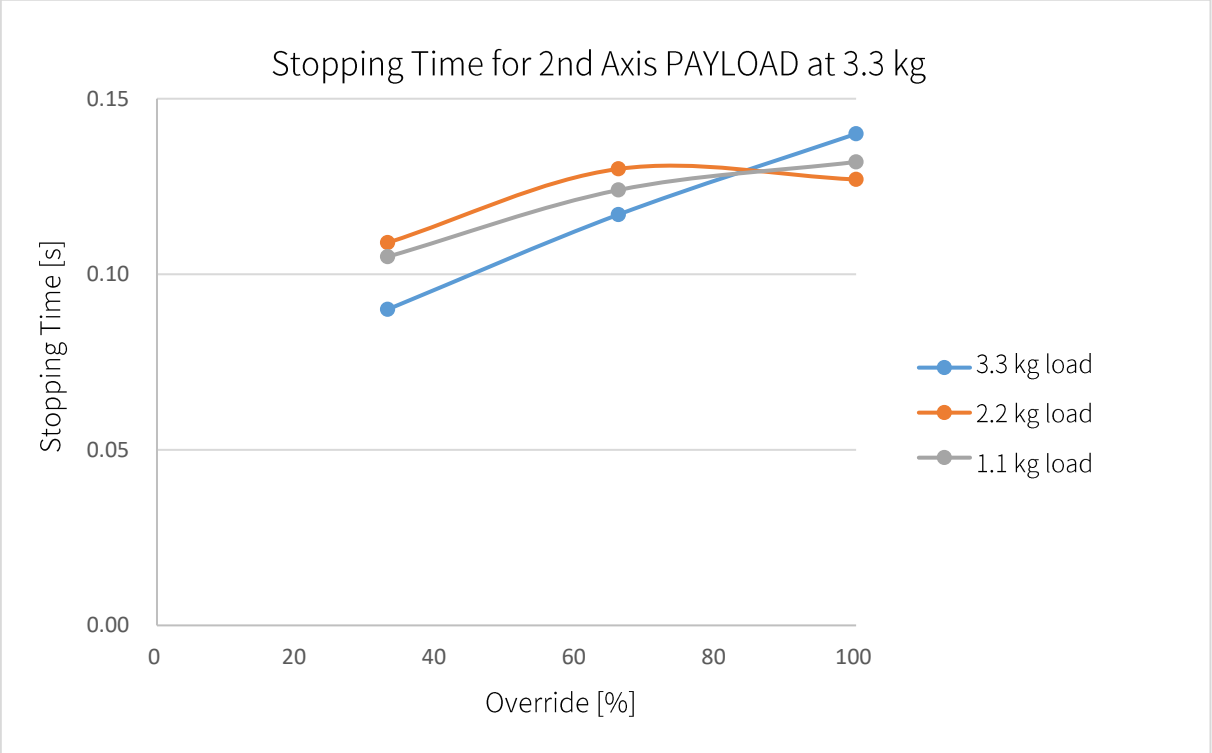


Fig. 20. 2nd Axis Stopping Time for Payload (3.3,0)

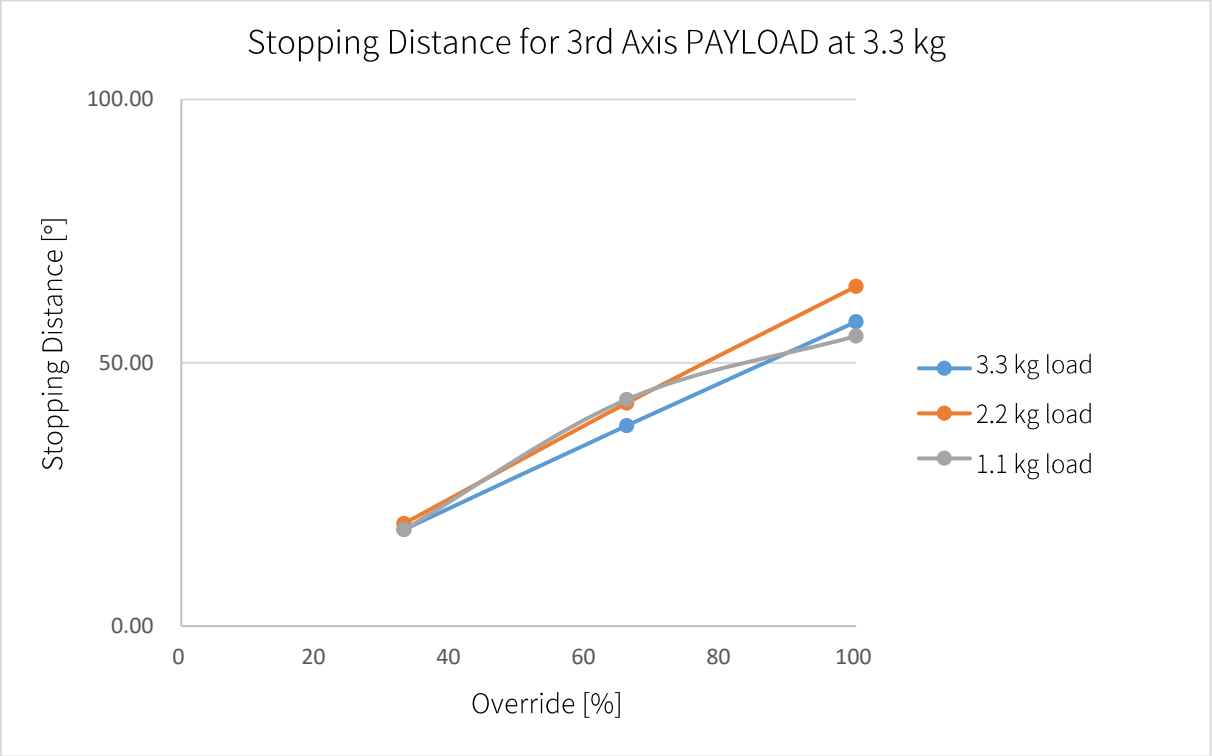


Fig. 21. 3rd Axis Stopping Angle for Payload (3.3,0)

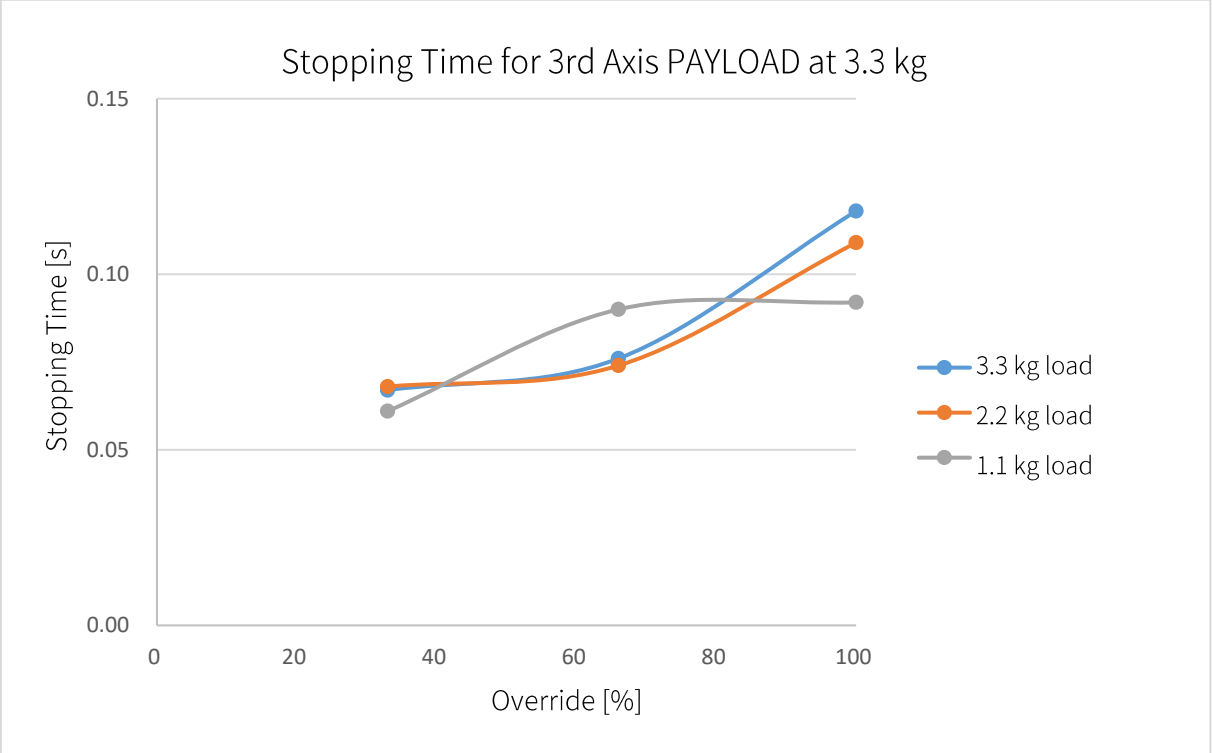


Fig. 22. 3rd Axis Stopping Time for Payload (3.3,0)

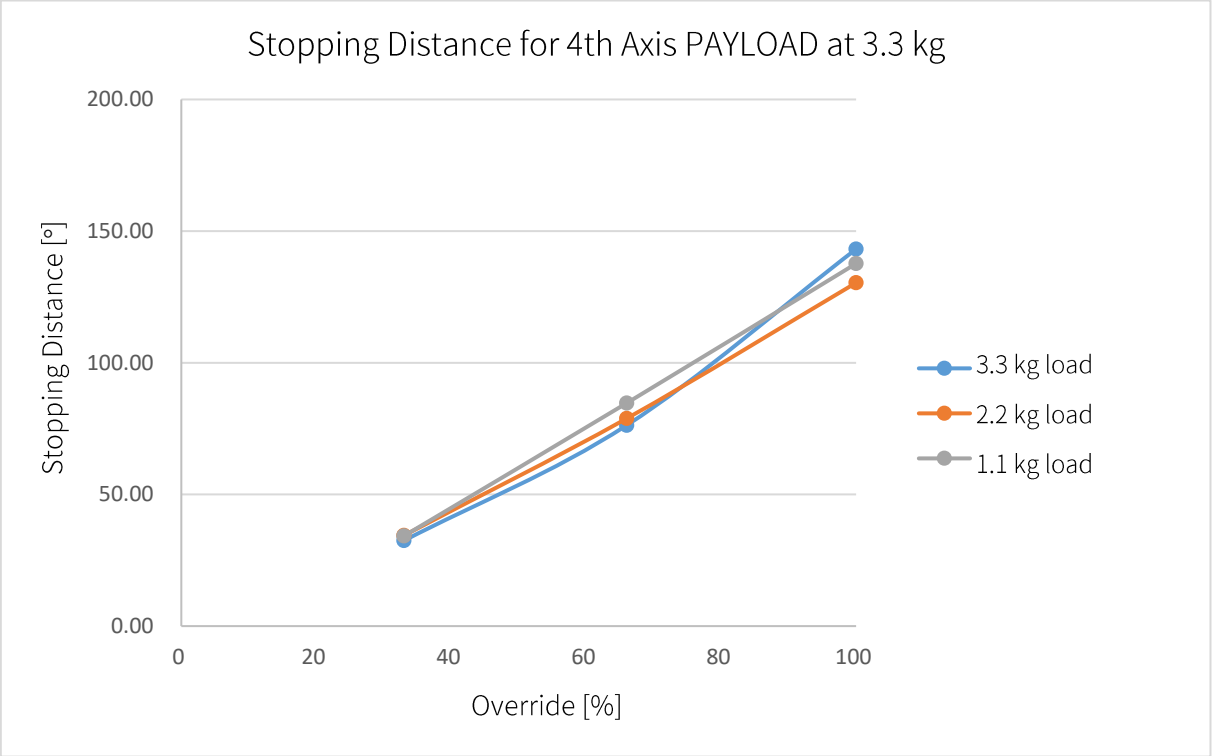


Fig. 23. 4th Axis Stopping Angle for Payload (3.3,0)

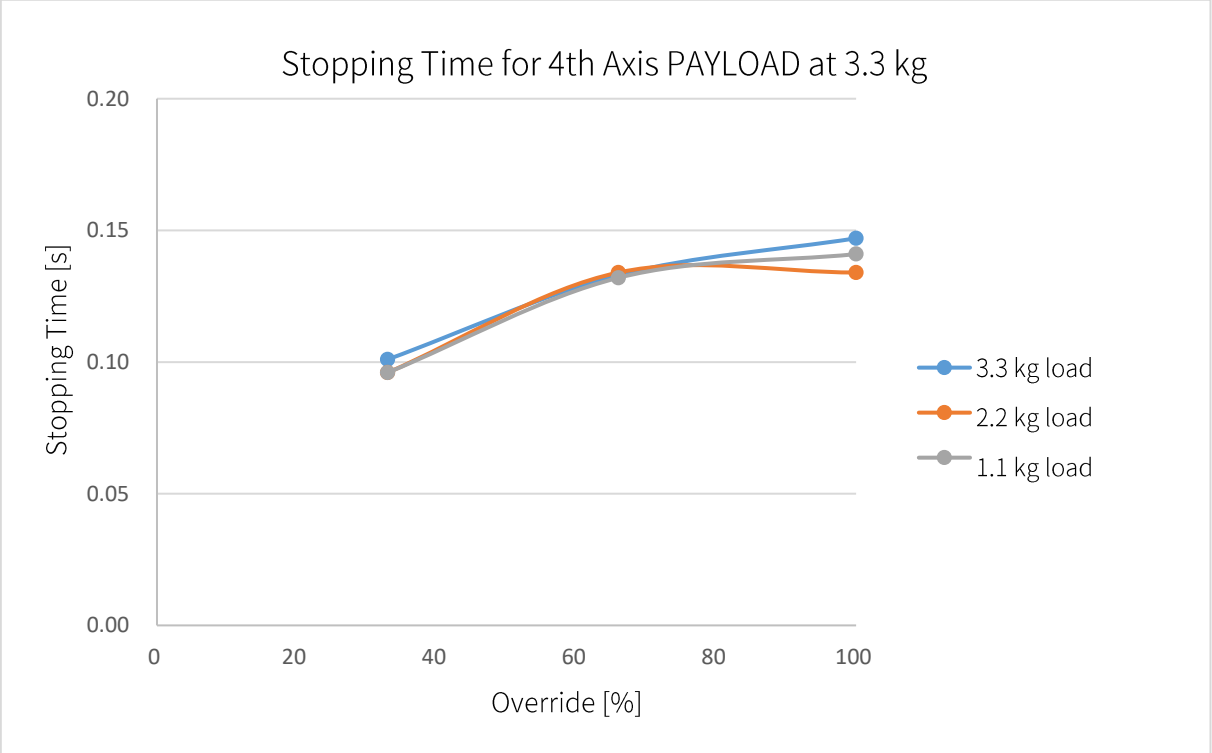


Fig. 24. 4th Axis Stopping Time for Payload (3.3,0)

2.2.2.3 Robot Behavior When Emergency Stop Is Applied (for Payload (1.7,0))

The stopping distances and stopping times for each joint axis when the PAYLOAD is set to "1.7 kg load with 0 mm offset" are shown below.

When operating 1st axis (for PAYLOAD (1.7,0))

The robot behavior when an emergency stop is applied during 3rd axis single-axis operation is shown below. Table 5 shows the operating conditions, and Figures 25 to 30 show the relationship between the stopping distance and stopping time when an emergency stop is applied for various operating conditions.

Table 5 Operating Conditions (1st Axis)

Operation axis	PAYLOAD setting value	3rd axis position	Approach start position	Stop signal issuing position	2nd axis angle	Override setting value	Load	Drawing
1st axis	(1.7,0)	0 mm (lower limit)	130°	0°	0° (100% of maximum displacement)	100%	1.7 kg	Fig. 25 Fig. 26
							1.2 kg	
							0.6 kg	
						66%	1.7 kg	
							1.2 kg	
							0.6 kg	
						33%	1.7 kg	
							1.2 kg	
							0.6 kg	
					49.3° (66% of maximum displacement)	100%	1.7 kg	Fig. 27 Fig. 28
							1.2 kg	
							0.6 kg	
						66%	1.7 kg	
							1.2 kg	
							0.6 kg	
						33%	1.7 kg	
							1.2 kg	
							0.6 kg	
					97.15° (33% of maximum displacement)	100%	1.7 kg	Fig. 29 Fig. 30
							1.2 kg	
							0.6 kg	
						66%	1.7 kg	
							1.2 kg	
							0.6 kg	
						33%	1.7 kg	
							1.2 kg	
							0.6 kg	

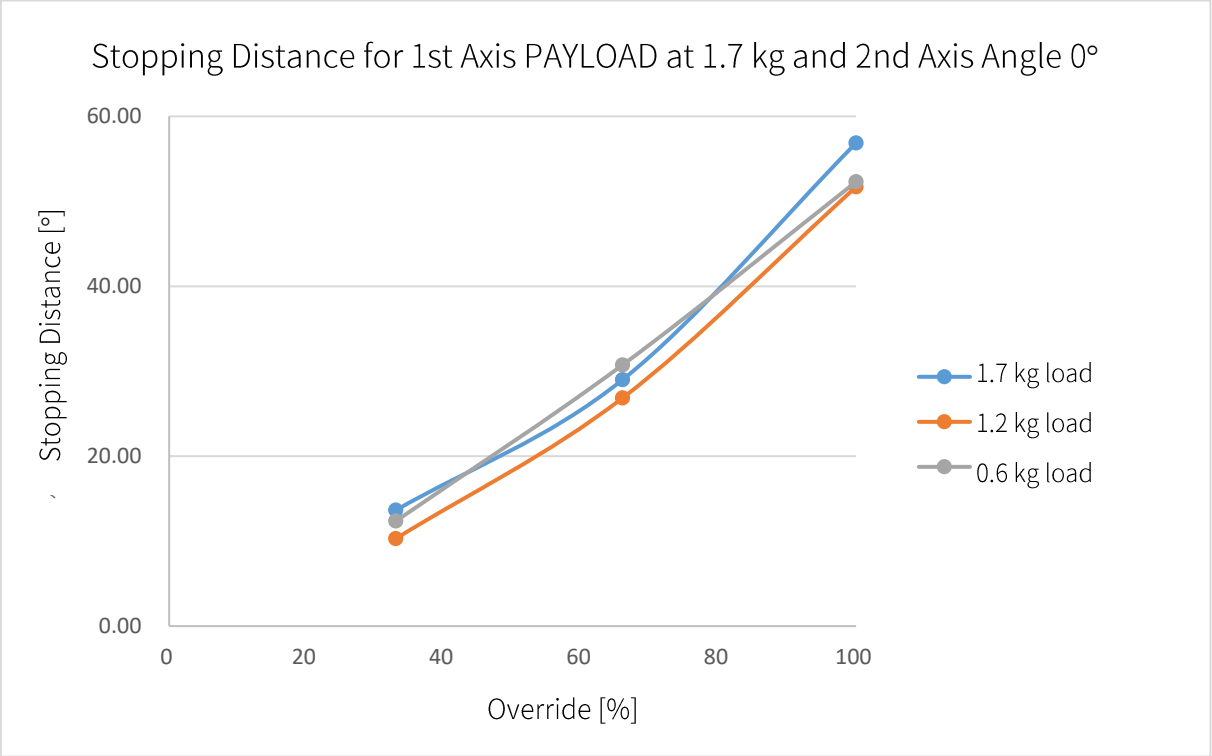


Fig. 25. 1st Axis Stopping Angle for Payload (1.7,0) and 2nd Axis Angle 0°

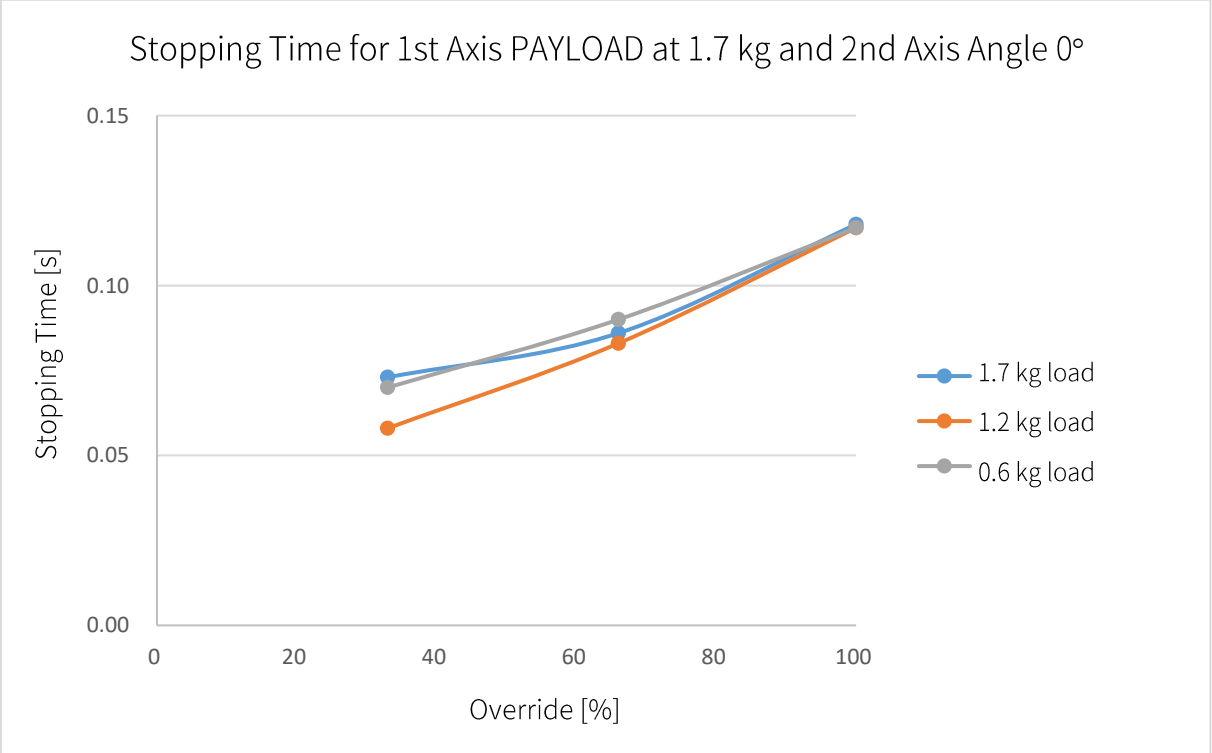


Fig. 26. 1st Axis Stopping Time for Payload (1.7,0) and 2nd Axis Angle 0°

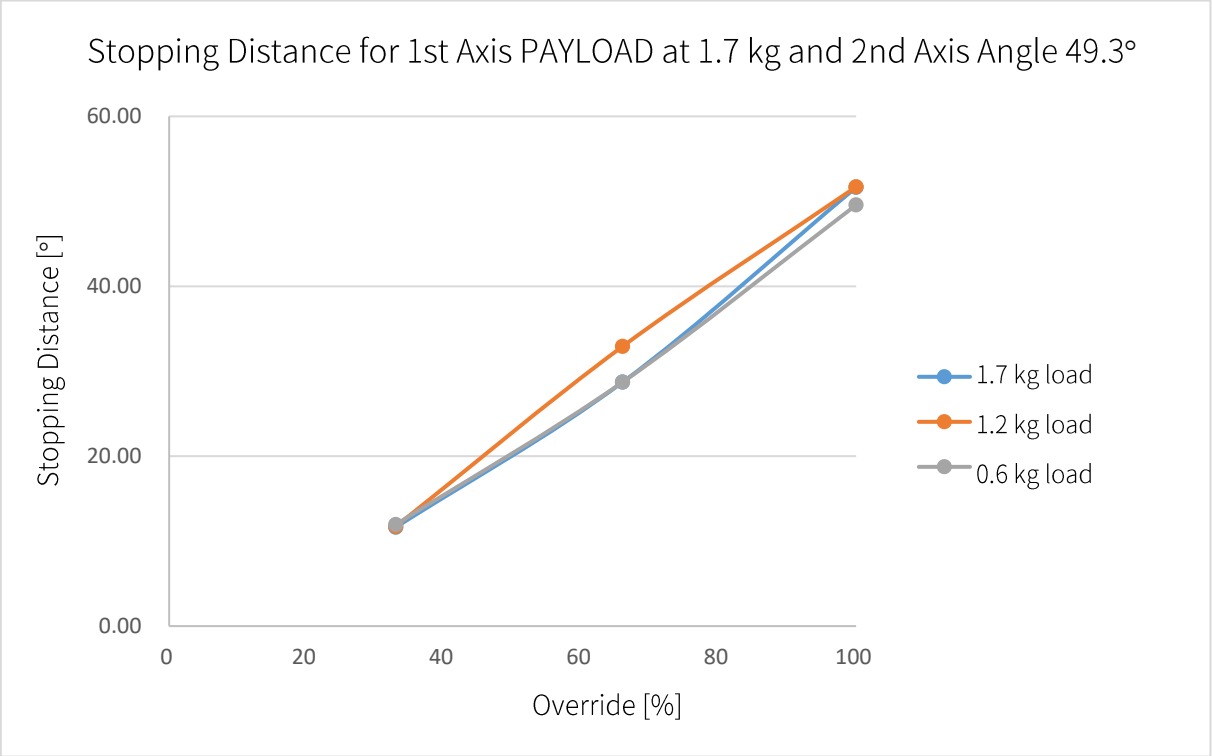


Fig. 27. 1st Axis Stopping Angle for Payload (1.7,0) and 2nd Axis Angle 49.3°

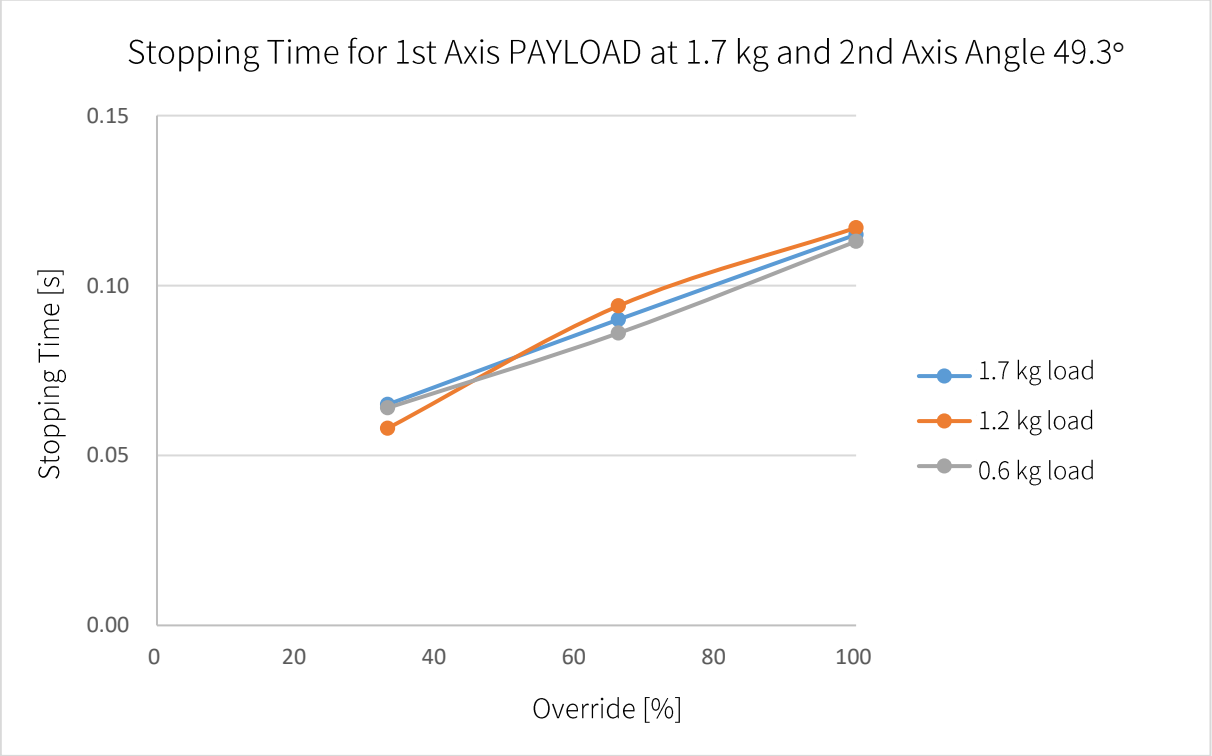


Fig. 28. 1st Axis Stopping Time for Payload (1.7,0) and 2nd Axis Angle 49.3°

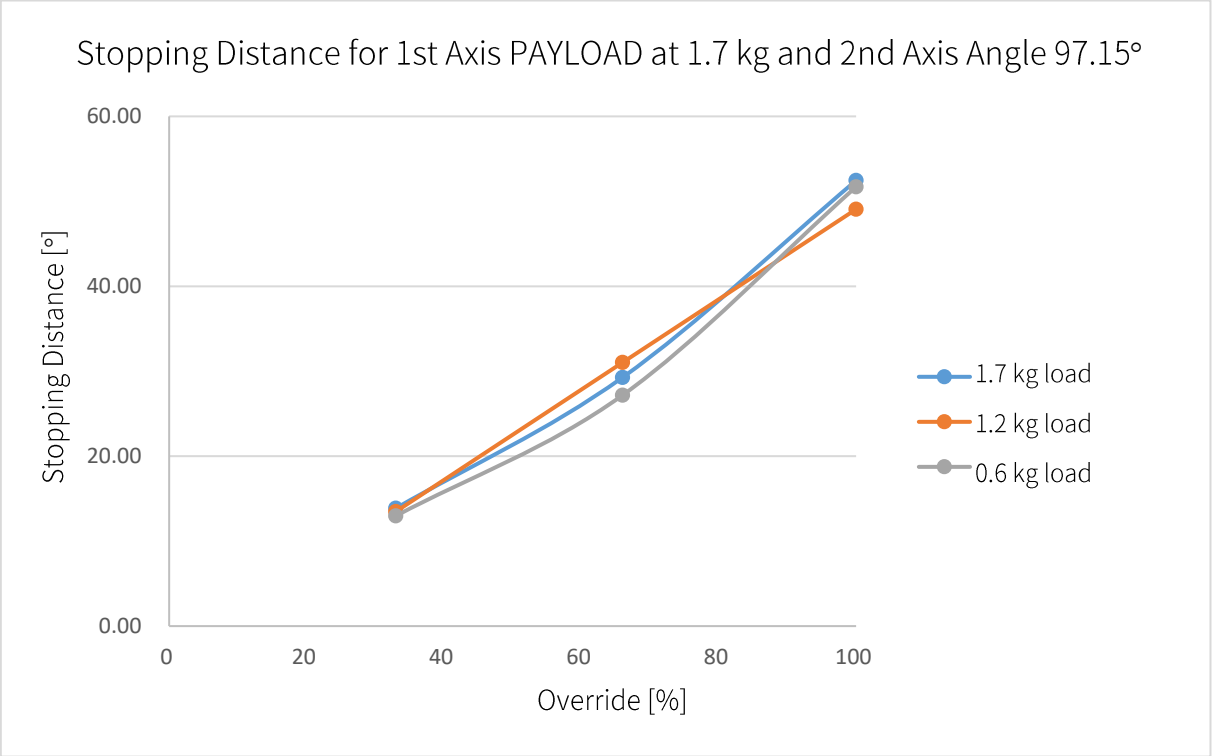


Fig. 29. 1st Axis Stopping Angle for Payload (1.7,0) and 2nd Axis Angle 97.15°

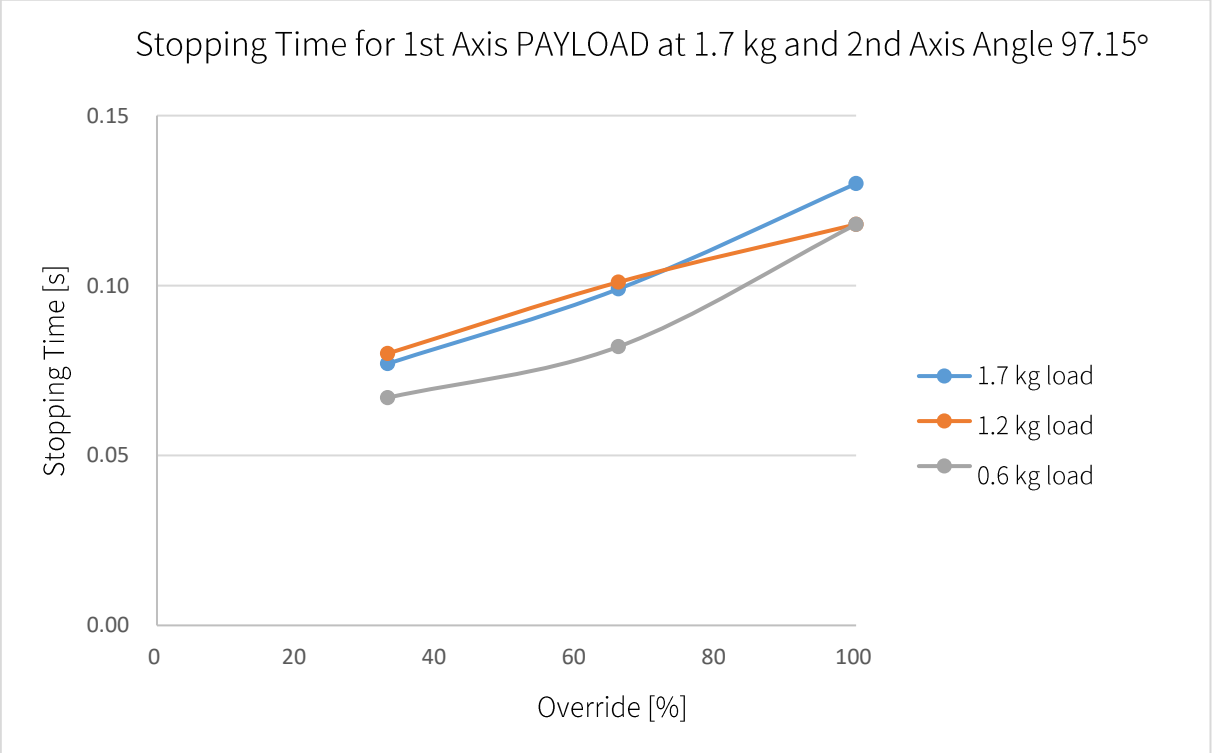


Fig. 30. 1st Axis Stopping Time for Payload (1.7,0) and 2nd Axis Angle 97.15°

When operating 2nd, 3rd, or 4th axis (for PAYLOAD (1.7,0))

The robot behavior when an emergency stop is applied during 2nd axis, 3rd axis, or 4th axis single-axis operation is shown below. Table 6 shows the operating conditions, and Figures 31 to 36 show the relationship between the stopping distance and stopping time when an emergency stop is applied for various operating conditions.

Table 6 Operating Conditions (2nd, 3rd, or 4th Axis)

Operation axis	PAYLOAD setting value	3rd axis position	Approach start position	Stop signal issuing position	Override setting value	Load	Drawing
2nd axis	(1.7,0)	0 mm	145°	0°	100%	1.7 kg	Fig. 31 Fig. 32
						1.2 kg	
						0.6 kg	
					66%	1.7 kg	
						1.2 kg	
						0.6 kg	
					33%	1.7 kg	
						1.2 kg	
						0.6 kg	
3rd axis			160 mm (Upper limit)	80 mm	100%	1.7 kg	Fig. 33 Fig. 34
						1.2 kg	
						0.6 kg	
					66%	1.7 kg	
						1.2 kg	
						0.6 kg	
					33%	1.7 kg	
						1.2 kg	
						0.6 kg	
4th axis		0 mm	180°	0°	100%	1.7 kg	Fig. 35 Fig. 36
						1.2 kg	
						0.6 kg	
					66%	1.7 kg	
						1.2 kg	
						0.6 kg	
					33%	1.7 kg	
						1.2 kg	
						0.6 kg	

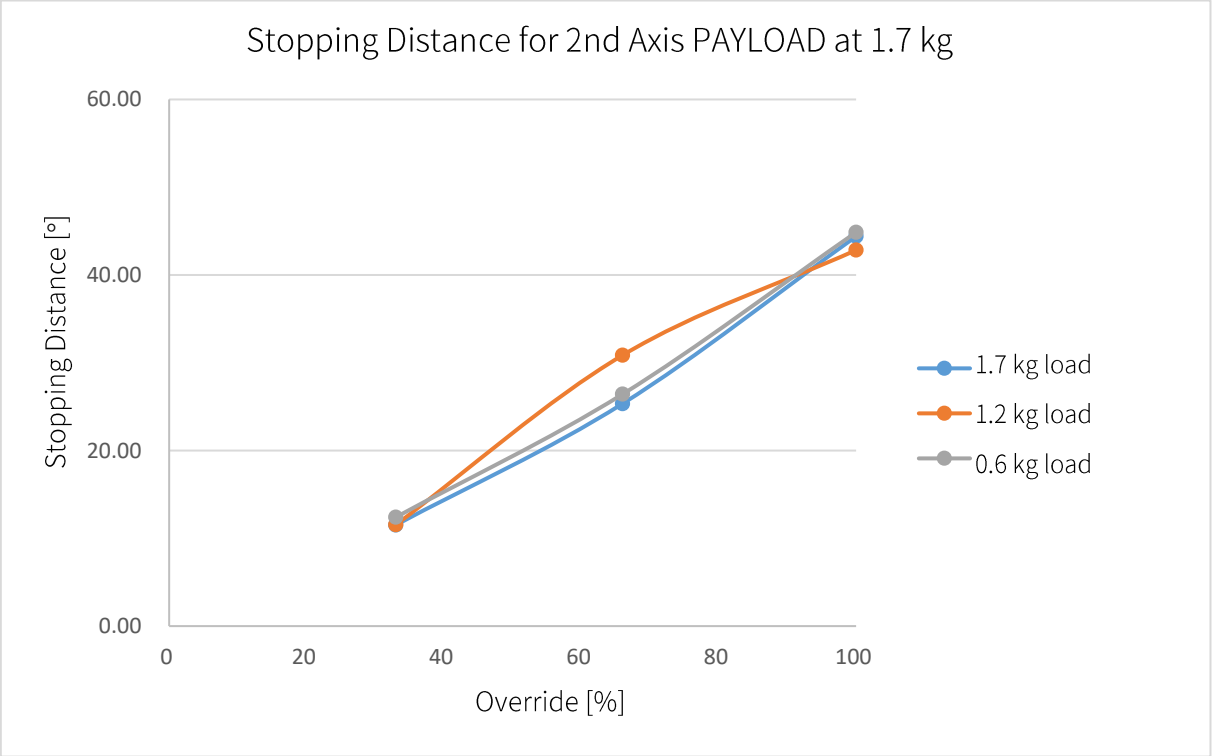


Fig. 31. 2nd Axis Stopping Angle for Payload (1.7,0)

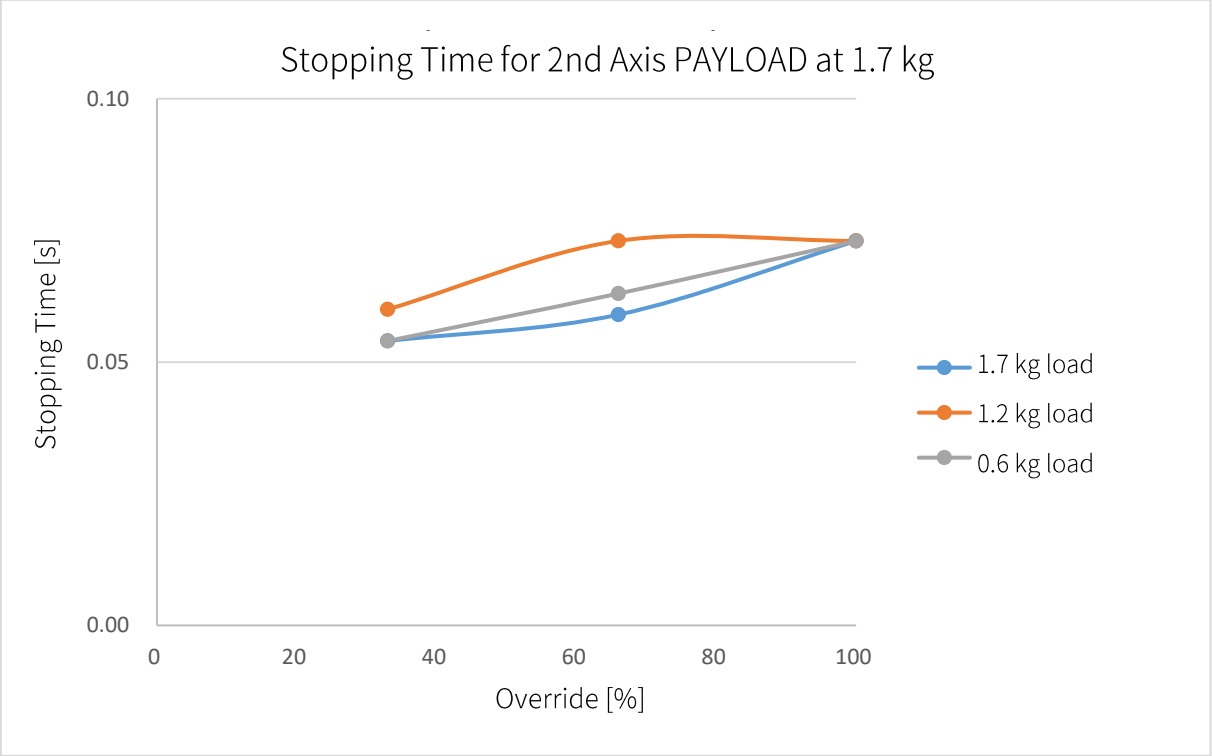


Fig. 32. 2nd Axis Stopping Time for Payload (1.7,0)

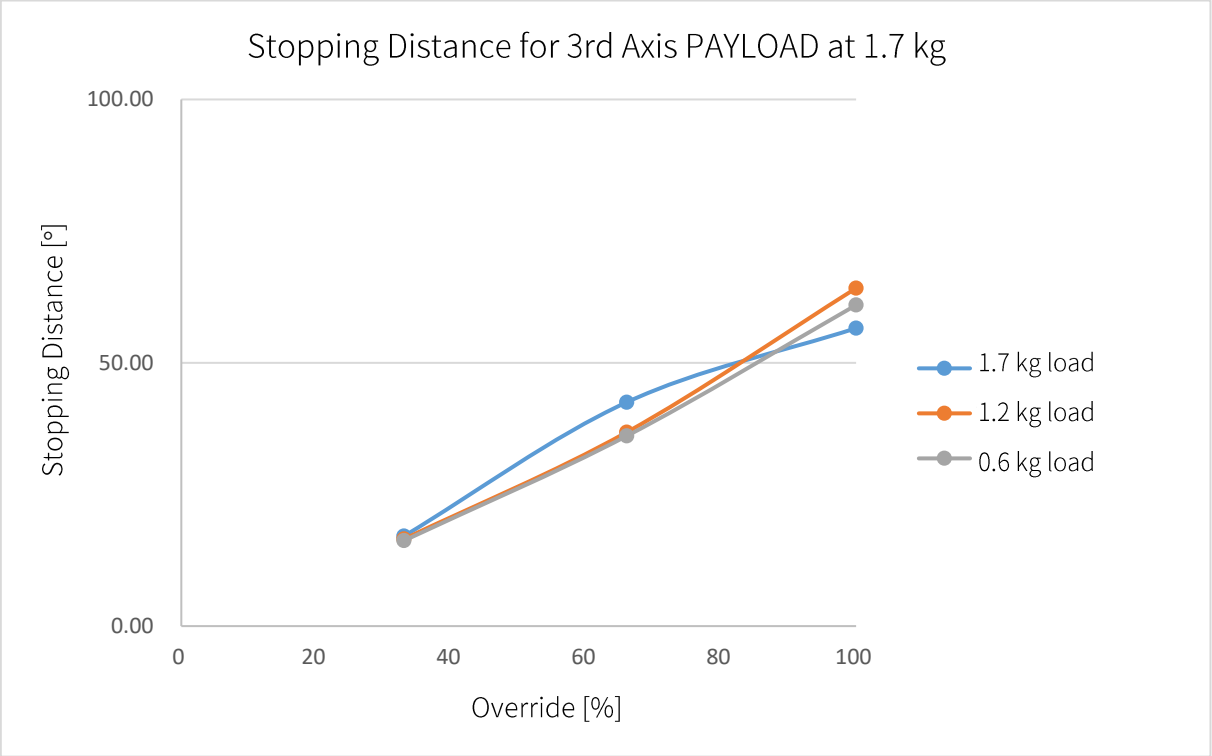


Fig. 33. 3rd Axis Stopping Angle for Payload (1.7,0)

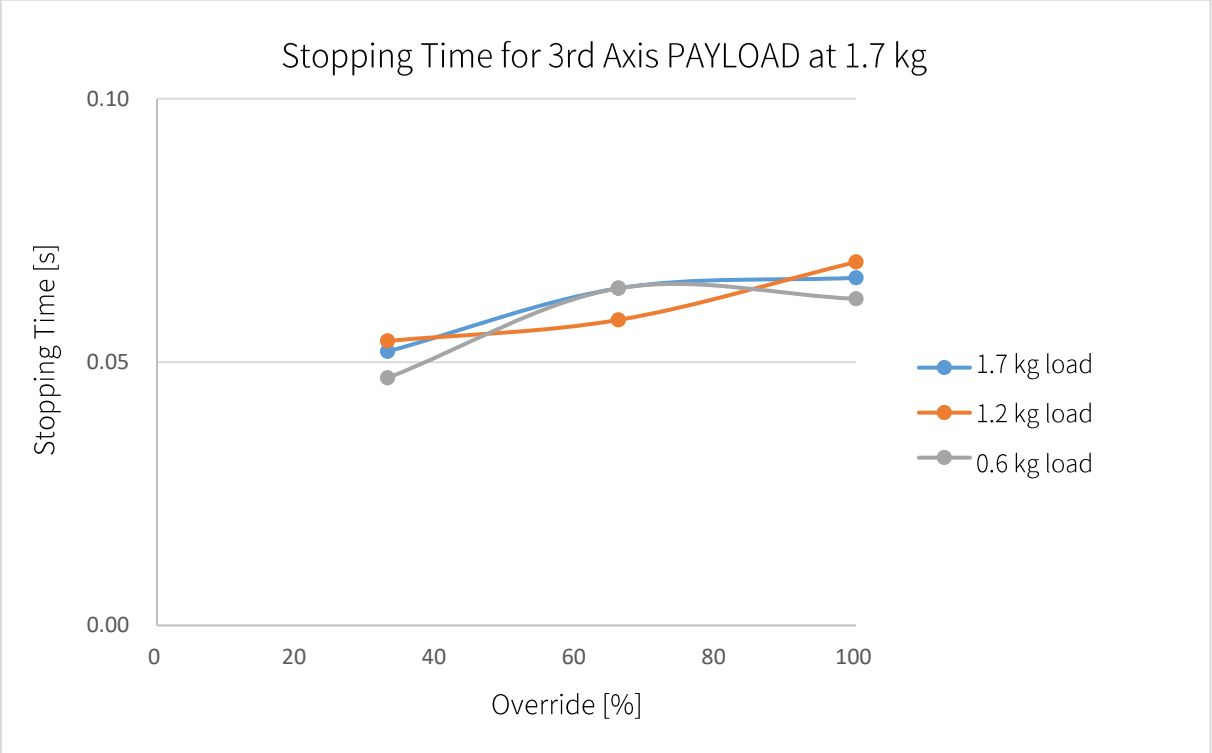


Fig. 34. 3rd Axis Stopping Time for Payload (1.7,0)

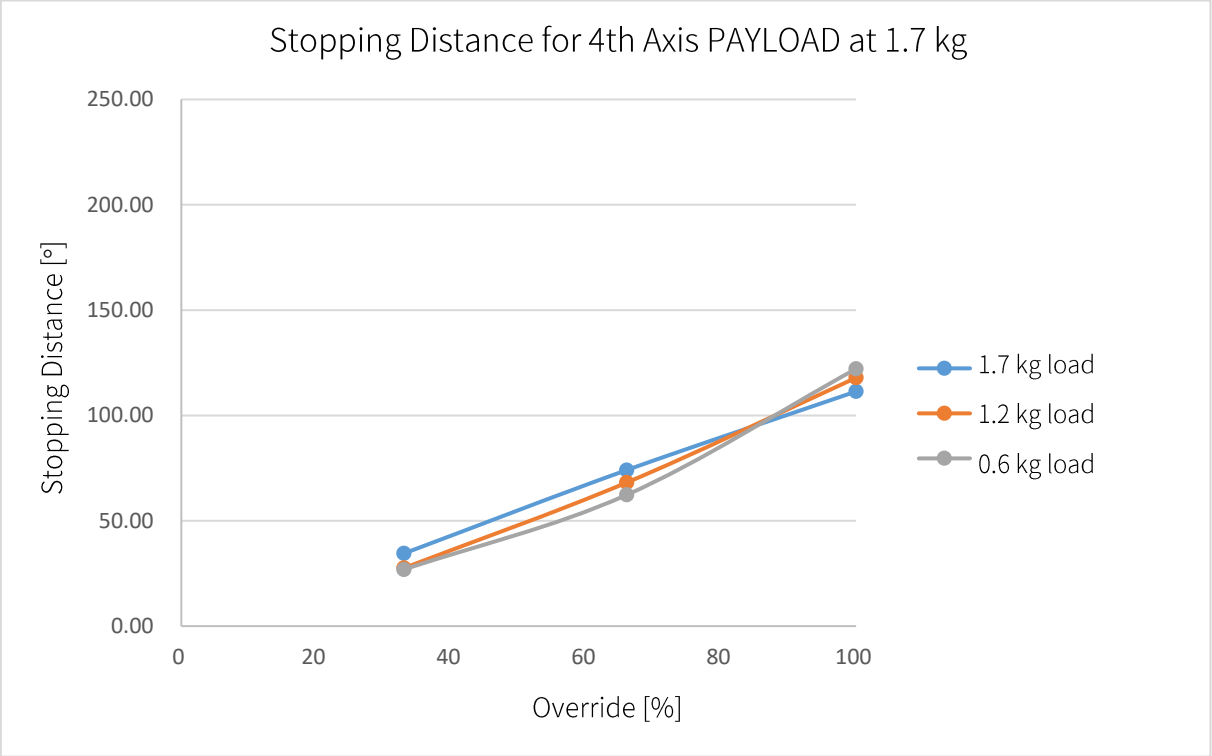


Fig. 35. 4th Axis Stopping Angle for Payload (1.7,0)

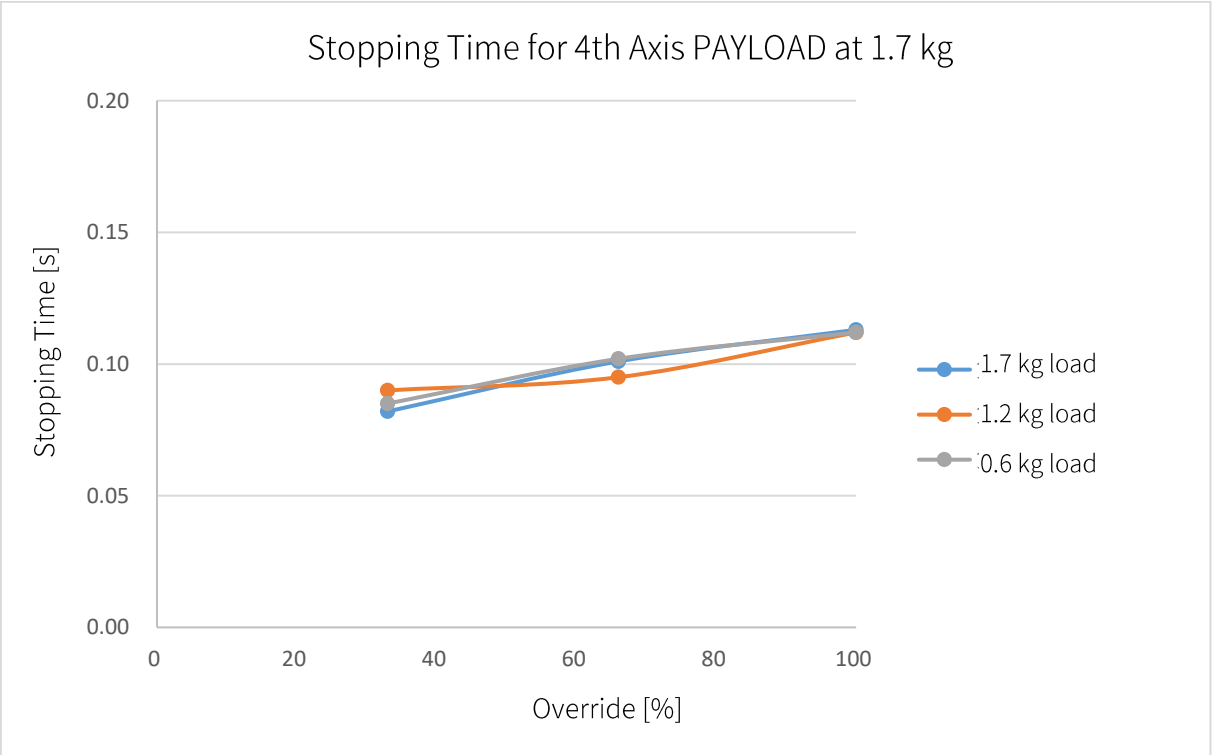
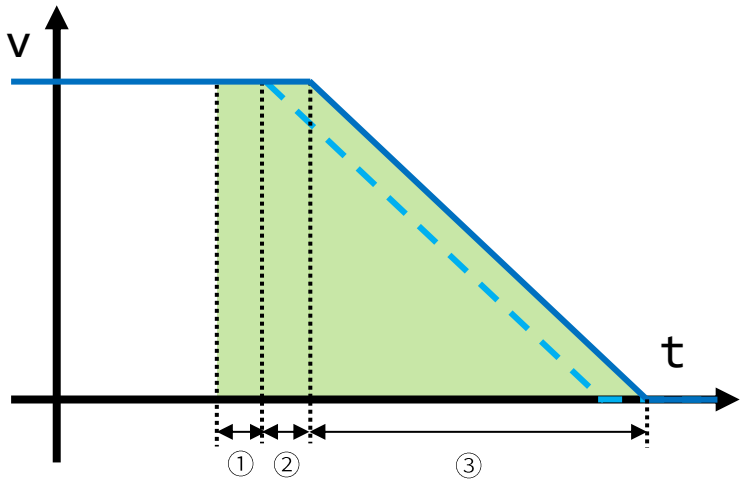


Fig. 36. 4th Axis Stopping Time for Payload (1.7,0)

2.2.2.4 Checking the Emergency Stop Time and Emergency Stop Distance

The time and distance from when an emergency stop is applied until the robot comes to a stop under the customer's operating conditions can be found by calculation or actual measurement.

Finding the Time and Distance Using Calculation



Velocity waveform when emergency stop is applied

The time from applying of the emergency stop signal until the robot comes to a stop includes three main elements.

① Time from applying the emergency stop signal until it is detected by the software
This is a maximum of approximately 20 ms.

② Servo tracking time delay
For the THE400, this is approximately 10 ms.

③ Deceleration time
The deceleration time can be calculated by the Velocity when the emergency stop is applied ÷ Emergency stop deceleration.
Calculate the velocity when an emergency stop is applied from the maximum velocity of each robot axis, override setting value, and SPEED setting value in the robot program.

THE400 Maximum velocity and emergency stop deceleration

	Maximum velocity	Emergency stop deceleration
1st axis	672 [°/s]	4800 [°/s^2]
2nd axis	780 [°/s]	9000 [°/s^2]
3rd axis	1120 [mm/s]	55000 [mm/s^2]
4th axis	1800 [°/s]	32000 [°/s^2]

From the values in ① to ③, the emergency stop time and emergency stop distance are calculated as follows.
Emergency stop time = ① + ② + ③
Emergency stop distance = (① + ② + ③ ÷ 2) × Velocity when emergency stop is applied

Calculation example: When 1st axis is operated at override 100% and SPEED = 50

Velocity when emergency stop applied = 672 [°/s] × 100/100 × 50/100 = 336 [°/s]
Deceleration time = 336 [°/s] ÷ 4800 [°/s^2] = 0.07 [s]

Emergency stop time = 0.02 [s] + 0.01 [s] + 0.07 [s] = 0.1 [s]
Emergency stop distance = (0.02+ 0.01+ 0.07 ÷ 2) × 336 [°/s] = 21.84 [°]

Finding the Time and Distance Using Actual Measurement

The HOST communication command EIM1 can be used to take an actual measurement of the stop time and stop distance.
For details on how to use the HOST communication command, refer to the Communication Manual.

After the robot is stopped by an emergency stop signal, the EIM1 command can be sent to obtain the following responses.

Send command
STXEIM1,[Axis number]ETX

Response command
STXOK[Operand length],[Emergency stop detection time],[Emergency stop detection position],[Emergency stop detection velocity],[Stop check time],[Stop check position]ETX

From these values obtained in this response, the stop time and stop distance can be found as follows.

Emergency stop time = [Stop check time] — [Emergency stop detection time]
Emergency stop distance = [Stop check position] — [Emergency stop detection position]

However, this value does not include the time in ①, and so to find the time and distance with more accuracy, add the time and distance for ① to the data obtained by the EIM1 command.

Emergency stop time = [Stop check time] — [Emergency stop detection time] + ①
Emergency stop distance = [Stop check position] — [Emergency stop detection position] + ① × Emergency stop detection velocity

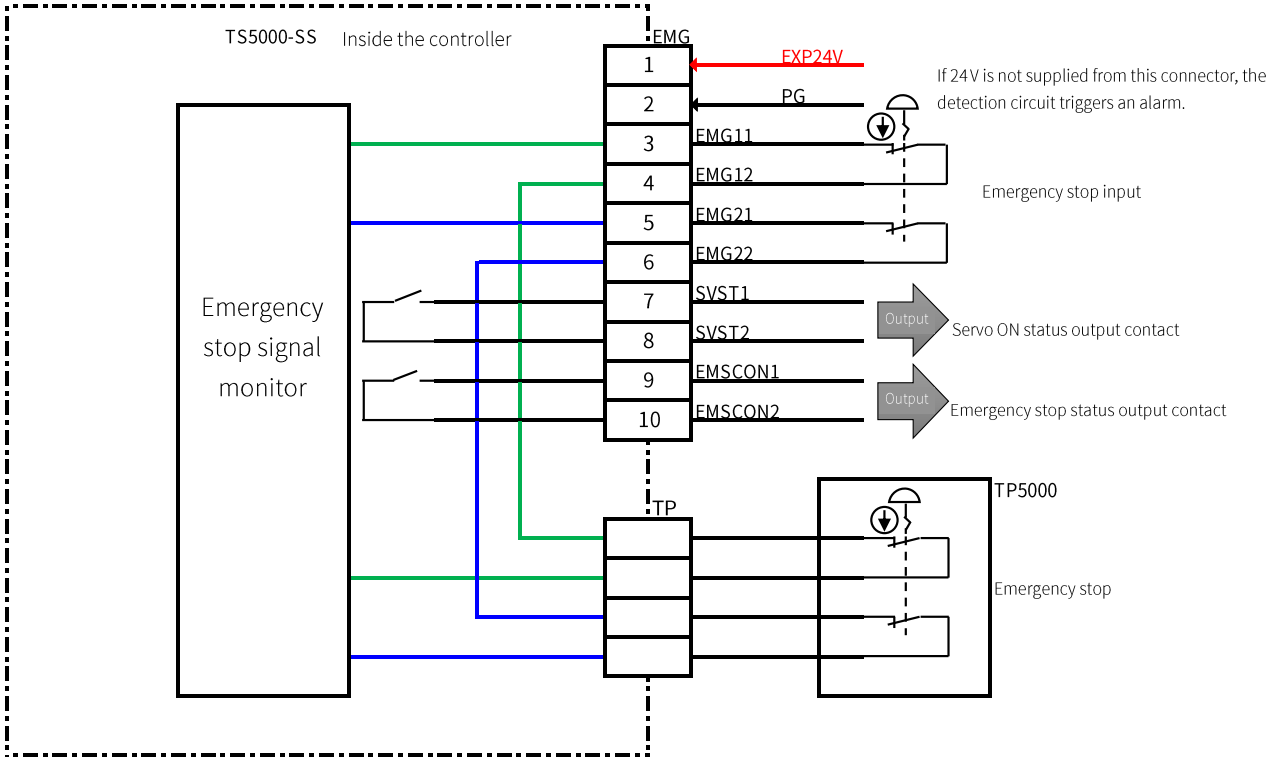
2.2.3 Safety Contact Input Function

The TS-5000-SS-**[1)] has a redundant safety contact input function that is separate from the emergency stop system. Connect an interlock switch such as a limit switch or foot switch that is linked to the door of the protective fence.

External emergency stop contacts are shown in the following figure.

Attach the emergency stop button to pins 3-4 and 5-6 of the EMG connector on the front of the controller to use this function.

1) For details on the model numbers, refer to STEA1237: TS5000-SS Controller Manual.

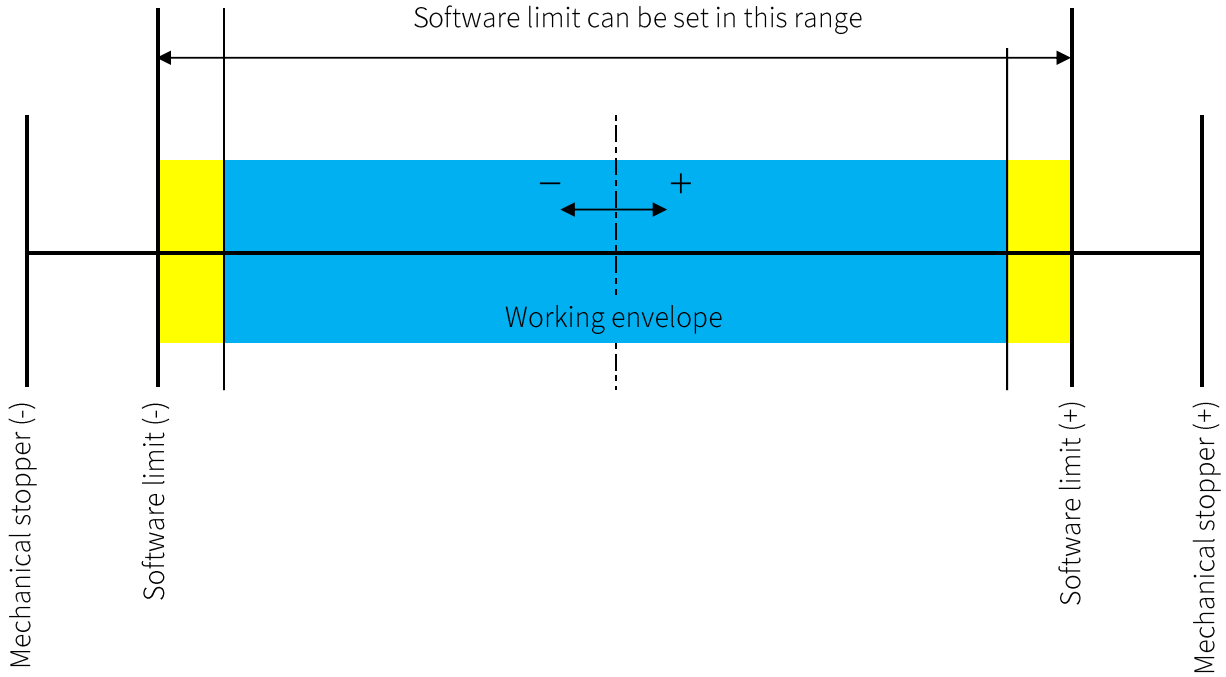


2.2.4 Work Envelope Limit Function

The robot is equipped with mechanical stoppers as a function to prevent overrun.
Mechanical stoppers are provided on the 1st, 2nd, and 3rd axes to prevent the robot moving parts from overrunning.
For some robot models, the positions of the mechanical stoppers can be changed for limiting the work envelope.
To change the positions of the mechanical stoppers, refer to "STEA1213: THE400 TS5000 Series Robot Manual: Installation".

A software limit can be set in the controller. The software limit is an auxiliary function for the mechanical stoppers that prevents collisions with the mechanical stoppers due to programming errors or mistaken operation during manual guidance. Set inside the robot work envelope that is limited by the mechanical stoppers.
For the procedure on changing the software limits, see “STEA1162: TP5000 Operation Manual”.

If the robot malfunctions, the software limit may be unable to limit the work envelope.
Use mechanical stoppers to implement safety protection functions for protecting the human body.

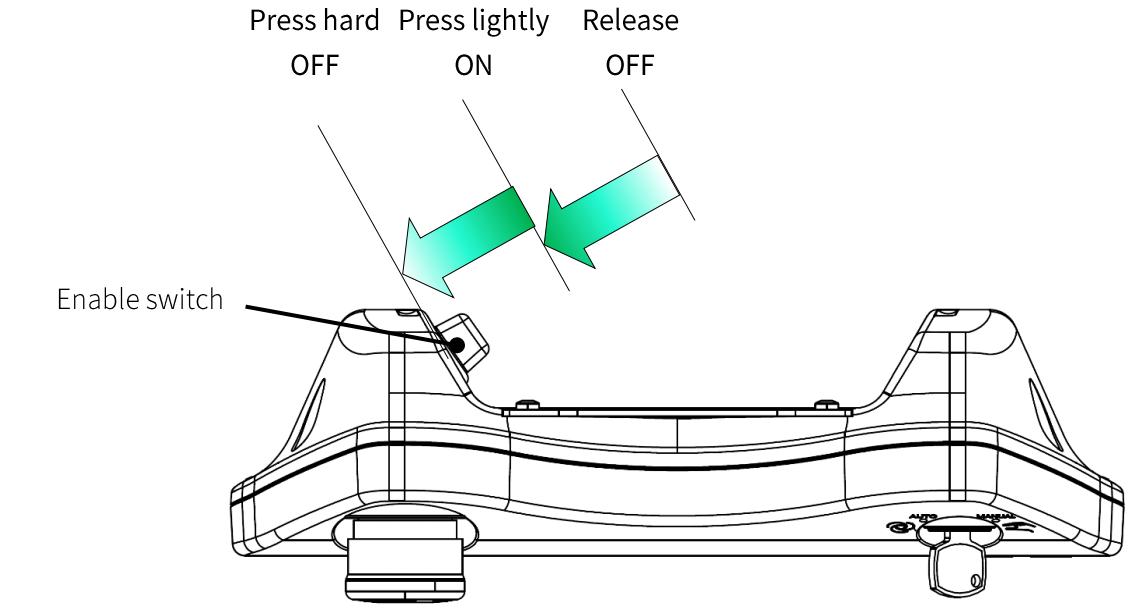


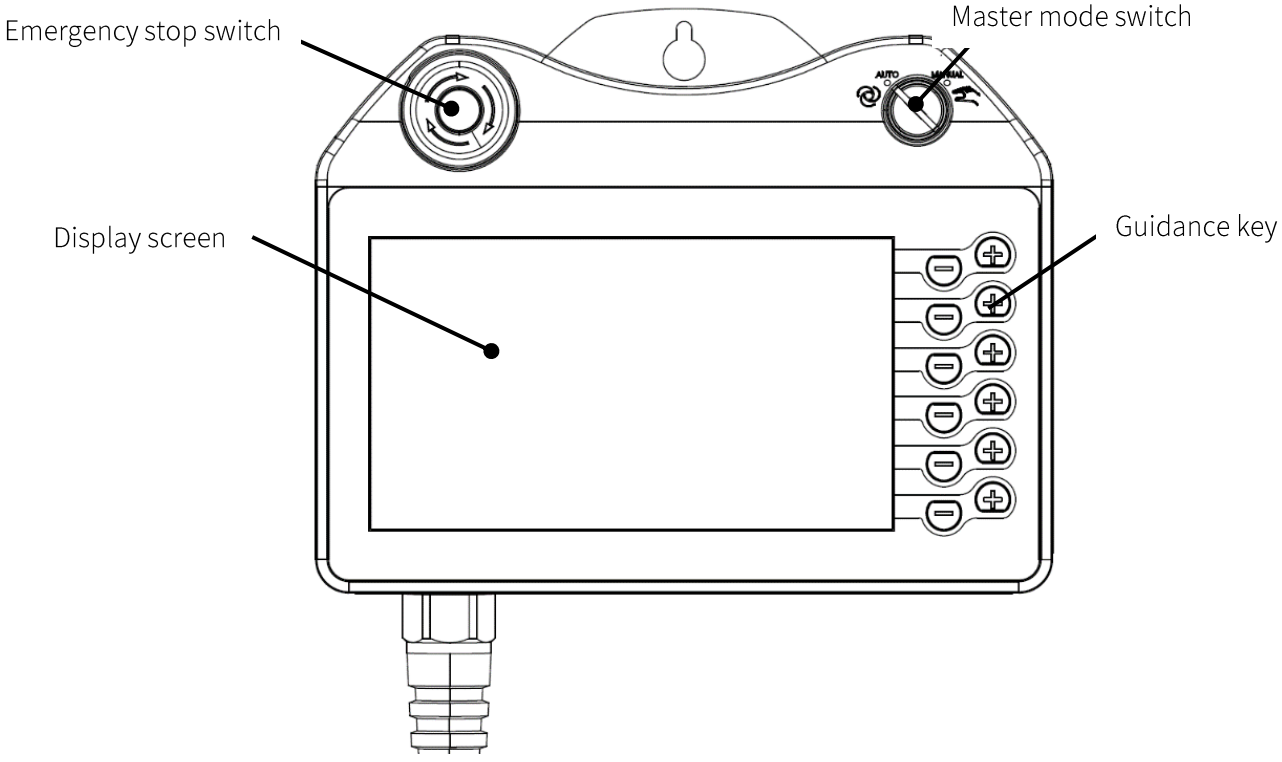
2.2.5 Enable Switch

An enable switch is provided on the teach pendant. When guiding the robot in manual mode, the operator must hold the teach pendant and keep the "press lightly" state for the enable switch to clear the emergency stop state and enable turning on of the servo power.

The enable switch is a three-position switch that can be "released" or "pressed hard" to quickly stop the robot and turn off the robot's power source.

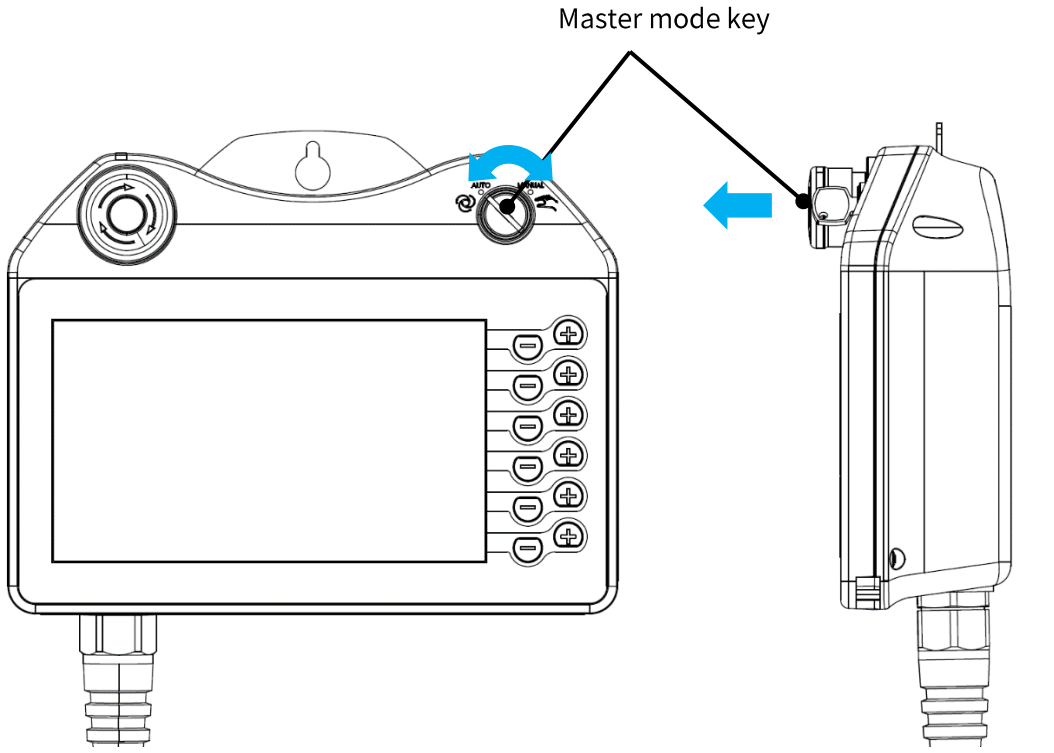
Also, the servo power cannot be turned on when the enable switch is set to OFF. When guiding the robot using the teach pendant, always keep the enable switch in the ON state.





2.2.6 Master Mode Key Switch

The TP5000 is provided with a selector switch for setting to MANUAL or AUTO modes. When the operator must enter the work envelope of the robot for teaching or inspection work due to unavoidable reasons, the operator can keep the key while performing the work to ensure that other operators cannot switch modes.



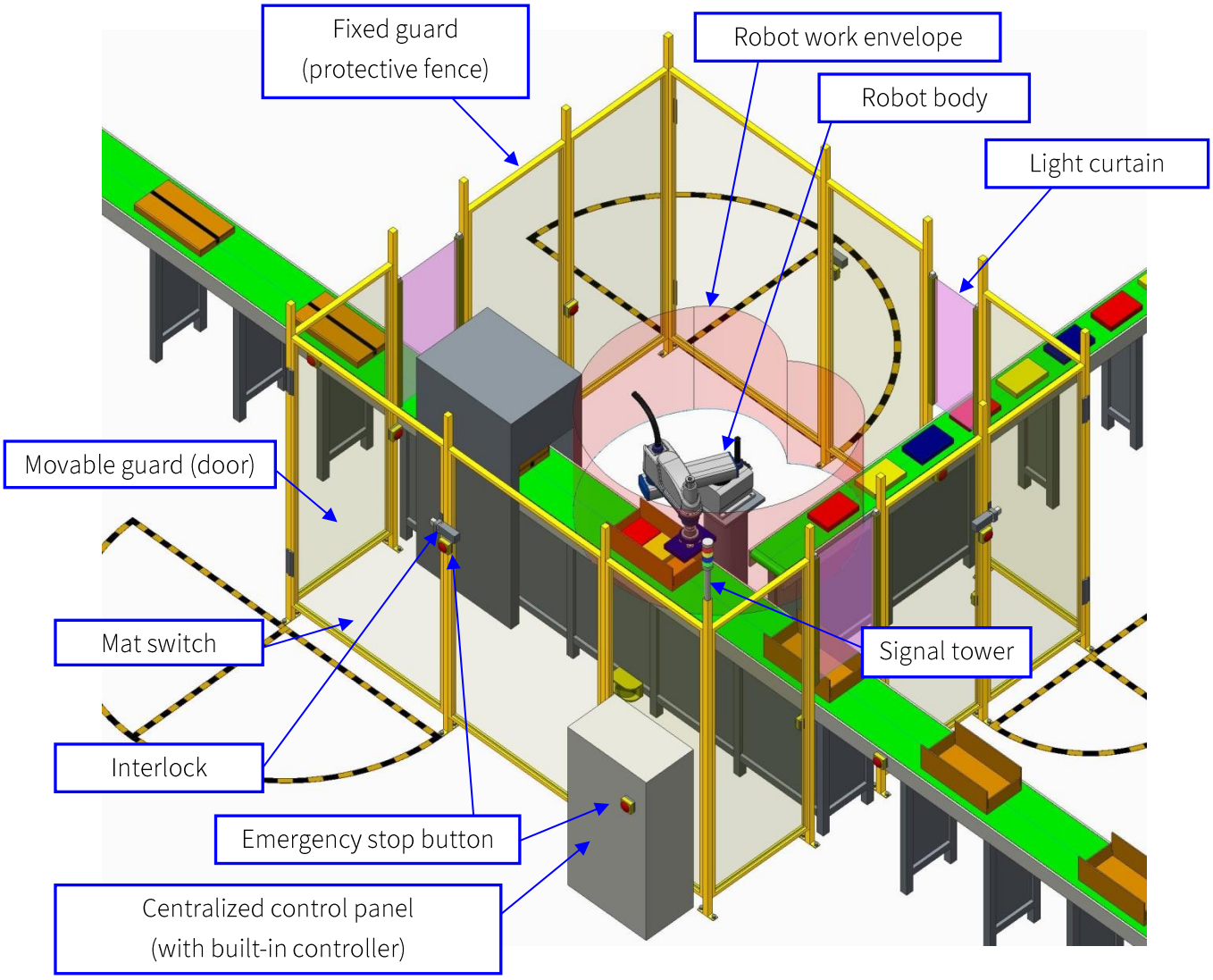
Master Switch Front View

Master Switch Side View

2.3 Examples of Safety Measures

Most accidents caused by robots arise due to unsafe human behavior. In order to use the robot, it is necessary to foresee when potential hazards may occur and avoid creating dangerous situations, and work must be performed only after all of the safety conditions are met. The following sections show examples of the main safety measures.

2.3.1 General Safety Information for Robot Operation



1) When installing the robot, a space must be provided for allowing work to be performed safely during and after installation (teaching, operation, maintenance, and disposal).
   - ISO 13854: Minimum gaps to avoid crushing of parts of the human body

2) Clearly delineate the hazard zones, and provide protective fences or similar barriers to prevent people from easily entering. A hazard zone is an area around a robot that may result in a dangerous situation if a person enters it.
   - ISO 14120: General requirements for the design and construction of fixed and movable guards
   - ISO13857: Safety distances to prevent hazard zones being reached by upper and lower limbs

3) Install the emergency stop switch in a location where it can be immediately pressed by a worker who determines that an abnormality has occurred inside or outside the protective fence or other barrier, and connect the switch to the external emergency stop contacts (EMS1B to EMS1C, EMS2B to EMS2C) of the controller.
   - ISO13850: Emergency stop function

4) The entry/exit door of the protective fence should be equipped with an interlock device such as an interlock switch, light curtain, or mat switch, and connected to the safety contact input of the controller so that the robot will stop when a person enters the hazard zone. The interlock device should be electrically isolated and be a Form B contact (normally closed during operation).
   - ISO14119: Interlocking devices associated with guards
   - IEC 61496-1,-2: Electro-sensitive protective equipment
   - ISO13855: Positioning of safeguards with respect to the approach speeds of parts of the human body
   - ISO13856-1: Pressure-sensitive protective devices

5) Install the controller outside the hazard zone at a location where the operator can oversee the robot's operation.

6) Carry out transportation, installation, teaching (teaching, programming, operation check), maintenance (inspection, adjustment, repair), and disposal in accordance with the laws, regulations, and standards specified in the respective country. The current status of the robot should be displayed at all times so that a third party cannot enter the area or operate the robot unintentionally.

7) Before starting work, perform the following pre-work inspection.
   Pre-work Inspection
   - There shall be no abnormality in the appearance of the robot body, controller, peripheral equipment, or cables.
   - There shall be no obstacles in or near the work envelope of the robot and peripheral equipment.
   - The emergency stop and other safety devices shall operate normally.
   - There shall be no abnormal noise or vibrations when the robot is operating.

8) Always use the teach pendant buttons while visually checking their operation.

2.3.2 Safety Information When Working

As a general rule, work such as teaching and inspection should be performed outside the hazard zone. When work must be performed in the hazard zone due to unavoidable reasons, be sure to follow the guidelines below.

1) Whenever possible, shut off the drive power (turn off the servo) for teaching and other work. Do not allow the drive power to be restored (servo turned on) automatically or by another operator. Whenever possible, shut off the primary power supply for inspections and other work. Do not allow the primary power to be restored automatically or by another operator.

2) Make sure that the switches for changing the robot operation status (start/stop switch, etc.) are not used by anyone other than the operator. (Display boards, locks on operation panels, assignment of supervisors, etc.)

3) Establish the following work regulations and follow them when performing work.
   ① Operation methods and procedures for starting up the robot, using switches, teaching, inspection, and their confirmation.
   ② Operating speeds appropriate for the situation (should be as slow as possible)
   ③ Signaling methods used between workers
   ④ What to do when an abnormality occurs
   ⑤ How to remedy abnormal conditions and confirm safety before restarting a robot that was stopped abnormally.
   ⑥ Other methods to prevent hazards caused by unintentional robot movements or mistaken operation, such as the following:
   - Methods to restrict the operation of switches that change the robot's operating status (start/stop switches, etc.) by non-operators (display boards, locks on operation panels, assignment of supervisors, etc.)
   - Methods of having operators in the hazard zone hold an emergency stop device (e.g. using a teach pendant)
   - Positions and orientations that ensure the safety of operators
   - Types of abnormalities and how to identify them (e.g., alarm check procedures)
   - Methods to prevent and reduce noise
   - Set-up changes and their confirmation (program change procedures, teaching data to be changed and its confirmation methods, etc.)
   - Methods of signaling between the operators of the robot and associated equipment

   * Work regulations should be developed by incorporating a wide range of opinions, including those of the operators and facility design engineers involved, to ensure that the regulations are appropriate for each site.

4) Perform work using at least two people. One person performs the work, and the other person monitors from outside the hazard zone. Each person works to prevent mistaken operation.

5) The operator should work in a position where the emergency stop button can be pressed at any time, and should also work in a location where, after fully checking the robot's work envelope and surrounding shielding, he or she can immediately evacuate in case of an abnormality.
Also, the operator must be careful to never turn his or her back to an operating robot.

6) The supervisor monitors the work from a location where the entire robot can be overseen and the emergency stop button can be pressed immediately in case of an abnormality. The supervisor also monitors that other people are kept away from the hazard zone.

2.3.3 Other Safety Information

1) Make sure that the grasping part of the robot has no protrusions or other hazardous parts, except for those necessary for work.
It is vital that the structure be able to grasp the target object stably even if the robot stops suddenly due to a power failure, breakdown, or emergency stop during operation.

2) If a malfunction occurs, turn off the power, eliminate the cause of the malfunction, service the peripheral equipment, and check operation at low speed once the equipment is fully recovered.
Even if the robot is stopped, do not immediately approach the hazard zone.

3) Turn off the power if the robot will not be used.

MEMO

The information contained in this section, "Safety Measures", is not necessarily sufficient and complete, and covers the most important points only due to space limitations. Before actually carrying out the work, read the documentation on safety measures compliant with the laws, regulations, and standards of the country where the robot will be used, or receive instructions from a consultant who is well-versed in the laws, regulations, and standards.
Also, conduct a risk assessment and implement risk reduction measures.

2.3.4 Recommended Protectors

Be sure to follow the standards, regulations, and laws of the respective country, and conduct a risk assessment before studying and selecting the product for use.

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with robot	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with robot	
Protective gloves	Protection part: Hand and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	
Ear plugs	Protection part: Ears Use: Protect from noise*1	

*1): For details on noise levels and measurement methods, refer to "STEA1213: THE400 TS5000 Series Robot Manual: Specifications". Based on the Guidelines for the Prevention of Noise-inducted Impairments (Notice No. 546) under Japanese law, we recommend wearing protective gear when working in environments where the robot alone generates noise of 85 dB or more.

3. Troubleshooting

This section describes what to do when an unexpected operation occurs.
If the problem persists even after implementing these remedies, the robot or controller may be malfunctioning. If you have purchased this robot in Japan, please contact the Shibaura Machine Service Department. If you have purchased this robot outside of Japan, please contact the company that you purchased it from, and follow the instructions from the person in charge.

Symptom	An alarm occurred
Cause	① The robot's self-diagnostic function was activated.
Remedy	① Tap the alarm button on the teach pendant to check the alarm code. Check the corrective action procedure for the alarm code that is found in the alarm manual.
Notes	Alarm causes and their remedies are shown in "STEA1165: Alarm Manual".

Symptom	The controller power does not turn on.
Cause	① The main power is not turned on. ② The power cable is not connected correctly.
Remedy	① Check whether the main power is turned on correctly. ② Refer to the controller manual, and check the connection for the AC power cable.
Notes	For details on the connection for the controller power cable, refer to "STEA1237: TS5000-SS Controller Manual, 4.1 Cable Wiring".

Symptom	The servo does not turn on.
Cause	① A stop signal is being input. ② A cable between the robot and controller is damaged, and an internal wire may be broken.
Remedy	① Check that the emergency stop switch, safety door, and other safety inputs are not left on. ② Check whether any cable between the robot and controller was damaged due to being stepped on or pinched. Replace any damaged cables.

Symptom	During program operation, the robot does not move to the location where it was taught to move.
Cause	① The origin position information or the teaching point data is shifted. ② The cables or tubes mounted to the robot are being pulled.
Remedy	① Set the origin position again, and perform teaching again. ② Turn off the servo for the robot, and check that there is no excessive tension on the cables or tubes when the arm is moved.
Notes	Refer to "STEA1213: THE400 TS5000 Series Robot Manual, 14. Robot Origin Position Detector Error", and set the origin position again.

Symptom	The robot tip (hand mounting section) vibrates.
Cause	① A hand that exceeds the allowable load conditions is being used. ② An acceleration/deceleration matching the payload conditions is not set.
Remedy	① Check the weight of the hand and workpiece and the load center of gravity offset to make sure that they meet the allowable load conditions. ② Vibrations may be eliminated by adjusting the speed and acceleration in the program. Re-examine the program.
Notes	For details on programming to adjust the allowable load conditions and acceleration/deceleration rates, refer to "STEA1213: THE400 TS5000 Series Robot Manual, 7.4 Allowable Load Requirements and Program Settings".

Symptom	The hand does not respond even when a signal is input.
Cause	① The hand is wired incorrectly. ② If you are using a pneumatic hand, the air supply may not be working properly. ③ A cable between the robot and controller is damaged, and an internal wire may be broken.
Remedy	① Refer to the Robot Manual about the hand wiring, and check the wiring. ② Check the tubes to make sure that they are connected correctly and are not crushed or damaged. ③ Check whether any cable between the robot and controller was damaged due to being stepped on or pinched. Replace any damaged cables.
Notes	For details on user wiring, refer to "STEA1213: THE400 TS5000 Series Robot Manual, 8. User Wiring", and for piping, refer to "STEA1213: THE400 TS5000 Series Robot Manual, 7.3 Hand Air Piping".

Symptom	The robot was stopped while the servo is turned on during operation.
Cause	The robot is waiting for an external signal. ① The external signal that is being input does not match the signal that the robot is waiting for. ② The device (sensor, etc.) that is inputting signals to the robot is malfunctioning.
Remedy	① Check the terminal arrangement of the user wiring and the created program, and modify it so that the external signals that are input will match the signals that the robot is waiting for. ② Operate the device that is inputting signals to the robot, and check on the teach pendant that the signals being input.
Notes	For details on user wiring, refer to "STEA1213: THE400 TS5000 Series Robot Manual, 8. User Wiring", and for details on checking the input signals, refer to "STEA1162: TP5000 Operation Manual, 6.3.2 Operations Performed on I/O Screen".

Symptom	The robot collided with a mechanical stopper.
Cause	① The software limit is set inside the limit range of the mechanical stoppers. ② The acceleration/deceleration is not set correctly.
Remedy	① If the limit range of the mechanical stoppers is narrowed from the factory defaults, the software limits must also be changed. Check the limit range of the software limits, and set the correct software limits. ② Do not operate under conditions exceeding the allowable load conditions. Check that the commands for adjusting the acceleration/deceleration for PAYLOAD and other commands are set correctly.
Notes	In the Shibaura Machine default settings, the software limit cannot be set outside the mechanical stoppers. For more information on setting the limit range, refer to "STEA1213: THE400 TS5000 Series Robot Manual, 6.3 Work Envelope". For details on programming to adjust the allowable load conditions and acceleration/deceleration rates, refer to "STEA1213: THE400 TS5000 Series Robot Manual, 7.4 Allowable Load Requirements and Program Settings".

Symptom	The power circuit breaker was tripped.
Cause	① The power cable may not be connected correctly. ② The robot is contacting a device or obstacles in the surrounding area. ③ The controller or robot may be malfunctioning.
Remedy	① Refer to the controller manual, and check the connection for the AC power cable. ② Check that the robot is not contacting a device or obstacle in the surrounding area. ③ The controller or robot must be replaced. Contact the Shibaura Machine Service Department.
Notes	For details on the connection for the controller power cable, refer to "STEA1237: TS5000-SS Controller Manual, 4.1 Cable Wiring".

Symptom	The I/O (INPUT connector/OUTPUT connector) is not operating normally.
Cause	① The input/output signal line connections may be incorrect. ② The ladder sequence may be incorrect.
Remedy	① Check the input/output signal line connections. ② Check that the standard ladder is inserted and that the ladder sequence is correct.
Notes	For details on connecting the input/output signal lines, refer to "STEA1225: Interface Manual, 5. Input Signal Line and 6. Output Signal Line". For details on the standard ladder sequence, refer to "STEA1168: TCmini (Internal PLC) Manual, 12.1 Standard Ladder Sequence".

Symptom	The alarm "8-014 Emergency stop switch ON" cannot be cleared.
Cause	① The emergency stop button on the TP5000 or the emergency stop for the safety device connected to the EMG connector (pins 3 to 6) may be still be on. ② The cable connected to the EMG connector (pins 3 to 6) may be broken. ③ The controller may have a broken wire, or the safety relay module may be malfunctioning.
Remedy	① Check the status of the emergency stop button. ② Check whether the cable connected to the EMG connector (pins 3 to 6) is broken. ③ If an alarm has occurred, make a note of it, and contact the Shibaura Machine Service Department.
Notes	For details on connecting the safety devices to an external emergency stop input, refer to "STEA1238: Interface Manual, 4. Input/Output External Power Supply Lines and Safety Input/Output Lines".

Symptom	The alarm "8-017 Safety switch ON" cannot be cleared.
Cause	① The enable switch on the TP5000 or the safety device connected to the EMG connector (pins 3 to 6) may not be in the enabled state. ② The cable connected to the EMG connector (pins 3 to 6) may be broken. ③ The controller may have a broken wire, or the safety relay module may be malfunctioning.
Remedy	① Check the status of the enable switch and safety devices. ② Check whether the cable connected to the EMG connector (pins 3 to 6) is broken. ③ If an alarm has occurred, make a note of it, and contact the Shibaura Machine Service Department.
Notes	For details on connecting the safety devices to a safety door input, refer to "STEA1238: Interface Manual, 4. Input/Output External Power Supply Lines and Safety Input/Output Lines".